



SACS



Dynamic Response

Version 24.00

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1 INTRODUCTION

1.1 OVERVIEW

The Dynamic Response program is designed to compute the dynamic responses of a structure subjected to dynamic excitation due to base motion such as in an earthquake, or dynamic forces due to periodic vibration or impact loads. The program can analyze base driven systems with input described either as a spectral input or as a time history input, and force driven systems with input described by a set of period forces or time history forces.

1.2 PROGRAM FEATURES

Dynamic Response analysis requires dynamic mode shape and mass files in addition to a Dynamic Response input file.

Some general features and capabilities of the program module are:

1. Ability to use a full structural model for use in Dynamic Response analysis.
2. Nonlinear fluid damping effects included automatically.

1.2.1 Earthquake/Base Driven Analysis

Both spectral earthquake and time history earthquake analyses are supported. Some of the seismic analysis capabilities follow:

1.2.1.1 *Spectral Earthquake*

1. API response spectra are built into the program.
2. Supports user defined response spectra.
3. Spectral motion can be described as acceleration, velocity, or displacement.
4. Modal combinations using linear, SRSS, peak plus SRSS, or CQC methods.
5. Ability to use a different response spectrum for each direction.
6. Combines seismic results with static results automatically.
7. Supports user defined power spectral densities.
8. Ability to generate response function for any joint degree of freedom.

1.2.1.2 *Time History Earthquake*

1. Includes earthquake time history libraries.
2. User defined input time histories.
3. Linear, quadratic, or cubic interpolation available for the time history input.
4. Variable time step integration procedure.

-
5. Automatic load case selection based on overturning moment, base shear, etc.
 6. Graphical representation of output variables.
 7. Create a force time-history file for the combined seismic and wave response analysis.

1.2.2 Force Driven Analysis

Force time history, Periodic and Engine vibration analyses are supported. The main capabilities and features for force driven analysis are detailed below:

1.2.2.1 Force Time History

1. Linear, quadratic, or cubic interpolation available for the time history input.
2. Input time histories may be saved to a file.
3. Automatic load case selection based on overturning moment, base shear, joint displacement, etc.
4. Variable time step integration procedure.
5. Time history plots including modal responses, overturning moments, base shear, etc.
6. Generation of equivalent static loads.
7. Generation of incremental loads for Collapse analysis

1.2.2.2 Periodic Vibration

1. Supports input forces and moments applied to any point at various frequencies and phase angles.
2. Automatic load case selection based on maximum joint displacement at a specific joint or at all joints.
3. Full plot capabilities including modal responses, overturning moments, base shear, etc.

1.2.2.3 Engine/Compressor Vibration

1. Supports mechanical unbalanced forces and gas torques in addition to reciprocating loads.
2. Linear and/or nonlinear interpolation of forces between running speeds.
3. User can select specific joints to monitor or monitor all joints.
4. Joint displacements can be compared and plotted versus D-line, SNAME and/or Military Specification allowables.
5. Allows user defined phasing of forces and moments within a load case.
6. Can automatically combine maximum response of various load cases.
7. Generates plots of input data versus time for any load case.

-
8. Calculates periodic forces amplitudes and periods from force versus time input.

1.2.3 Spectral Wind Analysis

The wind spectral fatigue and extreme wind analyses are supported. Some of the spectral wind analysis capabilities are as follows:

1.2.3.1 Extreme Wind

1. Determines dynamic amplification factors automatically.
2. Generates common solution file containing internal loads, stresses, reactions and displacements multiplied by its own dynamic amplification factor.
3. Includes cross correlation of modal responses using the Complete Quadratic Combination (CQC) modal combination technique.
4. Plots generalized force spectrum and response spectrum for each wind speed.
5. Uses Harris Wind spectrum.

1.2.3.2 Wind Fatigue

1. Uses Harris Wind spectrum.
2. Optionally creates Fatigue input file automatically.
3. Distributes wind speed utilizing a Weibull distribution.
4. Assumes Rayleigh distribution of RMS stresses.
5. Handles multiple wind directions in same analysis execution.

1.2.4 Ice Force Analysis

1.2.4.1 Ice Vibration

The ice vibration analysis capability includes the following features:

1. Automatically includes ice stiffness.
2. Maximum and minimum peak selection.
3. Automatic cycle count for fatigue analyses.
4. Creates fatigue input data automatically.
5. Full plot capabilities including ice forces, modal responses, overturning moments, base shear, etc.
6. Variable time step integration procedure.

1.2.5 Dynamic Impact Analysis

The dynamic impact analysis capability includes the following features:

1. Dynamic ship impact and dropped object analysis capabilities
2. Time history plots including modal responses
3. Generation of equivalent static loads for static analysis
4. Generation of incremental loads for Collapse analysis

1.3 PROGRAM STRUCTURE

The Dynamic Response program can be used to solve base motion time history or force driven systems.

1.3.1 Base Driven Systems

The base motion time history solution utilizes a variable step integration procedure that determines the largest time step size allowed for each situation such that results are within a specified accuracy while analysis execution time is optimized. This procedure allows the program to use small time steps only where required such as at points of rapid changes. The process can also account for fluid damping for submerged structures by using an equivalent fluid damping as an alternative to calculating the actual fluid forces at each step during the integration process.

1.3.2 Force Driven Systems

For force driven systems, the Dynamic Response program can predict the responses due to a set of periodic forces and moments applied to multiple points on the structure. These forces can be at different frequencies and phases with respect to each other. For time history force input, the same variable step integration procedure utilized for base driven systems is used to calculate the responses.

1.3.3 Impact Analysis

The Dynamic Response module can predict the response of structure resulting from an impact from a vessel or a dropped object. The process can also account for fluid damping for a submerged portion of the structure by using either an equivalent fluid damping or alternatively the program can calculate the actual fluid forces at each step during the integration process.

2 ANALYSIS PROCEDURE

The Dynamic Response program is generally used to modal responses in the form of velocity, acceleration, displacement or stress. This section details the analysis procedure used to determine the modal responses for the following:

- A. Base Driven Systems
- B. Force Driven Systems
- C. Spectral Wind
- D. Ice Force
- E. Dynamic Impact Analysis

The Dynamic Response program requires a Dynpac mode shape file, Dynpac mass file and a Dynamic Response input file. The following details the input for the various types of dynamic analyses.

2.1 BASIC ANALYSIS OPTIONS

Regardless of the analysis type, basic analysis options must be specified in the Dynamic Response input file.

2.1.1 Analysis Type

The analysis type is entered in columns 7-10 on the DROPT line. Enter 'SPEC', 'TIME', 'VIBR', 'WIND' or 'ENG' for spectral earthquake, time history earthquake, force driven periodic or time history, spectral wind or engine vibration analysis, respectively. Enter 'TCLP' to generate incremental loads for force/time history collapse analysis, enter 'SHIP' for dynamic ship impact analysis or enter 'DROP' for dynamic dropped object analysis.

2.1.2 Damping

Damping factors can have a profound effect on analysis results. The program has the ability to consider both structural and fluid damping.

2.1.2.1 Structural Damping

Structural damping input is required for any response analysis and is input using the SDAMP line. For single pass analyses, the structural damping value input on the SDAMP line should include all sources of damping including fluid damping if applicable.

Note: Fluid damping may optionally be specified or may be calculated automatically using the FDAMP line. When fluid damping is either specified or calculated by the program, the damping values on the SDAMP line should not include any damping due to the fluid.

If all modes have the same damping, the overall damping as a percent of critical is input in columns 11-15 on the SDAMP line and columns 21-70 should be left blank. The following shows total critical damping of 3.0% for all modes:

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	DROPT SPEC	15	+Z					
2	SDAMP	3.0						

If the damping value is different for various modes, the damping value for each mode must be specified in the appropriate columns. Damping values must be specified for each mode and must be expressed as a nonzero positive number. The sample below shows various damping values for the 15 modes to be included in the analysis.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	DROPT SPEC	15	+Z					
2	SDAMP		3.0	3.0	3.0	3.0	2.8	2.8
3	SDAMP		2.5	2.5	2.5	2.5	2.5	2.5

2.1.2.2 Fluid Damping

Fluid damping may be optionally considered during most dynamic response analysis. The program has the ability to calculate fluid modal damping automatically or to use damping values input by the user. Fluid damping options are specified on the FDAMP line following the SDAMP line.

2.1.2.2.1 Calculating Fluid Damping Automatically

When fluid damping is calculated by the program, the values are based on the nonlinear forces on the structure. For spectral analysis, an equivalent damping ratio is determined based on a particular amplitude.

Enter 'PC' in columns 7-8 if the program is to calculate fluid damping automatically. If nonlinear damping is to be used, enter 'NL' in columns 9-10. For time history analyses, enter the amplitude in columns 16-20 if a specific amplitude is to be used to calculate fluid damping.

2.1.2.2.2 Specifying Fluid Damping Directly

Fluid damping values may be specified directly by the user. If all modes have the same fluid damping, the overall damping as a percent of critical is input in columns 11-15 on the FDAMP line and columns 21-70 should be left blank. The following shows fluid damping of 2.0% for all modes:

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	DROPT SPEC	15	+Z					
2	SDAMP	3.0						
3	FDAMP	2.0						

If the damping value is different for various modes, the damping value for each mode must be specified in the appropriate columns. Damping values must be specified for each mode and must be expressed as a nonzero positive number.

Note: For single pass analysis, fluid damping must be included in the value specified for structural damping on the SDAMP line.

2.1.3 Mode Selection

By default, the response of all modes is considered in the dynamic response analysis. If the response of some modes is to be ignored, the number of modes to consider should be stipulated in columns 11-14

on the DROPT line. When the number of modes 'n' is specified, the program assumes that the first 'n' modes are to be considered unless mode numbers are designated using the MODSEL input line. For example, the following designates that modes 1-10 and modes 16-20 are to be considered in the analysis.

Note: The number of modes specified on the MODSEL line must be equal to the number of modes designated on the DROPT line.

	1				2				3				4				5				6				7				8			
	12345678901	23456789012	34567890123	45678901234	56789012345	67890123456	78901234567	89012345678	90123456789	01234567890	12345678901	23456789012	34567890123	45678901234	56789012345	67890123456	78901234567	89012345678	90123456789	01234567890	12345678901	23456789012	34567890123	45678901234	56789012345	67890123456	78901234567	89012345678	90123456789			
1	DROPT SPEC 15 +Z																															
2	SDAMP 3.0																															
3	MODSEL 1 2 3 4 5 6 7 8 9 10 16 17 18 19 20																															

2.1.4 Vertical Coordinate

The positive vertical coordinate axis (-X, +X, -Y, +Y, -Z or +Z) is entered in columns 17-18 on the DROPT line.

2.2 BASE DRIVEN ANALYSIS

The Dynamic Response program can be used to determine stresses, joint velocities, joint accelerations and joint displacements for both spectral and time history earthquake.

2.2.1 Spectral Earthquake

The Dynamic Response program can be used to determine response due to a response spectrum. Seismic analysis type, seismic load data and analysis output options are designated in the Dynamic Response input file in addition to the basic analysis options.

2.2.1.1 Analysis Type

Enter 'SPEC' in columns 7-10 on the DROPT line to designate a spectral earthquake analysis.

2.2.1.2 Seismic Load Data

For spectral earthquake analysis, the seismic load data is input after the LOAD header line in the form of a response spectrum or a power spectral density function. The program contains an automated API spectral analysis facility designated by the SPLAPI line along with a general response spectral analysis facility designated by the Spload line.

Note: Each seismic input load requires either a SPLAPI or a SPLOAD line.

2.2.1.2.1 Automated API Spectral Analysis

The automated facility contains API Soil Type A, B and C response spectra in addition to supporting user defined normalized response spectra. Each seismic load to be defined by one of the API spectra is input using a SPLAPI line as follows:

Enter the response factor or 'G' factor which defines the ratio of effective horizontal ground acceleration to gravitational acceleration in columns 11-15. The soil type or the ID of the user defined response spectrum and the Directionality factor to be applied to the 'G' factor must be specified for the X, Y and Z directions in columns 16-36. The method used to combine modal results is designated in

columns 38-41. Enter 'SRSS', 'PEAK', 'PRMS' or 'CQC' for square root of the sum of the squares, linear addition of absolute values, peak plus SRSS or complete quadratic combination, respectively.

Note: Structural damping only is assumed when using the automated API spectral analysis. Also, the ID of the user defined normalized response spectrum may be input in place of the soil type.

2.2.1.2.2 General Spectral Response Analysis

The general spectral analysis capabilities allow seismic loading to be defined using API or user defined input spectrum. API spectra may be referenced or response spectrum or power spectral density function data may be specified. Regardless of whether API spectra or user defined data is used, general load options and load data must be specified on the SPLOAD line as follows:

Specify the damping type 'SDO' structural damping only, 'FDS' equivalent fluid damping at specified amplitude or 'FDA' equivalent fluid damping at actual amplitude in columns 21-23. For 'FDS' damping, enter the damping amplitude in columns 39-44 if different from the value specified on the FDAMP line.

Note: Options 'FDA' and 'FDS' require that fluid damping input be specified using the FDAMP line.

The method used to combine modal results is designated in columns 25-28. Enter 'SRSS', 'PEAK', 'PRMS' or 'CQC' for square root of the sum of the squares, linear addition of absolute values, peak plus SRSS or complete quadratic combination, respectively.

Enter the response factor or 'G' factor which defines the ratio of effective horizontal ground acceleration to gravitational acceleration in columns 45-50. The directionality factor to be applied to the 'G' factor must be specified for the X, Y and Z directions in columns 51-56, 57-62 and 63-68, respectively.

2.2.1.2.3 API Spectrum

Enter the spectrum source, either 'API', 'APIA', 'APIB', or 'APIC' for API spectra in columns 9-12 on the SPLOAD line. Leave columns 15-18 blank.

2.2.1.2.4 User Defined Spectra

User defined data may be specified in the form of a normalized response spectrum, a general response spectrum or a power spectral density function.

Normalized response spectra are used in conjunction with the SPLAPI line by specifying the ID of the spectrum instead of a soil type in columns 22, 29 or 36.

Normalized user defined response spectrum data may be used to define additional soil types to be used in conjunction with the API spectral earthquake options. The spectrum data is specified using RSPU1 and RSPU2 lines immediately before the SPLAPI line. The first RSPU1 line requires the number of damping values (maximum of 3) in column 10 along with the ID of the spectrum in column 8. The spectrum data is entered on the RSPU2 line and includes the damping ratio as a percent of critical damping in columns 9-14, then the period and normalized spectrum value of each point of the spectrum in columns 21-80.

Note: Up to fifteen spectrum points may be defined by repeating the RSPU2 line with the same damping ratio specified on each line.

The following describes a user defined response spectrum to be used as soil type 'F' in the X direction with 5 percent critical damping and is defined by 5 sets of periods and values.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	RSPU1	F	1					
2	RSPU2	5.0	0.04	1.0	0.05	1.0	0.13	2.0
3	SPLAPI	0.2	1.0F	1.0A	0.5A	CQC		N

The general response spectrum and the power spectral density function are used with the SPLOAD line by entering the spectrum source in columns 9-12 on the SPLOAD line as 'LINE' if the spectrum data is defined on subsequent input lines, or 'PREV' if the spectrum is to be used from the previous seismic load case. Enter the spectrum type, 'RSP' for response spectrum or 'PSD' for power spectral density in columns 15-17 on the SPLOAD line. Acceleration 'A', velocity 'V' and displacement 'D' spectra are supported. Specify the spectrum form by entering the appropriate letter in column 18 if the user defined data is in the form of a response spectrum.

User defined response spectrum data is specified using RSPSPC lines immediately following the SPLOAD line. The first RSPSPC line requires the number of damping values in columns 7-10. Enter the number of points defined on the curve in columns 7-10 on the second RSPSPC line along with the critical damping for the curve in columns 11-16. The first two points on the curve, defined by a period and a response value are entered in columns 21-60. Additional points on the curve are defined in pairs in columns 21-60 on subsequent RSPSPC lines.

For example, the following describes a user defined response spectrum defined by 5 sets of periods and accelerations (acceleration form) with 5 percent critical damping.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	SPLOAD	LINE	RSPA	SDO	CQC	0.2	1.0	1.0
2	RSPSPC	1						
3	RSPSPC	5	5.0	0.04	1.0	0.05	1.0	
4	RSPSPC			0.13	2.5	0.50	2.5	
5	RSPSPC			5.0	0.275			

User defined power spectral density data is specified using PSDSPC lines immediately following the SPLOAD line. The first PSDSPC line requires the number of frequency values in columns 7-10. The first two points on the curve, defined by a frequency and a spectral density value are entered in columns 21-60. Additional points on the curve are defined in pairs in columns 21-60 on subsequent PSDSPC lines.

For example, the following describes a user defined power spectral density function defined by 5 sets of frequencies and spectral density values.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	SPLOAD	LINE	PSD	SDO	CQC	0.2	1.0	1.0
2	PSDSPC	5						
3	PSDSPC		0.1	0.00005	1.00	0.0005		
4	PSDSPC		1.50	0.001	2.00	0.0012		
5	PSDSPC		5.00	0.0025				

2.2.1.3 Output Options

By default, seismic load cases are created when performing a spectral earthquake analysis. The program also has the ability to output load combinations consisting of seismic and static results, equivalent static

loads for nonlinear analysis, response functions and joint data including displacement, velocity and acceleration.

2.2.1.3.1 Static + Seismic Combinations

The Dynamic Response has the ability to optionally combine seismic results with static results as part of the earthquake analysis. When using this feature, the program creates four seismic+static load combinations, two for element check and two for joint can check, for each seismic load case as follows:

LC	Combine Type	Description
1	PRST	Element code check case, seismic axial tension
2	PRSC	Element code check case, seismic axial compression
3	PRST	Joint can check case, seismic axial tension
4	PRSC	Joint can check case, seismic axial compression

Note: This feature requires that the static solution file exist prior to execution of the seismic analysis. It also requires that all seismic load cases are full seismic load cases containing the responses for all directions (i.e. X, Y and Z responses). When using seismic load cases containing only part of the seismic response, these load cases must be combined (using 'SRSS') by the user prior to manually combining with static solutions.

The seismic and static combination information is input using the STCMB line. Enter the factor to be applied to the seismic loads when combined with the static loads for the purpose of member and plate element check in columns 8-12. The factor to be applied to seismic loads when combined with static loads for joint check is input in columns 13-17. Enter each of the static load cases to be combined with the seismic load cases and the load factor to be applied. Since spectral earthquake results are valid only at the joints of the structure, it is recommended that the JO option is used in columns 27-28 of the OPTIONS line when generating the static solution file.

For example, 105% of load cases 8 and 9 contained in the static solution file are to be combined with the seismic solution. For element check and joint can check, seismic stresses are to be factored by 1.0 and 2.0, respectively.

	1	2	3	4	5	6	7	8
1	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
2	DROPT SPEC	15	+Z					
3	SDAMP	5.0						
4	STCMB	1.0	2.0	8	1.05	9	1.05	
5	LOAD							
6	SPLOAD	APIA	SDO	CQC	0.2	1.0	1.0	0.5 N

Note: The STCMB line should follow the SDAMP, FDAMP and MODSEL lines in the input file.

2.2.1.3.2 Equivalent Static Loads

Equivalent static loads used to simulate earthquake loads for nonlinear analysis may be created using the EQKLOD line. The load case can be created to represent either the actual base or actual overturning moment by designating 'S' or 'M' in column 8.

By default, modal results are added together such that the corresponding load represents either base shear or overturning moment. The load case may also be generated with the sign reversed to simulate load reversal by specifying 'R' in column 10. To obtain one load case corresponding to the standard loading and an additional loading representing the reversal, enter 'B' in column 10. To obtain one load case corresponding to loading in "all" directions, enter 'A' in column 10. In this case the number of directions must be specified in columns 14-16, with a default value of 20 and a maximum of 100.

The load cases created may be appended to an existing model or structural data file. If the existing file contains loading to be used by the subsequent nonlinear analysis, the load case to assign to the generated loads may be assigned by designating the number of existing load cases to skip in columns 11-13.

The program prints the response in the X (0.0 degree) and Y (90.0 degree) directions. When the structure is responding primarily in the X direction, these responses do not occur at the same time. The equivalent static load procedure assumes that the primary structure response may occur in any direction during the earthquake event (not only along the X or Y axes). The response of the structure is therefore calculated for 20 directions (every 18 degrees). For each of these twenty directions, the base shear in that direction and the moment about that direction are determined. Equivalent static loads are then generated for the highest base shear.

A separate load case may be output for each mode by entering 'M' in column 17 of the EQKLOD input line.

The seismic load cases created have the earthquake EQS loading plus the load cases specified on the STCMB line included in each load case created. The load cases on the STCMB line are factored by the appropriate load factor indicated on the STCMB line. The joint and member load cases factors on the STCMB line are ignored.

2.2.1.3.3 Response Functions

A frequency or period response functions may be generated at specific locations on the structure using the RSFUNC line. Up to six functions may be generated for each RSFUNC line designated. Enter the joint name, the degree of freedom and the damping to be used in columns 7-16. Additional functions may be generated by specifying the joint, DOF and damping in columns 17-66. The number of points used to define the functions is designated in columns 67-69 while the function type is designated by 'P' (period) or 'F' (frequency) in column 70. Plots options are specified in columns 73-77.

As many RSFUNC lines as required to designate the desired number of functions may be used.

2.2.1.3.4 Joint Results

Joint results such as velocity, acceleration and displacement may be reported for a particular seismic load case by entering 'V', 'A' or 'D' in columns 29-31 on the corresponding SPLOAD line. Results may also be reported for all seismic load cases by specifying the print selections in columns 25-27 on the DROPT line.

2.2.1.3.5 Low Level Earthquake Analysis

Low level earthquake analysis based on API-WSD or API-LRFD may be accomplished using the methods from the previous paragraphs. To specify low level earthquake analysis, the API code requires description of a rare, intense earthquake for analysis. The following sample specifies conditions for a rare, intense earthquake and the proper load combinations for use in low seismic activity zones per API.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	DROPT SPEC 15 +Z							
2	SDAMP 5.0							
3	STCMB 1.0.0001 1 1.0							
4	LOAD							
5	RSPU1 X 1							
6	RSPU2 5.0	0.04	1.0	0.05	1.0	0.13	2.5	0.3
7	RSPU1 Y 1							
8	RSPU2 5.0	0.04	1.0	0.05	1.0	0.13	2.5	0.49
9	RSPU1 Z 1							
10	RSPU2 5.0	0.04	1.0	0.05	1.0	0.13	2.5	0.61
11	SPLAPI 0.25	1.0X	1.0Y	0.5Z	CQC			N

The STCMB line specifies an element load case factor of 1.0 and a punching shear factor of .0001. This effectively eliminates seismic loads from load cases 3 and 4 generated by the SPLAPI line. Load case 3, which is effectively a dead load case, will be used subsequently in joint can low level earthquake analysis. See Joint Can manual for implementation of low level earthquake analysis in joint strength check.

2.2.1.4 Combining with Static Results

The program creates a common solution file containing end forces, stresses, reactions, and displacements for each seismic load set specified in the Dynamic Response input file. Because these results are obtained by combining modal results using RMS techniques, end forces, stresses, etc. could have either positive or negative values. Therefore, when manually combining spectral earthquake results with static results, the PRST and PRSC combine options must be used to ensure that the maximum response is considered.

2.2.2 Time History Earthquake

The Dynamic Response program can be used to determine response due to a base driven time history. Up to three separate time histories may be used for any analysis. Analysis type, seismic load data and analysis output options are designated in the Dynamic Response input file in addition to the basic analysis options.

2.2.2.1 Analysis Type

Enter 'TIME' in columns 7-10 on the DROPT line to designate a time history earthquake analysis.

2.2.2.2 Load Options

For time history earthquake analysis, input loading and load options are input after the LOAD header line using the THLOAD, THFACT, TIME and THBEGIN input lines.

Note: Each time history load is defined by using a separate set of these lines.

2.2.2.2.1 Damping Method

Note: For nonlinear fluid damping, the fluid forces are calculated at every time step during the integration. This option requires the program calculated fluid damping option 'PC' on the FDAMP line.

2.2.2.2.2 Interpolation Scheme

2.2.2.2.3 Directionality Factors

2.2.2.2.4 Integration Parameters

For example, the following describes a time history function specified in the input file. Structural damping only is used in conjunction with linear interpolation as designated on the THLOAD line. One time history function is used with directionality factors of 1.0, 1.0 and 0.5 applied to it for the X, Y and Z directions, respectively. The start time is 0 seconds and end time 25 seconds. Output is requested at every 0.25 seconds.

	1				2				3				4				5				6				7				8			
1	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890					
2	LOAD																															
3	THLOAD	LINE				SDO				LN																						
4	THFACT	1.0				1.0				0.5																						
5	TIME	0.0				25.0				0.25																						

Time history data may be specified in the Dynamic response file or may be read from an external data file. The source of the time history data is designated in columns 9-12 on the THLOAD line. Enter 'LINE' if the time history data is defined on subsequent input lines, 'FILE' if the data is defined in an external file or 'PREV' if the time history data is to be used from the previous seismic load case.

Specifying data in the input file requires that overall parameters be specified on the THBEGIN line. Up to 3 separate time histories may be defined for a particular input load. Enter the number of time histories to be defined in the file in columns 8-10 and the name identifying the time history in columns 22-25. The

type, either acceleration, velocity, displacement or gravity acceleration is designated by 'A', 'V', 'D' or 'G' in column 30, respectively.

2.2.2.3.2 Time History Load Data

The time history data may be entered in standard format, compressed format or via an external input file.

Standard Format

Specify 'STD' in columns 14-20 on the THBEGIN line for standard input format.

Time history load data is specified using a THDATA line for each time point. For any time point, enter the time in columns 11-20 and the value in columns 21-30. If more than one time history is to be defined, enter the value for the second function corresponding to this time in column 31-40. The value for the third function is input in columns 41-50. The following illustrates one acceleration time history input using standard format.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	LOAD							
2	THLOAD	LINE	SDO	LN				
3	THFACT	1.0	1.0	0.5				
4	TIME	0.0	25.0	0.25				
5	THBEGIN	1STD	TH1	A				
6	THDATA		1.0	0.0				
7	THDATA		2.0	40.0				
8	THDATA		2.5	0.0				
9	THDATA							

Note: The first time point of the time history function is assumed to be zero. The first time point entered by the user must be greater than zero. The time history is terminated by using a THDATA line with all field left blank.

Compressed Format

The time history data may be entered in compressed format by specifying 'CMP' in columns 14-20 on the THBEGIN line.

Time history load data is specified using a THCOMP line for each time point. Compressed data is assumed to be specified with the constant time interval specified in columns 14-20 on the THBEGIN line. Enter the time history value in columns 11-70. The THCOMP line with 'END' designated in columns 8-10 signifies the end of the input data.

The following illustrates one acceleration time history input using compressed format. The constant time interval is 0.25 seconds as designated on the THBEGIN line.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	LOAD							
2	THLOAD	LINE	SDO	LN				
3	THFACT	1.0	1.0	0.5				
4	TIME	0.0	25.0	0.25				
5	THBEGIN	1CMP	0.25	TH2	A			
6	THCOMP	0.0	40.0	0.0				
7	THCOMP							

External File Input

When specifying time history load data for time history earthquake, data which would normally be specified using THDATA lines is input in an external input file without data labels. The external file begins with a single record. This record begins with a four-character name as specified on the THLOAD data record. The next four columns, columns 5-8, specify the number of time history functions as specified in columns 8-10 of the THBEGIN data record. The type, either acceleration, velocity, displacement or gravity acceleration is designated by 'A', 'V', 'D' or 'G' in column 10, respectively. The input units inches, feet, centimeters, millimeters and meters are input by specifying 'IN', 'FT', 'CM', 'MM' and 'ME' in columns 12-13 respectively.

The following records are the equivalent of THDATA records in standard format, but are input with specific field widths and no data labels. Columns 1-12 specify the time point; this is equivalent to columns 11-20 of the THDATA record. Columns 13-24, 25-36 and 37-48 consist of the time history values for the first, second and third function, respectively. The number of time history function values specified must correspond to the value in columns 5-8 of the first record. Subsequent records specify other time history points. All time history points specified must have time points greater than zero. The final record has a time point of 0.0.

The following is an example of external file input. In order for this file to be used, the dynamic response input file must have a THLOAD record with time history input source field (columns 9-12) of 'MXCT'. The input specifies three time history function values are specified in the forthcoming records. The input time values of 0.00, 0.02, 0.04, 0.06, 0.08 and 0.10 are data in the first twelve columns; the three time history function values are specified in columns 13-24, 25-36 and 37-48. Typical data input would consist of many more records. The last data record has a time point of 0.0.

	1	2	3	4	5	6	7	8
1	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
2	MXCT	3						
3	0.000000E+00	-.551180E+000	.118110E+000	.944880E+00				
4	0.200000E-01	-.425196E+010	.748030E+00	-.905510E+01				
5	0.400000E-01	-.397637E+010	.267716E+01	-.108267E+02				
6	0.600000E-01	-.346456E+010	.114173E+01	-.156299E+02				
7	0.800000E-01	-.374015E+010	.114173E+01	-.153543E+02				
8	0.100000E+00	-.472440E+010	.212598E+01	-.236220E+01				
9	0.000000E+000	.000000E+000	.000000E+000	.000000E+00				

2.2.2.4 Output Options

The time history earthquake analysis creates load cases, prints and plots modal responses, base shear and overturning moment in addition to joint accelerations, velocities and displacements. Analysis output options are designated in the output options fields in columns 33-59 on the THLOAD line.

2.2.2.4.1 Load Case Creation

The Dynamic Response program has the ability to create a load case corresponding to the time point having maximum overturning moment and/or maximum base shear by specifying 'MXM' or 'MXS' in the output options fields on the THLOAD line, respectively. Enter 'ALL' if load cases are to be created at for all time points.

2.2.2.4.2 Create Wave Response Force Time-History Input File

The Dynamic Response program can create the time-history input file for the Wave Response program to analyze the combined effects of the seismic loads and the wave loads on the model. To create the time-history input file for all time steps, enter 'ALL' and 'WVR' in the output options fields on the THLOAD line. The base-driven joint loads are printed to the Wave Response time-history input file (wvrthi.*) which can be imported for the wave time-history analysis.

2.2.2.4.3 Modal Response Data

Modal responses versus time may be printed and/or plotted by specifying 'PRT' and 'PLT', respectively, in the output options fields on the THLOAD line.

2.2.2.4.4 Base Shear and Overturning Moment Plots

Base shear and overturning moment plots may be generated by entering 'PLM' and 'PLS' in one of the output option fields located on the THLOAD line.

2.2.2.4.5 Joint Results

Joint results including acceleration, velocity and displacement may be plotted and listed for up to sixteen joints. Joint plot options are specified in the output options fields on the THLOAD line.

Joint acceleration options include:

'JMA' Prints maximum and minimum values for joint acceleration for each direction.

'JPA' Same as JMA plus plots acceleration time history

'JTA' Same as JPA plus prints acceleration time history data

Note: Joint acceleration options are mutually exclusive. Only one of the options may be selected.

Joint velocity options include:

'JMV' Prints maximum and minimum values for joint velocity for each direction.

'JPV' Same as JMV plus plots velocity time history

'JTV' Same as JPV plus prints velocity time history data

Note: Joint velocity options are mutually exclusive. Only one of the options may be selected.

Joint displacement options include:

'JMD' Prints maximum and minimum values for joint displacement for each direction.

'JPD' Same as JMD plus plots displacement time history

'JTD' Same as JPD plus prints displacement time history data

Note: Joint displacement options are mutually exclusive. Only one of the options may be selected.

Any number of joints may be designated for plots and reports. Joints to be output are specified using JTNUM lines immediately following the THLOAD line.

The following input illustrates some of the output options. Two load cases, one corresponding to the time of maximum base shear and one corresponding to the time of maximum overturning moment, are created. Base and overturning moment time histories are to be plotted in addition to joint acceleration and displacement plots for joints 601, 603, 605 and 607.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	LOAD							
2	THLOAD	LINE	SDO	LN	MXMMXS	PLMPLS	JPAJPD	
3	JTNUM	601	603	605	607			

2.3 FORCE DRIVEN ANALYSIS

The Dynamic Response program can be used to determine stresses, joint velocities, joint accelerations and joint displacements for structures subjected to periodic forces, force time history or engine/compressor vibration.

For periodic, time history or engine/compressor vibration analysis, the analysis type, load data and analysis output options are designated in the Dynamic Response input file in addition to the basic analysis options.

2.3.1 Force Time History

2.3.1.1 Analysis Type

The analysis label 'VIBR' must be entered in columns 7-10 on the DROPT line for any forced response analysis. For force time history analysis, enter 'THIS' in columns 7-10 on the FVIB line.

2.3.1.2 Load Options

Load options and input loading is specified following the LOAD header line using the FVIB, TIME, THFORCE and LOAD lines. Basic load options are designated on the FVIB line while integration parameters are specified on the TIME line.

Note: Each time history load requires a separate set of FVIB, TIME, THFORCE and LOAD lines.

2.3.1.2.1 Damping Method

Specify the damping type 'SDO' structural damping only, 'LFD' linearized fluid damping or 'NFD' for nonlinear fluid damping in columns 17-19 of the FVIB line.

Note: For nonlinear fluid damping, the fluid forces are calculated at every time step during the integration. This option requires the program calculated fluid damping option 'PC' on the FDAMP line.

For linearized fluid damping, the damping amplitude used to calculate the equivalent linear fluid damping may be overridden by specifying a value in columns 20-27.

2.3.1.2.2 Interpolation Scheme

The method used to interpolate between input values is designated in columns 28-39 on the FVIB line. Enter 'LN' for linear, 'QD' for quadratic or 'CU' for cubic.

Note: In general, linear interpolation is applicable for step, ramp or spike functions. The quadratic and cubic interpolation methods smooths out the input function.

For example, the following describes a time history function specified in the input file. Structural damping only is used in conjunction with linear interpolation of the force time history input as designated on the FVIB line.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	LOAD							
2	FVIB	THIS LINE	SDO	LN				

2.3.1.2.3 Integration Parameters

Integration parameters are stipulated on the TIME line. Enter the start for the beginning of the time history integration in columns 11-20. If the analysis is to terminate before the end of the time history input, enter the end time in columns 21-30. The output time interval, minimum integration step and the tolerance factor are designated in columns 31-40, 41-50 and 51-60, respectively.

For example, the following describes a time history function specified in the input file. Structural damping only is used in conjunction with linear interpolation as designated on the FVIB line. The start time is 0 seconds and end time 25 seconds. Output is requested at every 0.25 seconds.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	LOAD							
2	FVIB	THIS LINE	SDO	LN				
3	TIME	0.0	25.0	0.25				

2.3.1.3 Time History Input

Time history input data must be specified in the Dynamic response file. The source of the time history data is designated in columns 9-12 on the FVIB line. Enter 'LINE' if the data is defined on subsequent input lines or 'PREV' if the data is to be used from the previous load case.

2.3.1.3.1 Input Parameters

When specifying force time history data in the input file, time history input parameters must be specified on the THFORCE line. Enter the total number of joints that force is applied in columns 8-10. Time history data may be input using a uniform time interval between points or may be specified for various time points spaced non-uniformly. The input format, either uniform or non-uniform must be designated by 'UNI' or 'NON' in columns 11-13 respectively. For uniform input, specify the time interval in columns 14-20. The time history name is input in columns 22-25. The following illustrates the input required for the non-uniform time history input named 'TEST' applied at one joint.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	LOAD							
2	FVIB	THIS LINE	SDO	LN				
3	TIME	1.0	1.5	0.10				
4	THFORCE	1NON	TEST					

2.3.1.3.2 Time History Data

The time history input data is specified using LOAD lines located after the THFORCE line. If joint displacements are to be entered use the DISP and THDISP lines instead. Enter the joint to which the load

is applied in columns 8-11. The time that the load is to be applied is entered in columns 12-16. If several times are specified in consecutive LOAD lines, the times must be in ascending order. The forces and moments acting on the joint at the specified time are designated in columns 17-59.

For example, a load of 10.0 is applied in the global X direction to joint 107 at time 1.0 seconds. The load remains constant for 0.25 seconds after which it is removed.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	LOAD							
2	FVIB	THIS LINE	SDO		LN			
3	TIME	1.0	1.5	0.10				
4	THFORCE	1NON	TEST					
5	LOAD	107	1.0	0.0				
6	LOAD	107	1.001	10.0				
7	LOAD	107	1.25	10.0				
8	LOAD	107	1.251	0.0				
9	LOAD	107	10.0	0.0				

Note: Notice that the third time point is defined at 1.001 seconds instead of 1.00 seconds so that the force is applied over a small time period rather than applied instantaneously.

2.3.1.3.3 Scaling Load Data

Load data may be factored by specifying a load factor in columns 59-65 on the FVIB line.

2.3.1.3.4 Load Case Time History Data

Time history input data may also be specified using LOADC lines. These lines are located after the THFORCE line. In this case, rather than specifying joints at which the load applies and supplying a time history, the loads from the SACS IV load case specified in columns 8-11 on the LOADC line will be applied at the time specified in columns 12-16. If several times are specified in consecutive LOADC lines, the times must be in ascending order. The scaling factor to be applied to the loads is specified in columns 17-23. This method is very useful for applying similar time-varying loads to many positions, as in blast loading. Notice also that this loading is not limited to joint loads only; distributed and pressure loads may be applied in this manner as well.

For example, load case B01 is applied with a scaling factor of 0.0 at time 1.0 seconds, a factor of 1.0 at time 1.001 seconds, a factor of 1.0 at time 1.25 seconds, a factor of 0.0 at time 1.251 seconds and a factor of 0.0 at time 10.0 seconds. If load case B01 specified a load at joint 107 of 10.0 in the global X direction, then this example would result in the same loading at joint 107 as the previous 'LOAD' example.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	LOAD							
2	FVIB	THIS LINE	SDO		LN			
3	TIME	1.0	1.5	0.10				
4	THFORCE	1NON	TEST					
5	LOADC	B01	1.0	0.0				
6	LOADC	B01	1.001	1.0				
7	LOADC	B01	1.25	1.0				
8	LOADC	B01	1.251	0.0				
9	LOADC	B01	10.0	0.0				

2.3.1.3.5 Time History Collapse Analysis

Incremental loads for force time history Collapse analysis can be generated by specifying TCLP in columns 7-10 on the DROPT input line. Also, incremental loads from a dynamic ship impact Collapse analysis can be generated by specifying 'SHIP' in columns 7-10 on the DROPT line together with 'CLP' in columns 33-35 on the THLOAD input line. Alternatively, equivalent static loads can be generated by specifying 'ESL' in columns 33-35 on the THLOAD input line. The user will need to add additional information in the next selection of the card. Load cases corresponding to the time point having maximum overturning moment, maximum base shear or both by specifying 'MXM', 'MXS' or 'MMS' in the output options fields on the THLOAD line, respectively, may be created. Enter 'ALL' to have a load case created at each time point of the analysis.

The example below refers to a dynamic ship impact analysis with incremental loads being generated for a subsequent Collapse analysis. The weight, speed and direction of impact is defined on the SHIP input line together with impacted joint name.

	1	2	3	4	5	6	7	8
1	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
2	DROPT	SHIP	EC+Z					
3	SDAMP	5.0						
4	LOAD							
5	SHIP	1250.	6.0	180.0	1.0		31P7	
6	THLOAD		SDO					
7	JTNUM	31P7	701					
8	TIME		2.00000	0.0100	1.0E-9	1.0000		

2.3.1.4 Output Options

The force time history analysis creates load cases, prints and plots modal responses, base shear and overturning moment in addition to joint accelerations, velocities and displacements. Analysis output options are designated in the output options fields in columns 32-58 on the FVIB line.

2.3.1.4.1 Load Case Creation

Load cases corresponding to each time point or the critical time points may be generated by the forced response analysis. Load cases corresponding to the time point having maximum overturning moment, maximum base shear or both by specifying 'MXM', 'MXS' or 'MMS' in the output options fields on the FVIB line, respectively, may be created. Enter 'ALL' to have a load case created at each time point of the analysis.

Note: Load options are mutually exclusive. Only one of the options may be designated.

2.3.1.4.2 Modal Response Data

Modal responses versus time may be printed and/or plotted by specifying 'PRT' and 'PLT', respectively, in the output options fields on the FVIB line. Enter 'PPT' to have modal responses printed and plotted.

Note: Modal response options are mutually exclusive. Only one of the options may be designated.

2.3.1.4.3 Base Shear and Overturning Moment Plots

Base shear and overturning moment plots may be generated by entering 'PLM' and 'PLS', respectively, in one of the output option fields located on the FVIB line.

2.3.1.4.4 Joint Results

Joint results including acceleration, velocity and displacement may be plotted and listed for up to sixteen joints. Joint plot options are specified in the output options fields on the FVIB line.

Joint acceleration options include:

‘JMA’ Prints maximum and minimum values for joint acceleration for each direction.

‘JPA’ Same as JMA plus plots acceleration time history

‘JTA’ Same as JPA plus prints acceleration time history data

Note: Joint acceleration options are mutually exclusive. Only one of the options may be selected.

Joint velocity options include:

‘JMV’ Prints maximum and minimum values for joint velocity for each direction.

‘JPV’ Same as JMV plus plots velocity time history

‘JTV’ Same as JPV plus prints velocity time history data

Note: Joint velocity options are mutually exclusive. Only one of the options may be selected.

Joint displacement options include:

‘JMD’ Prints maximum and minimum values for joint displacement for each direction.

‘JPD’ Same as JMD plus plots displacement time history

‘JTD’ Same as JPD plus prints displacement time history data

Note: Joint displacement options are mutually exclusive. Only one of the options may be selected.

Up to sixteen joints may be designated for plots and reports. Joints to be output are specified using the JTNUM line immediately following the THLOAD line.

The following input illustrates some of the output options. Two load cases, one corresponding to the time of maximum base shear and one corresponding to the time of maximum overturning moment, are created. Base and overturning moment time histories are to be plotted in addition to joint acceleration and displacement plots for joints 601, 603, 605 and 607.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	LOAD							
2	FVIB	THIS LINE SDO	LN	MXMMXS	PLMPLS	JPAJPD		
3	JTNUM	601 603 605 607						

2.3.2 Periodic Vibration

2.3.2.1 Analysis Type

The analysis label ‘VIBR’ must be entered in columns 7-10 on the DROPT line for any forced response analysis. For periodic vibration analysis, enter ‘PERI’ in columns 7-10 on the FVIB line.

2.3.2.2 Load Options

Periodic loading and options for that loading is defined using the FVIB and LOAD input lines specified after the LOAD header line. General load options are specified on the FVIB line.

Note: Each periodic loading defined requires a separate set of FVIB and LOAD input lines.

2.3.2.2.1 Damping Method

Specify the damping type 'SDO' structural damping only, 'LFD' linearized fluid damping or 'NFD' for nonlinear fluid damping in columns 17-19 on the FVIB line.

Note: For nonlinear fluid damping, the fluid forces are calculated at every time step during the integration. This option requires the program calculated fluid damping option 'PC' on the FDAMP line.

For linearized fluid damping, the damping amplitude used to calculate the equivalent linear fluid damping may be overridden by specifying a value in columns 20-27.

2.3.2.2.2 Time Parameters

For periodic vibration the time span that the vibration is to be monitored is input in columns 72-77 on the FVIB line. In general, this time span is the shortest time that the vibration is repeatable. The number of time points that the time span is to be divided is specified in columns 78-80.

Note: The number of time points should be sufficient to pick up the highest frequency of interest.

For example, the following describes a periodic function specified in the input file. Structural damping only is used in conjunction with quadratic interpolation of the periodic input. The analysis time span is 1.0 second and the analysis is to be broken up into 50 time points.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	LOAD							
2	FVIB	PERI	SDO				603DX	1.0 50

2.3.2.3 Periodic Input

Periodic load data must be specified in the Dynamic Response input file using LOAD lines located after the FVIB line. Input load data may be scaled automatically by entering a load scaling factor.

2.3.2.3.1 Periodic Load Data

Enter the joint to which the load is applied in columns 8-11. The forces and moments acting on the joint are designated in columns 17-59.

Enter the period that the set of forces is acting in columns 69-74 along with the phase angle in columns 75-80.

Note: Forces are applied as $F \cdot \cos(2\pi T / (T+a))$ where T is the period and a is the phase angle.

For example, a periodic force of 10.0 is applied in the global X direction to joint 107. The period is 0.20 seconds and the phase angle is 90 degrees.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							

2.3.2.3.2 Scaling Load Data

2.3.2.4 Output Options

2.3.2.4.1 Load Case Creation

Note: The above load options are mutually exclusive. Only one of the options may be designated.

Note: When creating a load case for a maximum joint displacement or rotation, no other load cases are created. Therefore, the 'MXM', 'MXS' and 'MMS' may not be used when using this feature.

2.3.2.4.2 Modal Response Data

Note: Modal response options are mutually exclusive. Only one of the options may be designated.

2.3.2.4.3 Base Shear and Overturning Moment Plots

2.3.2.4.4 Joint Displacements

'DSM' - prints max. absolute sum of X, Y and Z displacement for each periodic load case to produce a maximum possible displacement.

Note: Joint displacement options are mutually exclusive. Only one of the options may be selected.

The following input illustrates some of the output options. Two load cases, one corresponding to the time of maximum base shear and one corresponding to the time of maximum overturning moment, are created. Base and overturning moment time histories are to be plotted in addition to reporting the maximum X, Y and Z joint displacements.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	LOAD							
2	FVIB	THIS LINE SDO	LN	MXMMXS	PLMPLS	MXD		

2.3.3 Engine/Compressor Vibration

The engine or compressor vibration analysis determines joint displacement due to unbalanced forces. Joint displacements can be compared versus various allowable deflection specifications and expressed as displacement unity check ratios.

2.3.3.1 Analysis Type

The analysis label 'ENGV' must be entered in columns 7-10 on the DROPT line for engine or compressor vibration analysis.

2.3.3.2 Load Options

Engine unbalanced loading is entered in the form of mechanical unbalanced forces, gas torques or general unbalanced forces. Loading and load options are defined using the ENGVIB, RSPEED and UNBAL lines following the LOAD header line.

Note: Each set of loading requires a separate set of RSPEED and UNBAL lines.

2.3.3.2.1 Damping Method

Only structural damping input on the SDAMP line is considered for engine vibration analysis.

Note: Because fluid damping is not supported, the FDAMP line should not be used for engine vibration.

2.3.3.2.2 Engine Speed Parameters

Engine speed parameters are designated on the ENGVIB line. The beginning speed (the lowest speed) and the ending speed (the highest speed) are specified in columns 7-13 and 14-20, respectively.

The running speed range is defined by the beginning and ending speeds. The program divides the speed range into increments for the purpose of the analysis using either constant increments or varying increments based on modal frequencies. Specify one of the following incrementation methods in columns 21-23:

'CON' - Constant incrementation

'MOD' - Increments varied so each modal frequency is included as an analysis speed

'MAH' - Same as 'MOD' except that each harmonic frequency is also included

'USR' - Analysis speeds defined by the user using USRSP lines

For constant increments, the speed increment value entered in columns 24-28 is the total number of speed points to analyze. For varying increments on the other hand, the value entered in these columns is used to determine the smallest speed increment allowed between modes.

The following shows the input for a begin speed of 100 rpm and ending speed of 500 rpm using constant speed increments.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	LOAD							
2	ENGVI	100.	500.CON	DL	PT			

User defined running speeds are specified using the USR speed option on the ENGVIB line and USRSP lines immediately following the ENGVIB line. For example, the following input designates analysis running speeds of 120, 150, 200, 300, and 400 rpm.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	LOAD							
2	ENGVI	100.	500.USR	DL	PT			
3	USRSP	120.	150.	200.	300.			
4	USRSP	400.						

2.3.3.2.3 Nonlinear Interpolation Power

By default, 2.0 is used as the nonlinear interpolation power. Enter the nonlinear interpolation power override to be used for any mechanical unbalanced forces interpolated nonlinearly between running speeds in columns 29-33. If the interpolation power is p , then the interpolation is accomplished as follows:

$$\frac{F - F_1}{F_2 - F_1} = \frac{\omega^p - \omega_1^p}{\omega_2^p - \omega_1^p}$$

where F are the forces and ω are the running speeds. This field may be left blank if linear interpolation is used for all unbalanced forces.

2.3.3.2.4 Calculation Points per Cycle

By default, 10 points are calculated for the highest harmonic determined. This value may be overridden by specifying the maximum number of points calculated for the highest harmonic in columns 37-39 on the ENGVIB line.

Twenty points are calculated for one cycle of the fundamental frequency by default. Enter the minimum number of points to be calculated per cycle in columns 34-36 if this value is to be overridden.

Note: If harmonics are encountered, the number of points calculated per cycle is the maximum of the number of points per fundamental and the number of points calculated per harmonic.

2.3.3.2.5 Allowable Option

The deflections determined by the program can be compared to published allowable deflections and expressed as displacement unity check ratios. The allowable option is specified in columns 40-41 as follows:

‘DL’ - D Line Allowable

'SN' - SNAME

'ML' - US Military Specification

'VE' - Maximum Velocity

'AC' - Maximum Acceleration

The following illustrates the input required for displacement unity check ratios to be determined using the D Line allowables.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	LOAD							
2	ENG VIB	100.	500.CON	DL	PT			

If the allowable option is 'VE' (maximum velocity), the maximum velocity allowed is entered in columns 42-46. Enter the maximum acceleration allowed if the allowable option is 'AC' (maximum acceleration).

2.3.3.2.6 Joint Selection

By default, all joints are monitored in the engine vibration analysis. Joints may be optionally selected to be included using the JNTSEL line. For example, the following designates that only joints 101, 103, 105, 107, 109 and 111 are to be monitored in the analysis.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	LOAD							
2	ENG VIB	100.	500.CON	DL	PT			
3	JNTSEL	101	103	105	107	109	111	

Note: As many JNTSEL lines as required may be used. If JNTSEL lines are specified, only joints specified are monitored during the analysis.

2.3.3.3 Unbalanced Force Input

Engine unbalanced forces may be expressed in terms of gas torques, mechanical unbalanced forces and/or general unbalanced periodic forces at various running speeds.

For a particular running speed, unbalanced forces can be input as separate load conditions where maximum response from each is added and/or may be specified in the same load condition if phase angles between unbalanced forces is known.

Each load condition that unbalanced forces are to be defined is designated by a RSPEED line with the running speed designated in columns 9-15.

2.3.3.3.1 Gas Torque Loading

Unbalanced forces due to gas torque may be expressed as maximum gas torque at various harmonics or in the form of a total gas torque curve.

When unbalanced forces due to gas torque are known for various harmonics, loading is specified in the form of a periodic loading using UNBAL lines specified immediately after the RSPEED line.

The joint to which the load is applied is designated in columns 8-11. The force type 'SIN' is used for load described by a single sine wave (amplitude and phase angle) and must be designated in columns 12-14. The forces acting on the joint are designated in columns 17-58.

Enter the phase angle in columns 59-65, the interpolation type, either 'LN' or 'NL', in columns 66-67 and the harmonic number in columns 68-69.

The program allows loading to be grouped and considers each load group to act independently. The maximum displacements resulting from each load group are summed together to determine the total displacement. The load group to which this force is assigned is stipulated in columns 70-71.

For example, a gas torque about the global X axis at joint 107 is known for the first 3 harmonics at a running speed of 300 rpm. Since phasing is known, each value is to be assigned to the same load group, load group 1. Linear interpolation is to be used between running speeds.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	LOAD							
2	ENG VIB	250.	720.CON	DL	PT			
3	RSPEED	300.0						
4	UNBAL	107SIN		100.		90.0LN	0	1
5	UNBAL	107SIN		62.		180.0LN	1	1
6	UNBAL	107SIN		90.		290.0LN	2	1

A total gas torque curve may be input as a series of equally spaced points in time using an UNBAL line and LDFACT lines immediately after the RSPEED line.

The joint to which the load is applied is designated in columns 8-11. The force option 'TIM', designating force input by a series of equally spaced time points, must be designated in columns 12-14. The applied forces acting on the joint are designated in columns 17-58.

Enter the interpolation type, either 'LN' or 'NL', in columns 66-67 and the highest harmonic number to be used from the Fourier series in columns 72-73. The load group to which this force is assigned is stipulated in columns 70-71.

Note: The harmonic number and phase angle fields must be left blank when inputting load described by equally spaced time points as designated by the 'TIM' force option.

The following example shows a total gas torque curve for moment about the global X axis at joint 107 defined at 300 rpm. The curve will be defined at 18 degree increments (20 points). Each value on the curve is to be assigned to the same load group, load group 1. Linear interpolation is to be used between running speeds and the highest harmonic number to be used is 10.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	LOAD							
2	ENG VIB	250.	720.CON	DL	PT			
3	RSPEED	300.0						
4	UNBAL	107TIM		100.		LN	110	
5	LDFACT	-0.587	0.0	0.587	0.951	0.951	0.587	0.0
6	LDFACT	-0.887	0.0	0.887	0.951	0.951	0.887	0.0

2.3.3.3.2 Mechanical Unbalanced Forces

Mechanical unbalanced forces are specified in the form of a periodic loading using UNBAL lines specified immediately after the RSPEED line.

The joint to which the forces are applied is designated in columns 8-11. The force type 'SIN' is used for load described by a single sine wave (amplitude and phase angle) and must be designated in columns 12-14. The forces acting on the joint are designated in columns 17-58.

Enter the phase angle in columns 59-65, the interpolation type, either 'LN' or 'NL', in columns 66-67 and the harmonic number in columns 68-69.

The program allows loading to be grouped and considers each load group to act independently. The maximum displacements resulting from each load group are summed together to determine the total displacement. The load group to which this force is assigned is stipulated in columns 70-71.

For example, a compressor has primary and secondary mechanical unbalanced forces that create moments about the global Y and Z axes which are phased 90 degrees apart. At 300 rpms, the primary and secondary unbalanced forces create 16000 in-kip and 2600 in-kip moments about the Y axis and 2400 in-kip and 750 in-kip moments about the Z axis respectively, applied at joint 107. Because the unbalanced forces are assumed to vary with the square of the running speed, nonlinear interpolation with a power of 2 is to be used.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	LOAD							
2	ENGVI	250.	720.CON	2	DL	PT		
3	RSPEED	300.0						
4	UNBAL	107SIN			16000.	0.0NL	1	
5	UNBAL	107SIN			2600.	90.0NL	1 1	
6	UNBAL	107SIN				2400.	0.0NL	
7	UNBAL	107SIN				750.	90.0NL	1

2.3.3.3 General Unbalanced Periodic Forces

General unbalanced forces may be input in the form of time history or periodic loading.

When unbalanced forces are known for various harmonics, loading is specified in the form sine waves of a known amplitude and phase angle using UNBAL lines specified immediately after the RSPEED line. Unbalanced forces may also be input in the form of a time history with equally spaced time points using an UNBAL line and LDFACT lines.

In either case, the joint to which the load is applied is designated in columns 8-11. The force type, either 'SIN' for single sine wave or 'TIM' for time history, must be designated in columns 12-14. The forces acting on the joint are designated in columns 17-58.

Single sine wave type loading requires the phase angle in columns 59-65 and the harmonic number in columns 68-69 while the time history type requires only the highest harmonic to be used from the Fourier series in columns 72-73.

The interpolation type, either 'LN' or 'NL', must be designated in columns 66-67. The load group to which this force is assigned is stipulated in columns 70-71.

Note: The program allows loading to be grouped and considers each load group to act independently. The maximum displacements resulting from each load group are summed together to determine the total displacement.

The following illustrates an unbalanced force along the global Z and about the global X axis at joint 107. The force is known for the first 3 harmonics at a running speed of 300 rpm and each value is to be assigned to the same load group, load group 1. Linear interpolation is to be used between running speeds.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	LOAD							
2	ENGVIB	250.	720.CON	DL	PT			
3	RSPEED	300.0						
4	UNBAL	107SIN	10.0	100.	90.0LN	1		
5	UNBAL	107SIN	6.2	62.	180.0LN	1	1	
6	UNBAL	107SIN	9.0	90.	290.0LN	2	1	

The following example shows a force time history input for moment about the global X axis at joint 107 defined at 300 rpm. The curve will be defined at 18 degree increments (20 points). Each value on the curve is to be assigned to the same load group, load group 1. Linear interpolation is to be used between running speeds and the highest harmonic number to be used is 10.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	LOAD							
2	ENGVIB	250.	720.CON	DL	PT			
3	RSPEED	300.0						
4	UNBAL	107TIM	100.	LN	110			
5	LDFACT	-0.587	0.0	0.587	0.951	0.587	0.0	-0.587
6	LDFACT	-0.887	0.0	0.887	0.951	0.887	0.0	-0.887

2.3.3.4 Output Options

The engine vibration analysis calculates generalized forces and joint displacements for the various conditions defined. Joint displacements may be compared to allowable displacement curves and expressed in terms of a displacement unity check ratio.

2.3.3.4.1 Generalized Forces

Generalized force print options are designated in columns 53-54 on the ENGVIB line. Enter 'PT' for the standard generalized force print or 'FL' for a full print.

2.3.3.4.2 Joint Results

Joint results for all joints that exceed the allowable displacement at any running speed may be printed by specifying 'PT' in columns 57-58 on the ENGVIB line.

Joint displacements may also be plotted by specifying 'PL' in columns 55-56. By default, all joints are plotted when the joint plot feature is instigated. Joints to be plotted may be designated using the JNTPLT line following the ENGVIB line.

For example, the following designates that joint results are to be plotted for joints 101, 103, 105 and 107.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	LOAD							
2	ENGVIB	250.	720.CON	DL	PT			
3	JNTPLT	101	103 105 107					
4	RSPEED	300.0						
5	UNBAL	107TIM	100.	LN	110			

2.3.3.4.3 Plot Options

Optional plot options may be designated using the PLTOPT line. Up to three allowable curves may be plotted on the joint displacement plots. Designate the allowable curves to be plotted in columns 11-16 as follows:

‘DL’ - D-Line Allowable

‘SN’ - SNAME

‘ML’ - US Military Specification

Enter ‘GR’ in columns 35-36 if grid lines are to be included on the plots. Plot size and character sizes may also be specified in columns 17-34.

The following illustrates the input to plot joint results for joints 101, 103, 105 and 107. D-Line and SNAME allowable curves are to be shown along with grid lines.

	1				2				3				4				5				6				7				8			
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
1	LOAD																															
2	ENGVI	B	250.			720.	CON						DL				PT															
3	JNTPLT		101		103		105		107																							
4	PLTOPT		DLSN																													
5	RSPEED		300.0																													
6	UNBAL		107TIM										100.											LN		110						

2.4 SPECTRAL WIND ANALYSIS

The Dynamic Response program has the ability to perform spectral wind analyses. The program has specialized features that generate solution files for extreme wind analysis and wind fatigue analysis.

2.4.1 Extreme Wind

The program can be used to analyze wind dynamically utilizing a Harris wind spectrum and create a solution file containing end forces, stresses, reactions and displacements for each wind velocity to be analyzed. These results contain dynamic amplification and can be combined with the static results due to self-weight, etc.

Generating the dynamic results requires that aerodynamic data and the wind velocities to be analyzed be specified in the Seastate input (or SACS model file) while all other data including analysis, spectrum and plot options are designated in the Dynamic Response input file.

2.4.1.1 General Model Options

The ‘JO’ option which designates that only stresses at the joints are to be contained in the solution file should be designated in columns 27-28 on the OPTIONS line in the SACS model file. For each element, the dynamic amplification factor is based on the stress in the element and is a function of the dynamic RMS stress and the static stress. Because each member internal load will be factored by a unique dynamic amplification factor applicable only to that particular internal load, internal loads are not consistent with each other nor are they consistent with the applied loading along the member. Therefore, stresses and unity check calculations are only valid at the member ends.

2.4.1.2 Aerodynamic and Wind Data

Aerodynamic and wind data must be specified in the Seastate input or SACS model file.

The 'WIN' Seastate option must be specified in columns 56-58 on the LDOPT line. Wind load data is specified after the LOAD line. Each load case defined should contain only wind loading with the mean wind velocity specified as the wind speed.

Note: Each wind should be specified as a separate load case. As many wind speeds and directions as desired may be specified.

The following sample input shows two wind cases with a mean velocity of 100 for the 0 degree and 90 degree directions.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	* Dynamic Spectral Wind							
2	LDOPT	+Z	1.03	7.85	-50.0	50.0	GLOBME WIN	NPNP K
3	FILE S							
4	CDM							
5	CDM AP							
6	GRPOV							
7	GRPOV	PL2NN		0.001	0.001	0.001		
8	LOAD							
9	LOADCN	1						
10	WIND							
11	WIND D	100.0	0.0	AP08				
12	LOADCN	2						
13	WIND							
14	WIND	100.0	90.0	AP08				
15	END							

2.4.1.3 Dynamic Response Options

Dynamic response options including analysis and plot options are designated in the Dynamic Response input file.

2.4.1.3.1 Analysis Type

The analysis label 'WIND' must be entered in columns 7-10 on the DROPT line for spectral wind analysis.

2.4.1.3.2 Damping Method

Only structural damping input on the SDAMP line is considered for spectral wind analysis.

Note: Because fluid damping is not supported, the FDAMP line should not be used for spectral wind analysis.

2.4.1.3.3 Spectral Wind Data

Spectral wind data is specified on the SPCWIN line immediately following the LOAD header. Designate the extreme wind analysis option 'EX' in columns 8-9.

By default, 600 seconds is used as the mean wind speed averaging time used to calculate the dynamic amplification factors. This value may be overridden by entering an averaging time override in columns 39-44.

For each wind speed to be analyzed, a Harris wind spectrum is created based on the wind velocity at the reference height along with the spectrum reference length and surface roughness parameters input by the user. The spectrum to be used can be specified by using WINSPC lines for each direction. If no WINSPC lines are provided, then Harris spectrum will be used. The program uses this generated spectrum to determine model responses. Enter the reference length and surface roughness to be used for the Harris spectrum in columns 45-50 and 51-56, respectively.

By default, the program calculates the spatial correlation constant, enter 'SK' in columns 35-36 if a spatial correlation constant is not to be used.

The following shows the input for a spectral extreme wind analysis. Default values for wind averaging time, Harris spectrum reference length and roughness coefficient are to be used.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	DROPT WIND							
2	SDAMP 2.0							
3	LOAD							
4	SPCWIN EX MN PL							
5	END							

2.4.1.4 Output Options

By default, the program creates a common solution file containing end forces, stresses, reactions and displacements for each wind load case specified in the Seastate input file.

The program also has the ability to plot a generalized force spectrum and/or a response spectrum for each wind speed. Enter 'PL' in columns 14-15 and 17-18 respectively to output generalized force and response spectra.

Enter the print level desired in columns 11-12 as follows:

'MN' - Minimum print containing one line of output for each wind speed analyzed

'MD' - Moderate print level containing one page of output for each wind analyzed

'MX' - Maximum print containing detailed output including spectrum for each wind analyzed.

2.4.1.4.1 Static + Wind Combinations

The Dynamic Response has the ability to optionally combine spectral wind results with static results as part of the extreme wind analysis. When using this feature, the program creates a wind + static combination for each wind load case. If a different joint check and member check factors are specified however, two combinations are created for each wind load case.

Note: This feature requires that the static solution file exist prior to execution of the spectral wind analysis.

The wind and static combination information is input using the STCMB line. Enter the factor to be applied to the wind loads when combined with the static loads for the purpose of member and plate element check in columns 8-12. The factor to be applied to wind loads when combined with static loads for joint check is input in columns 13-17. Enter each of the static load cases to be combined with the seismic load cases and the load factor to be applied.

For example, 105% of load cases 8 and 9 contained in the static solution file are to be combined with the wind solution. For same factor wind load case factor is used for element check and joint can check.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	DROPT WIND	15	+Z					
2	SDAMP	5.0						
3	STCMB	1.0	1.0	8 1.05	9 1.05			

Note: The STCMB line should follow the SDAMP, FDAMP and MODSEL lines in the input file.

2.4.1.5 Combining with Static Results Manually

The program creates a common solution file containing end forces, stresses, reactions and displacements for each wind load case specified in the Seastate input file. Because these results are obtained by combining modal results using RMS techniques, end forces, stresses, etc. have no sign associated and are taken as all positive values. Therefore, when combining spectral wind results with static results manually, the PRST and PRSC combine options must be used.

2.4.2 Wind Fatigue

The Dynamic Response program can be used to perform spectral wind fatigue analysis utilizing a Harris wind spectrum. The program creates a Fatigue input file containing fatigue load data in conjunction with the mode participation factors and executes the Fatigue module automatically.

Generating the dynamic results requires that aerodynamic and wind information be specified in the Seastate input (or SACS model file) while all other data including analysis, spectrum, fatigue and plot options are designated in the Dynamic Response input file.

2.4.2.1 Aerodynamic and Wind Data

Aerodynamic and wind data must be specified in the Seastate input or SACS model file.

The 'WIN' Seastate option must be specified in columns 56-58 on the LDOPT line. Wind load data is specified after the LOAD line. Each load case defined should contain only wind loading with the mean wind velocity specified as the wind speed. Wind load cases should be specified in order of increasing wind speed, with all wind cases for a particular direction specified followed by all wind load cases for the next direction.

The wind loads specified are used to determine the fatigue damage. A stress range is calculated for each wind speed specified. The Harris spectrum is then used to determine the probability of occurrence of that speed.

Note: Each wind should be specified as a separate load case. As many wind speeds and directions as desired may be specified.

The following sample input shows wind load cases with speed ranging from 2 to 20 for two different directions.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	* Dynamic Spectral Wind							
2	LDOPT	+Z	1.03	7.85	-50.0	50.0	GLOBME WIN	NPNP K
3	FILE S							
4	CDM							

5	CDM AP				
6	GRPOV				
7	GRPOV	PL2NN		0.001 0.001 0.001	
8	LOAD				
9	LOADCN	1			
10	WIND				
11	WIND D	2.0	0.0	AP08	
12	LOADCN	2			
13	WIND				
14	WIND D	5.0	0.0	AP08	
15	LOADCN	3			
16	WIND				
17	WIND D	10.0	0.0	AP08	
18	LOADCN	4			
19	WIND				
20	WIND D	15.0	0.0	AP08	
21	LOADCN	5			
22	WIND				
23	WIND D	20.0	0.0	AP08	
24	LOADCN	6			
25	WIND				
26	WIND	2.0	90.0	AP08	
27	LOADCN	7			
28	WIND				
29	WIND	5.0	90.0	AP08	
30	LOADCN	8			
31	WIND				
32	WIND	10.0	90.0	AP08	
33	LOADCN	9			
34	WIND				
35	WIND	15.0	90.0	AP08	
36	LOADCN	10			
37	WIND				
38	WIND	20.0	90.0	AP08	
39	END				

Note: All wind cases are specified in the 0 degree direction before the 90 degree wind cases are input.

2.4.2.2 Dynamic Response Options

Dynamic response options including analysis, fatigue and plot options are designated in the Dynamic Response input file.

2.4.2.2.1 Analysis Type

The analysis label 'WIND' must be entered in columns 7-10 on the DROPT line for spectral wind analysis.

2.4.2.2.2 Damping Method

Only structural damping input on the SDAMP line is considered for spectral wind analysis.

Note: Because fluid damping is not supported, the FDAMP line should not be used for spectral wind analysis.

2.4.2.2.3 Spectral Wind Data

Spectral wind data is specified on the SPCWIN line immediately following the LOAD header. Designate the wind fatigue analysis option 'FT' in columns 8-9.

By default, 600 seconds is used as the mean wind speed averaging time used to calculate the dynamic amplification factors. This value may be overridden by entering an averaging time override in columns 39-44. The spectrum to be used can be specified by using WINSPEC lines for each direction. If no WINSPEC lines are provided, then Harris spectrum will be used. The program uses this generated spectrum to

determine model responses. Enter the reference length and surface roughness to be used for the Harris spectrum in columns 45-50 and 51-56, respectively.

The following shows the input for a spectral wind fatigue analysis. Default values for wind averaging time, Harris spectrum reference length and roughness coefficient are to be used.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	DROPT	WIND						
2	SDAMP	2.0						
3	LOAD							
4	SPCWIN	FT	MN	PL				

2.4.2.2.4 Output Options

The program creates a Fatigue input file containing fatigue load data for each wind direction specified in the Seastate input file.

The program also has the ability to plot a generalized force spectrum and/or a response spectrum for each wind speed. Enter 'PL' in columns 14-15 and 17-18 respectively to output generalized force and response spectra.

Enter the print level desired in columns 11-12 as follows:

'MN' - Minimum print containing one line of output for each wind speed analyzed

'MD' - Moderate print level containing one page of output for each wind analyzed

'MX' - Maximum print containing detailed output including spectrum for each wind analyzed.

2.4.2.2.5 Fatigue Input Data

The Dynamic Response program creates the input file required by the Fatigue program module. Fatigue input options are specified directly in the Dynamic Response input file following the SPCWIN line. All Fatigue input is supported and may be specified up to the point of defining fatigue load case data. Fatigue load case data is created by the program automatically based on wind spectrum options specified by the user on the WINSPC lines.

For each wind direction, wind spectrum data used to create the modal participation input and the fatigue load case input is specified on the corresponding WINSPC line. The wind direction is designated in columns 7-13 along with the fraction of time that wind from this direction occurs specified in columns 14-20.

The Weibull spectrum label 'WEI' is entered in columns 22-24 along with the distribution parameters 'K' and 'A' in columns 26-32 and 33-39, respectively.

The following illustrates the input required to generate the Fatigue input for two wind directions, 0 and 90 degrees. Winds from 0 degrees occur 45% and winds from 90 degrees occur 55% as designated on the WINSPC lines.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	DROPT	WIND						
2	SDAMP	2.0						
3	LOAD							

4	SPCWIN	FT	MN	PL															
5	FTOPTG	20.0				2.0	SMUSR		EX			2.0					KAW		
6	FTOPT2	PT																	
7	GRPSCF	0.0	ST1	ST2															
8	S-N	87.0	1000.	3.33200.E6	3.33800.E6														
9	JSLC	107	105	201	115	251	117												
10	WINSPC	0.0	0.45	WEI	2.0	32.8HR													
11	WINSPC	90.0	0.55	WEI	2.0	32.8HR													
12	END																		

Note: Each wind direction is treated as a separate fatigue load condition and must be designated with a WINSPC line.

2.5 ICE FORCE ANALYSIS

2.5.1 Ice Fatigue

The Dynamic Response program can be used to determine fatigue due to vibrations caused by ice forces. The program creates a Fatigue input file containing fatigue load data in conjunction with the mode participation factors and executes the Fatigue module automatically.

2.5.1.1 Analysis Type

The analysis label 'VIBR' must be entered in columns 7-10 on the DROPT. For ice vibration analysis, enter 'ICE' in columns 7-10 on the FVIB line.

2.5.1.2 Load Options

Ice loading and load options are defined using the FVIB and ICE input lines specified after the LOAD header line. Effective diameter overrides used to calculate ice loading may be specified using GRPMD and MEMMD lines.

2.5.1.2.1 Damping Method

Specify the damping type 'SDO' structural damping only, 'LFD' linearized fluid damping or 'NFD' for nonlinear fluid damping in columns 17-19 on the FVIB line.

Note: For nonlinear fluid damping, the fluid forces are calculated at every time step during the integration. This option requires the program calculated fluid damping option 'PC' on the FDAMP line.

For linearized fluid damping, the damping amplitude used to calculate the equivalent linear fluid damping may be overridden by specifying a value in columns 20-27.

2.5.1.2.2 Effective Diameter Overrides

The effective diameter of members that penetrate the ice sheet may be overridden using the MEMMD or the GRPMD lines following the FVIB line.

For member groups, enter the group name and the effective diameter on the GRPMD line. For members to be modified, enter the start joint, end joint and effective diameter on the MEMMD line.

For example, group PL1, PL2 and PL3 represent piles inside of the jacket leg that penetrate the ice sheet. The effective diameter is modified to 0.001 so that no ice loading is applied to members assigned to these groups.

1	2	3	4	5	6	7	8
---	---	---	---	---	---	---	---

	1234567890123456789012345678901234567890123456789012345678901234567890
1	LOAD
2	FVIB ICE SDO
3	GRPMD PL1 0.001PL2 0.001PL3 0.001

2.5.1.2.3 Integration Parameters

Integration parameters are stipulated on the TIME line. Enter the start for the beginning of the time integration in columns 11-20 and the end time in columns 21-30. The output time interval, minimum integration step and the tolerance factor are designated in columns 31-40, 41-50 and 51-60, respectively.

For example, the following describes an ice function specified in the input file. The start time is 0 seconds and end time 25 seconds. Output is requested at every 0.25 seconds.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	LOAD							
2	FVIB ICE SDO							
3	TIME 0.0 25.0 0.25							

2.5.1.3 Fatigue Input Data

The Dynamic Response program creates the input file required by the Fatigue program module. Fatigue input options are specified directly in the Dynamic Response input file following the FVIB or TIME line. All Fatigue input is supported and may be specified up to the point of defining fatigue load case data. Fatigue load case data is created by the program automatically based on ice data specified by the user on the ICE lines.

The following shows fatigue input options copied into the Fatigue input file created by the program.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	DROPT VIBR							
2	SDAMP 2.0							
3	LOAD							
4	FVIB ICE SDO							
5	TIME 0.0 25.0 0.25							
6	FTOPTG 20.0 2.0 SMUSR EX 2.0 KAW							
7	FTOPT2 PT							
8	GRPSCF 0.0 ST1 ST2							
9	S-N 87.0 1000. 3.33200.E6 3.33800.E6							
10	JSLC 107 105 201 115 251 117							

2.5.1.4 Ice Data

For each ice floe, the ice data used to create the modal participation input and the fatigue load case input is specified on the corresponding ICE line. Each ice floe is treated as an individual fatigue load case. The ice thickness, elastic modulus, static crushing strength and top of ice elevation are specified in columns 8-13, 14-19, 20-25 and 32-37, respectively.

The ratio of total length to elastic length must be designated in columns 26-31 while the floe density is designated in columns 50-55. The ice stiffness parameter used to estimate the stiffness of the ice is input in columns 56-61. The default value 0.0315 represents an infinite sheet of ice flowing past a vertical cylinder. Enter the ice velocity or the velocity of the first step if using multiple steps, in columns 44-49. If using multiple steps to obtain a variation of results with ice velocity, enter the velocity step size in columns 62-67. The number of steps should be stipulated in columns 68-70. The time duration

entered in columns 71-76 is the duration of the floe and is used to determine the number of cycles for damage calculations.

The following example, shows 5.0 thick ice floe in the -40.0 degree direction with velocities ranging from 0.10 to 0.35.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	DROPT	VIBR						
2	SDAMP	2.0						
3	LOAD							
4	FVIB	ICE	SDO					
5	MEMMD	501	601					
6	TIME	0.0	15.0	0.05	1.0	E-81000.		
7	FTOPT	30.0	1.0	2.0	SMAPI	EX	2.0	KAW
8	ICE	5.0	.2040.01854	28.16	0.00	-40.	0.10 56.0 .00051	0.084
9	ICE	5.0	.2040.01854	28.16	0.00	-40.	0.20 56.0 .00051	0.084
10	ICE	5.0	.2040.01854	28.16	0.00	-40.	0.25 56.0 .00051	0.084
11	ICE	5.0	.2040.01854	28.16	0.00	-40.	0.30 56.0 .00051	0.084
12	ICE	5.0	.2040.01854	28.16	0.00	-40.	0.35 56.0 .00051	0.084
13	END							

2.5.1.5 Output Options

The ice vibration analysis creates a Fatigue input file in addition to optional modal response output. Optional output options are designated in columns 32-58 on the FVIB line.

2.5.1.5.1 Modal Response Data

Modal responses versus time may be printed and/or plotted by specifying 'PRT' and 'PLT', respectively, in the output options fields on the FVIB line. Enter 'PPT' to have modal responses printed and plotted.

Note: Modal response options are mutually exclusive. Only one of the options may be designated.

2.6 DYNAMIC IMPACT ANALYSIS

The Dynamic Response program can be used to determine the transient response of a structure resulting from accidental impact loading. Accidental impact loading resulting from a floating vessel and dropped objects are considered. The program can output equivalent static loads at discreet time steps to be used for a subsequent static analysis or incremental loads for a subsequent Collapse analysis. Analysis type and analysis output options are designated in the Dynamic input file in addition to the basic analysis options.

2.6.1 Analysis Type

Enter 'SHIP' in columns 7-10 on the DROPT line to designate a ship impact analysis or enter 'DROP' to designate a dropped object analysis.

2.6.2 Load Options

For s ship impact or a dropped object analysis the input loading and load options are input following the LOAD header line using the SHIP, DRPOBJ and THLOAD input lines.

2.6.3 Ship Impact Analysis

For a ship impact analysis, enter the ship parameters including the ship weight, initial velocity, direction, distance before impact, impact angle, coefficient of friction between the ship and the structure and the impact joint number on the SHIP input line together with 'SHIP' in columns 9-12 of the THLOAD line to designate a time history ship impact analysis.

2.6.4 Dropped Object Analysis

To conduct a dropped object analysis, enter the object weight, initial velocity, distance before impact and the impact joint name on the DRPOBJ input line together with 'DROP' in columns 9-12 of the THLOAD line to designate time history dropped object analysis.

2.6.5 Damping Method

General time history options are designated on the THLOAD line immediately following the LOAD header. Specify the damping type 'SDO' structural damping only, 'LFD' linearized fluid damping or 'NFD' for nonlinear fluid damping in columns 18-20.

Note: For nonlinear fluid damping, the fluid forces are calculated at every time step during the integration. This option requires the program calculated fluid damping option 'PC' on the FDAMP line.

For linearized fluid damping, the damping amplitude used to calculate the equivalent linear fluid damping may be overridden by specifying a value in columns 21-28.

2.6.6 Interpolation Scheme

The method used to interpolate between time history input values is designated in columns 29-30 on the THLOAD input line. Enter 'LN', 'QD' or 'CU' for linear, quadratic or cubic interpolation, respectively.

2.6.7 Integration Parameters

Integration parameters are stipulated on the TIME line. Enter the start for the beginning of the time history integration in columns 11-20. If the analysis is to terminate before the end of the time history input, enter the end time in columns 21-30. The output time interval, minimum integration step and the tolerance factor are designated in columns 31-40, 41-50 and 51-60, respectively.

For example, the following describes a time history function specified in the input file. Structural damping only is used in conjunction with linear interpolation as designated on the THLOAD line. One time history function is used with directionality factors of 1.0, 1.0 and 0.5 applied to it for the X, Y and Z directions, respectively. The start time is 0 seconds and end time 25 seconds. Output is requested at every 0.25 seconds.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	LOAD							
2	DROPOBJ	5.	0.0	5.0		3218		
3	THLOAD	DROP	SDO	PLTPLMPLSPRTALLMXSCLPJPD				
4	THFACT	1.0	1.0	0.5				
5	TIME	0.0	25.0	0.25				

2.6.8 Output Options

The time history earthquake analysis creates load cases, prints and plots modal responses, base shear and overturning moment in addition to joint accelerations, velocities and displacements. Analysis output options are designated in the output options fields in columns 33-59 on the THLOAD line.

2.6.9 Load Case Creation

The Dynamic Response program has the ability to create a load case corresponding to the time point having maximum overturning moment and/or maximum base shear by specifying 'MXM' or 'MXS' in the output options fields on the THLOAD line, respectively. Enter 'ALL' if load cases are to be created at for all time points, enter 'ESL' to generate equivalent static loads for a subsequent static analysis, enter 'CLP' to generate incremental loads for a Collapse analysis.

Note: the 'ESL' and the 'CLP' options are mutually exclusive. Only one of the options may be designated.

2.6.10 Modal Response Data

Modal responses versus time may be printed and/or plotted by specifying 'PRT' and 'PLT', respectively, in the output options fields on the THLOAD line.

2.6.11 Base Shear and Overturning Moment Plots

Base shear and overturning moment plots may be generated by entering 'PLM' and 'PLS' in one of the output option fields located on the THLOAD line.

2.6.12 Joint Results

Joint results including acceleration, velocity and displacement may be plotted and listed for up to sixteen joints. Joint plot options are specified in the output options fields on the THLOAD line.

Joint acceleration options include:

'JMA' Prints maximum and minimum values for joint acceleration for each direction.

'JPA' Same as JMA plus plots acceleration time history

'JTA' Same as JPA plus prints acceleration time history data

Note: Joint acceleration options are mutually exclusive. Only one of the options may be selected.

Joint velocity options include:

'JMV' Prints maximum and minimum values for joint velocity for each direction.

'JPV' Same as JMV plus plots velocity time history

'JTV' Same as JPV plus prints velocity time history data

Note: Joint velocity options are mutually exclusive. Only one of the options may be selected.

Joint displacement options include:

'JMD' Prints maximum and minimum values for joint displacement for each direction.

'JPD' Same as JMD plus plots displacement time history

'JTD' Same as JPD plus prints displacement time history data

Note: Joint displacement options are mutually exclusive. Only one of the options may be selected.

Any number of joints may be designated for plots and reports. Joints to be output are specified using JTNUM lines immediately following the THLOAD line.

The following illustrates a typical dynamic response input for a ship impact analysis. The analysis option is set to 'SHIP' on the DROPT line. Structural damping of 5 percent is assigned on the SDAMP input line. The SHIP line describes a 1250 tonne ship with an initial velocity of 6 knots travelling in a 180 degree direction. The distance between the ship and the structure is given as 1 meter. The output load option on the THLOAD line is set to generate incremental loading for a Collapse analysis. The results are output for joints 31P7 and 701 on the JTNUM line. The analysis start time is set to 0 seconds and the end time is set to 2.0 seconds on the TIME input line. The results are output at every 0.01 seconds.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	DROPT	SHIP	EC+Z					
2	SDAMP	5.0						
3	LOAD							
4	SHIP	1250.	6.0	180.	1.0		31P7	
5	THLOAD	SHIP	SDO		PLTPLMPLSPRTALLMXSCLPJPD			
6	JTNUM	31P7	701					
7	TIME	2.00000	0.0100	1.0E-9	1.0000			
8	END							

The following example illustrates a typical dropped object analysis. The analysis option is set to 'DROP' on the DROPT line. Structural damping of 5 % is assigned on the SDAMP input line. The weight of the object is defined as 5 tonnes on the DRPOBJ line. The initial velocity of the object is defined as 0 meters per seconds and the distance before impact is assigned as 5 meters. The impact joint is defined as 3218.

The output load option is set to generate incremental loading for a Collapse analysis by entering 'CLP' in columns 51-53 of the THLOAD line. Results are output for joint 3218 on the JTNUM line.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	DROPT	DROP	EC+Z					
2	SDAMP	5.0						
3	LOAD							
4	DRPOBJ	5.0	0.0	5.0		3218		
5	THLOAD	DROP	SDO		PLTPLMPLSPRTALLMXSCLPJPD			
6	JTNUM	3218						
7	TIME	2.00000	0.0100	1.0E-9	1.0000			
8	END							

3 COMMENTARY

3.1 BASE DRIVEN SYSTEM

The primary purpose of the deflection driven system is to calculate the structural response due to earthquakes. For this purpose, all support points are assumed to be moving with the ground. Since a modal analysis is being used, each mode can be considered to act independently of the other modes and can be shown to act as a single degree of freedom system as follows:

The force - deflection relation for an elastic linear structure can be expressed as:

$$\{F\} = [K]\{d\} \quad (1)$$

where $\{F\}$ is the force vector, $[K]$ is linear stiffness matrix and $\{d\}$ is the deformation vector.

Separating the free and the reaction degrees of freedom, the force - deflection relation can be expressed as:

$$\begin{Bmatrix} F_F \\ F_R \end{Bmatrix} = \begin{bmatrix} K_{FF} & K_{FR} \\ K_{RF} & K_{RR} \end{bmatrix} \begin{Bmatrix} d_F \\ d_R \end{Bmatrix} \quad (2)$$

where the F and R subscripts differentiate the free and reaction degrees of freedom.

For a base driven system, the loading in the free degrees of freedom is due to the inertia and can be expressed as:

$$\{F_F\} = -[M_{FF}]\{\ddot{d}_F\} \quad (3)$$

where d_F are the accelerations of the free degrees of freedom and M_{FF} is the mass matrix.

From equation (2),

$$\{F_F\} = [K_{FF}]\{d_F\} + [K_{FR}]\{d_R\} \quad (4)$$

which becomes:

$$-[M_{FF}]\{\ddot{d}_F\} = [K_{FF}]\{d_F\} + [K_{FR}]\{d_R\} \quad (5)$$

The deformation of a free degree of freedom can be expressed in terms of deformation due to external loads and deformation due to movement of the supports, so that

$$\{d_F\} = \{d_{FE}\} + \{d_{FS}\} \quad (6)$$

where d_{FE} is due to external loads and d_{FS} is due to movement of the supports. Equation (4) becomes

$$\{F_F\} = [K_{FF}]\{d_{FE}\} + [K_{FF}]\{d_{FS}\} + [K_{FR}]\{d_R\} \quad (7)$$

by definition

$$\{d_{FE}\} = [K_{FF}]^{-1}\{F_F\} \quad (8)$$

so that

$$[K_{FF}]\{d_{FS}\} + [K_{FR}]\{d_R\} = 0 \quad (9)$$

or

$$\{d_{FS}\} = -[K_{FF}]^{-1}[K_{FR}]\{d_R\} \quad (10)$$

Substituting (10) into (6) yields the following:

$$\{d_F\} = \{d_{FE}\} - [K_{FF}]^{-1}[K_{FR}]\{d_R\} \quad (11)$$

Differentiating both sides twice with respect to time, produces

$$\{\ddot{d}_F\} = \{\ddot{d}_{FE}\} - [K_{FF}]^{-1}[K_{FR}]\{\ddot{d}_R\} \quad (12)$$

Substituting equation (11) and (12) into equation (5),

$$[M_{FF}]\{\ddot{d}_{FE}\} + [K_{FF}]\{d_{FE}\} = [M_{FF}][K_{FF}]^{-1}[K_{FR}]\{\ddot{d}_R\} \quad (13)$$

The deformation of the free degrees of freedom with the base fixed can be expressed in terms of the normal vibration modes of the restrained structure such that;

$$\{d_{FE}\} = [\theta_{FF}]\{\xi\} \quad (14)$$

where θ_{FF} are mode shapes and ξ represents modal coordinates. Substituting equation (14) into equation (13) yields the following:

$$[M_{FF}][\theta_{FF}]\{\ddot{\xi}\} + [K_{FF}][\theta_{FF}]\{\xi\} = [M_{FF}][K_{FF}]^{-1}[K_{FR}]\{\ddot{d}_R\} \quad (15)$$

Noting that the eigenvalues/vector relation of the mass matrix,

$$[M_{FF}]^{-1}[K_{FF}][\theta_{FF}] = [\theta_{FF}]\omega_n^2 \quad (16)$$

where ω_n are the natural frequencies of the restrained structure, substituting (16) into (15) and multiplying by the mode shapes,

$$[\theta_{FF}]^T[M_{FF}][\theta_{FF}]\{\ddot{\xi}\} + [\theta_{FF}]^T[M_{FF}][\theta_{FF}]\{\xi\}\omega_n^2 = [\theta_{FF}]^T[M_{FF}][K_{FF}]^{-1}[K_{FR}]\{\ddot{d}_R\} \quad (17)$$

Noting that the generalized mass is expressed as:

$$[\overline{M}] = [\theta_{FF}]^T[M_{FF}][\theta_{FF}] \quad (18)$$

equation (17) becomes:

$$[\overline{M}]\{\ddot{\xi}\} + [\overline{M}]\omega_n^2\{\xi\} = [\theta_{FF}]^T[M_{FF}][K_{FF}]^{-1}[K_{FR}]\{\ddot{d}_R\} \quad (19)$$

or

$$\{\ddot{\xi}\} + [\omega_n^2]\{\xi\} = [\overline{M}]^{-1}[\theta_{FF}]^T[M_{FF}][K_{FF}]^{-1}[K_{FR}]\{\ddot{d}_R\} \quad (20)$$

Defining the parameter R as

$$[R] = -[\overline{M}]^{-1}[\theta_{FF}]^T[M_{FF}][K_{FF}]^{-1}[K_{FR}] \quad (21)$$

equation (20) may be expressed as:

$$\{\ddot{\xi}\} + [\omega_n^2]\{\xi\} = -[R]\{\ddot{d}_R\} \quad (22)$$

Modal damping can be added so that

$$\{\ddot{\xi}\} + [2c\omega_n]\{\dot{\xi}\} + [\omega_n^2]\{\xi\} = -[R]\{\ddot{d}_R\} \quad (23)$$

where c is the modal critical damping ratio. Equation (23) then represents the modal equation of motion for a structure subjected to base driven excitation. Each of the modes on the left side of the equation are uncoupled so that each mode can be analyzed as an equivalent single degree of freedom system.

3.1.1 Responses Due to Sinusoidal Input

If the base motion can be described by sinusoidal input such that

$$\{d_R\} = \{d_{RO}\} \sin \omega t \quad (24)$$

then the equation of motion becomes

$$\{\ddot{\xi}\} + [2c\omega_n]\{\dot{\xi}\} + [\omega_n^2]\{\xi\} = [R]\{d_{RO}\}\omega^2 \sin \omega t \quad (25)$$

Note that the R matrix is an NM x NR matrix where NM is the number of modes and NR is the number of reaction degrees of freedom. If the reaction degrees of freedom are not moving together as if rigidly connected, there will be stresses induced due to the relative support point motions. These effects can be accounted for by using equation (10) and substituting into the elemental stress-deformation relations. For the purpose of earthquake response, it is assumed that there is no significant relative motion between supports and that those resulting stresses would be insignificant.

The response of the modal degrees of freedom in equation (25) can be obtained by similarity with single degree of freedom systems such that the maximum steady state response amplitude can be expressed as:

$$\xi_{\max} = \frac{R d_{RO} \Omega^2}{\sqrt{(1 - \Omega^2)^2 + (2c\Omega)^2}} \quad \text{where} \quad \Omega = \frac{\omega}{\omega_n} \quad (26)$$

3.1.2 Responses Due to Time History Base Motion

If the base motion consists of a history of accelerations versus time, then the equation of motion becomes:

$$\{\ddot{\xi}\} + [2c\omega_n]\{\dot{\xi}\} + [\omega_n^2]\{\xi\} = -[R]\{\ddot{d}_R(t)\} \quad (27)$$

where the base motion is represented by $d_R(t)$.

In this case, the results are obtained by integrating each of the equivalent single degree of freedom modal equations in the time domain to obtain the deflections, velocities, and accelerations versus time. This approach is very useful when measured data is available.

The effective modal damping is extremely important in computing the dynamic response of a structure. In the case of a structure immersed in a fluid, the effects of the fluid on the structure can be to effectively increase the modal damping in addition to the effective increase in mass. There are two options available in the Dynamic Response program for this type of fluid damping consideration. The first is to assume a value of effective fluid damping and compute the responses accordingly. This approach requires less computer time, but requires more knowledge about the effects of fluid on the structure than is normally available. In the other case, the program will calculate the fluid forces on the structure at each integration time step and use these forces in the equations of motion.

3.1.3 Responses Due to Spectral Input

The output power spectral density (PSD) is related to the input PSD by the mechanical admittance function as follows:

$$\Psi = H^2 \Phi \quad (28)$$

where Ψ is output the power spectral density, Φ is the input power spectral density and H is the mechanical admittance function. For a single degree of freedom system, the H function is:

$$|H| = \frac{1}{\sqrt{(1 - \Omega^2)^2 + (2c\Omega)^2}} \quad (29)$$

Extending the single degree of freedom system results to a multi-degree of freedom system, the mechanical admittance function becomes:

$$|H| = R |H_S| \quad (30)$$

The mean squared response of a single degree of freedom system due to spectral input is:

$$\xi_S^2 = \int_0^\infty \Psi d\omega = \int_0^\infty H_S^2 \Phi d\omega \quad (31)$$

For most structural applications, the damping is large enough to result in a narrow banded transfer function. In that case, the input PSD can be considered constant (white noise) in the vicinity of the natural frequency resulting in a response of:

$$\xi_S = \sqrt{\frac{\omega_n \Phi}{8c}} \quad (32)$$

For multi-degree of freedom systems, the response becomes:

$$\xi^2 = \int_0^\infty \Psi d\omega = \int_0^\infty R^2 H_S^2 \Phi d\omega \quad (33)$$

Therefore:

$$\xi = |R| \xi_S \quad (34)$$

In some cases, the input PSD is not supplied, but the response of a single degree of freedom system versus frequency (response spectrum) is supplied. In that case, the R matrix can be used as shown in equation (34) to compute the response of each mode.

Equation (34) gives the response of a mode for base motion input in a particular direction. The next consideration is how to combine the modes to obtain the total response for excitation in one direction and how to combine the results for multi-directional driving motions. Normally, the modes are assumed to have sufficient frequency separation such that the responses can be considered to be uncorrelated. If that assumption is valid, then the modes can be combined using the SRSS (Square Root of the Sum of the Squares) method. In the case where there is insufficient separation, it may be necessary to consider correlation between modal responses. In that case, the modes can be combined using the CQC (Complete Quadratic Combination) method.

In either case, the base driving input from different directions are assumed uncorrelated so that the combination of responses for the different directions are done using the SRSS approach.

As in the case of the time history analysis, damping effects are very important and in the case where the structure is immersed in a fluid, the damping due to fluid is a nonlinear effect. For the spectral analysis, the Dynamic Response program has three approaches to the fluid damping calculations. The first and most simplistic is to assume a linear modal damping value and proceed with the normal response calculations. Since the fluid forces are nonlinear and the basis for the spectral approach is linear, the Dynamic Response program has two additional options for calculating the nonlinear damping. The basic approach is to calculate an equivalent linear damping ratio that will dissipate the same amount of energy in one cycle as the actual nonlinear fluid damping. The amount of energy dissipated depends on the amplitude of the response. One option allows the user to specify the response amplitude to be used in the equivalent linear damping calculations. The next option is to allow the program to calculate the amplitude through an iteration technique as follows: First the response is calculated using the fluid damping based on an assumed amplitude. Then based on that response, the equivalent fluid damping is recalculated for the next response calculation. This process is repeated until the response amplitude agrees with the amplitude used in the equivalent fluid damping calculation. This iteration procedure does not use a large amount of computer time since the response calculations only involve a few modal degrees of freedom.

3.1.4 Equivalent Static Load Generation

The program has the ability to generate a set of equivalent static loads for the earthquake event.

The equivalent static load procedure assumes that the primary structure response may occur in any direction during the earthquake event (not only along the X or Y axis). The response of the structure is therefore calculated for 20 directions (every 18 degrees). For each of these twenty directions, the base shear in that direction and the moment about that direction are determined. Equivalent static loads are then generated using the direction with highest base shear or overturning moment, based on the output option selected.

Note: The program prints the response in the X (0.0 degree) and Y (90.0 degree) directions. Because these responses do not occur at the same time, the equivalent static load may be generated for a different direction.

3.2 Force Driven System

There are many applications for force driven systems such as vibrations due to operating machinery, impact loadings, wind gust loadings, etc. The equation to describe the force driven system is simpler than the base driven system. In generalized coordinates (modal form), the equation is:

$$[M]\{\ddot{\xi}\} + [C]\{\dot{\xi}\} + [K]\{\xi\} = \{F(t)\} \quad (35)$$

where $[M]$ is the generalized mass, $[C]$ is generalized damping, $[K]$ is generalized stiffness, $\{F\}$ is the generalized force vector and $\{\xi\}$ is modal displacement.

Since these equations are uncoupled, each mode can be analyzed separately and the results then combined in a linear manner as required. The calculation of the generalized forces consists simply of summing the force multiplied by the normalized modal displacements for all the forces applied. This results in an analysis directly comparable to a single degree of freedom analysis.

Basically, the force driven applications can be divided into two categories, periodic loadings and time history loadings. In the case of periodic loadings, the equation describing these applications can be expressed as:

$$\ddot{\xi} + 2c\omega_n\dot{\xi} + \omega_n^2\xi = \frac{F}{M}\sin(\omega t) \quad (36)$$

where ω is the driving frequency, ω_n is the natural frequency and c is the damping ratio.

The steady state response can be expressed as:

$$\xi = \xi_{\max} \sin(\omega t + \phi) \quad (37)$$

(Amplitude)

$$\xi_{\max} = \frac{F}{\omega_n^2 M} \left(\frac{1}{\sqrt{(1-\Omega^2)^2 + (2c\Omega)^2}} \right) \quad (38)$$

(Phase Angle)

$$\Omega = \frac{\omega}{\omega_n} \quad (39)$$

$$\psi = \tan^{-1} \left(\frac{2 c \Omega}{1 - \Omega^2} \right) \quad (40)$$

The response frequency is the same as the driving frequency.

The program allows any number of forces and moments to be applied at any number of joints. The modal responses are calculated for all modes and for all forcing functions. The resulting deflections, internal loads, stresses, etc. are then available for any subsequent post-processing.

One of the most common applications for the periodic vibration analysis capability is to prevent machinery from causing unacceptable vibration levels in living quarters. In this case, the governing criterion is that deflection or acceleration does not exceed a maximum allowable for any translation degree of freedom for the engine/compressor operating speed range. The program searches through all the translation degrees of freedom and determine the maximum period specified for all conditions.

For time history analyses, the generalized force is calculated as a function of time based on the input load time histories supplied by the user. Any number of load time histories can be input which the program will use to calculate accelerations, velocities, and displacements by integration techniques. The program uses a variable step integration procedure which efficiently changes step size to maintain error control regardless of how slow or fast the variables are changing in value.

The input loading is described at time points throughout the complete time history. For the intermediate time points, the program must use some form of interpolation scheme. This is an option for the user which can be very significant. The options available are linear, quadratic, and cubic interpolations. Linear interpolation allows the user to represent sudden changes in loading such as step functions, ramps, impulses, etc. without having the interpolation function smoothing out the desired changes. On the other hand, the user can represent a smooth function with fewer points and without introducing any roughness by using higher order interpolation schemes.

3.2.1 Allowable Displacement for Reciprocating Machinery

The following is what we have found for allowable displacement in mils (mil = one thousandth of an inch):

SNAME (The Society of Naval Architects and Marine Engineers):

$$A(\text{mils}) = \frac{V_{\max}}{2\pi f}$$

where $V_{\max}=293$ mils/sec

Reference:

Technical and Research Bulletin 2-25

Ship Vibration and Noise Guidelines

Prepared by Panel HS-7 of the Hull Structure Committee

Published by The Society of Naval Architects and Marine Engineers

One World Trade Center, Suite 1369, New York, N.Y. 10048

January 1980

D-Line is from CDG "D" Line from Shell:

$$A(\text{mils}) = \frac{V_{\max}}{2\pi f}$$

where $V_{\max}=149$ mils/sec

Military Spec is from "L-Exp" Military Long Exposure Allowable:

For $f \leq 1200\text{cpm}$

$$A(\text{mils}) = 762.645 \left(\frac{60}{f} \right)^2$$

For $5400 \geq f > 1200\text{cpm}$

$$A(\text{mils}) = 3.625 \left(\frac{60}{f} \right)^{0.21447}$$

For $f > 5400\text{cpm}$

$$A(\text{mils}) = 11185 \left(\frac{60}{f} \right)^2$$

3.3 Ice Vibration

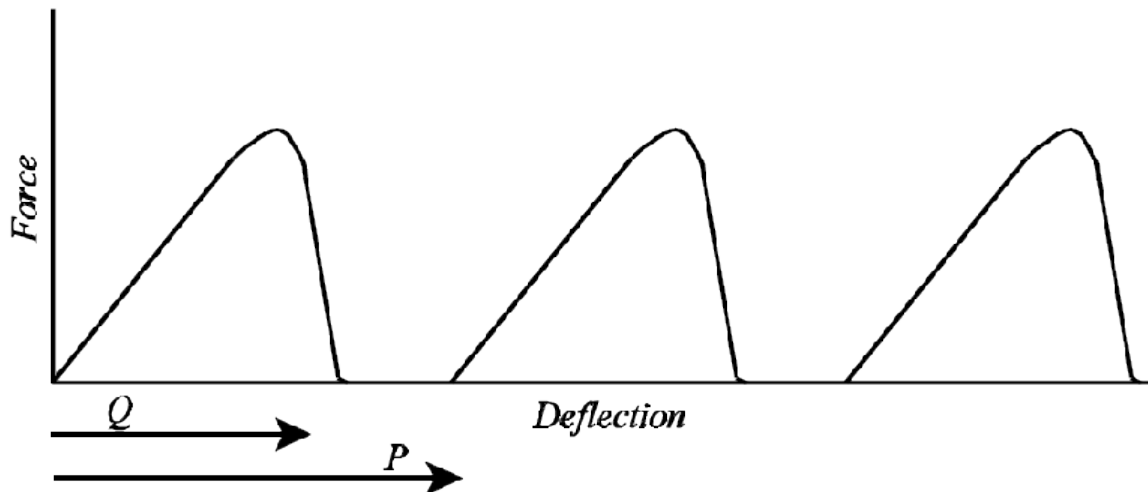
When a sheet of ice moves past a structure, the ice breaks as it encounters the structural members that penetrate the ice sheet. After breaking, the ice builds up to breaking load and breaks again. This breaking and buildup cycle is shown in the figure on the following page where Q is the buildup distance and P is the overall breaking distance. The deflection is the relative deflection between the ice and the structure at the point of contact. The slope of the buildup is based on the ice deflection as analyzed by the model of a simulated infinite ice sheet. From this analysis, the ice stiffness has been parameterized to be a function of ice thickness, ice modulus of elasticity, and the width of penetration.

Since the ice force is a function of relative displacement, a time history analysis is performed. The relative displacement is calculated by:

$$U = V_{\text{ice}} \times t - d_{\text{struct}} - d_{\text{crush}}$$

where U is the relative displacement, V_{ice} is the ice velocity, t is time, d_{struct} is the structural deflection in ice floe direction and d_{crush} is the crushed ice distance from start of analysis.

The time history analysis should be conducted until the startup transients have dissipated. For the fatigue analysis, the program uses the mode having the largest response to monitor the cycles. It uses the last half of the time history and calculates the maximum and minimum values. It then counts the number of times that the mode crosses the average of the maximum and minimum, and divides by two to calculate the number of cycles. This value is then divided into one half of the time span to get an effective period.



3.4 Spectral Wind

3.4.1 Generalized Force Spectrum

The generalized force spectrum used for wind spectral analysis is a Harris wind spectrum with gust effects spatial correlation and mean wind velocity variation. For any mode, the generalized force spectrum $S_{qi}(f)$ is taken as:

$$S_{qi}(f) = \frac{4}{f} F_i^2 J_{ai}(f) J_{ri}^2(f) S_V(f)$$

where F_i is the generalized force and $S_V(f)$ is the Harris spectrum given by:

$$S_V(f) = \frac{4k\eta_1(f)}{(2 + \eta_1^2(f))^{\frac{5}{6}}}$$

In which

$$\eta_1 = f \left(\frac{L_H}{v_{10}} \right)$$

Where k is the roughness parameter (default 0.0025) L_H is the Harris spectrum reference length (default 1800 m) and v_{10} is the velocity at the reference height (10m). The terms J_{ai} and J_{ri} are the mean wind velocity variation function and the gust effects spatial correlation function, respectively.

3.4.2 Mean Wind Velocity Variation Function J_a

The mean wind variation models the mean wind speed variation with height. It is expressed as:

$$J_{ai} = \left[\frac{\int_0^{z_0} \gamma_i(z) dz}{\int_0^{z_0} \gamma_i(z) \frac{V(z)}{V_{10}} dz} \right]^2$$

Where z_0 is the distance to the top of the structure

$V(z)$ is the wind velocity variation with height given by:

$$V(z) = V_{10} \left[1.0 + 0.137 \ln \left(\frac{z}{z_0} \right) \right]$$

In which z_0 is the reference height in meters (10m) and

$$\gamma(z) = V(z) [C_d b] \mu_i(z)$$

Where

C_d is the drag coefficient

b is the cross-sectional dimension (OD for a tubular)

and $\mu_i(z)$ is the i^{th} mode shape

3.4.3 Gust Effects Spatial Correlation Function J_r

The J_r function expresses the correlation of gust effects over the structure and is given by:

$$J_{ri}(n)^2 = \frac{\int_0^{z_0} \int_0^{z_0} \gamma_i(z) \gamma_i(z') \text{Coh}(r, n) dz dz'}{\left[\int_0^{z_0} \lambda_i(z) dz \right]^2}$$

Where z' is used to differentiate z and z' as being two separate points in space and r is the separation difference.

The spatial correlation of the wind velocity is modeled by the Davenport coherence function taken as:

$$\text{Coh}(r, n) = \exp \left[\frac{-n}{V_{10}} (C_z^2 d_z^2 + C_y^2 d_y^2)^{1/2} \right]$$

Where:

d_y, d_z are the horizontal and vertical coordinate differences between the two points normal to the direction of the mean wind.

C_y, C_z are decay factors usually taken as 12 and 8 respectively.

3.4.4 Modal Response Spectrum R_q

The modal response spectrum is given by:

$$R_{qi}(f) = S_{qi}(f) H_i(f)^2$$

where

$$H_i(f) = \frac{1}{K_i} \left(\left[1 - \left(\frac{f}{f_n} \right)^2 \right]^2 + \left[2c \left(\frac{f}{f_n} \right)^2 \right] \right)^{\frac{1}{2}}$$

$H_i(f)$ is known as the mechanical transfer function in which K_i is the generalized stiffness matrix:

$$K_i = (2\pi f_n)^2 M_i$$

f_n is the natural frequency and c is the percent critical damping.

3.4.5 Modal Response

The RMS response for the i^{th} mode may be expressed as follows:

$$\sigma_{RMS_i} = \sqrt{\int_0^\infty H_i^2(f) S_i(f) df}$$

The response of each mode is then combined using the CQC (complete quadratic combination) method to obtain the total response σ_{jtot} using the expression below:

$$\sigma_{jtot}(f) = \sqrt{\sum_i \sum_k P_{ik} \sigma(f)_{ij} \sigma(f)_{kj}}$$

where $i, k = 1..$ number of modes and P_{ik} is the modal correlation coefficient:

$$P_{ik} = \frac{2\sqrt{c_i c_k} [(w_i + w_k)^2 (c_i + c_k) + (w_i^2 - w_k^2)(c_i - c_k)]}{4(w_i - w_k)^2 + (c_i + c_k)^2 (w_i + w_k)^2}$$

where c_i and w_i are respectively damping factor and frequency for mode i

In the case of spectral wind fatigue, the modal responses are passed to fatigue as fatigue load cases to calculate the fatigue damage.

3.4.6 Dynamic Amplification Factor and Peak Member Forces

In the case of extreme wind, the dynamic amplification factor is calculated for all members by the following expression:

$$DAF_j = 1 + G_j \frac{\sigma_{jtot}(f)}{F_{Mj}}$$

where

$$G_j = \sqrt{2 \ln(T_a v_j)} + \frac{0.577}{\sqrt{2 \ln(T_a v_j)}}$$

In which

j is an iterator for each force component for each member

T_a is the applied averaging time

F_{mj} is the static force response due to a mean wind load

v_j is the resulting zero crossing frequency given by:

$$v_j = \frac{\sqrt{\sum_i (\sigma_{ij}(f) v_i)^2}}{\sigma_{jtot}(f)}$$

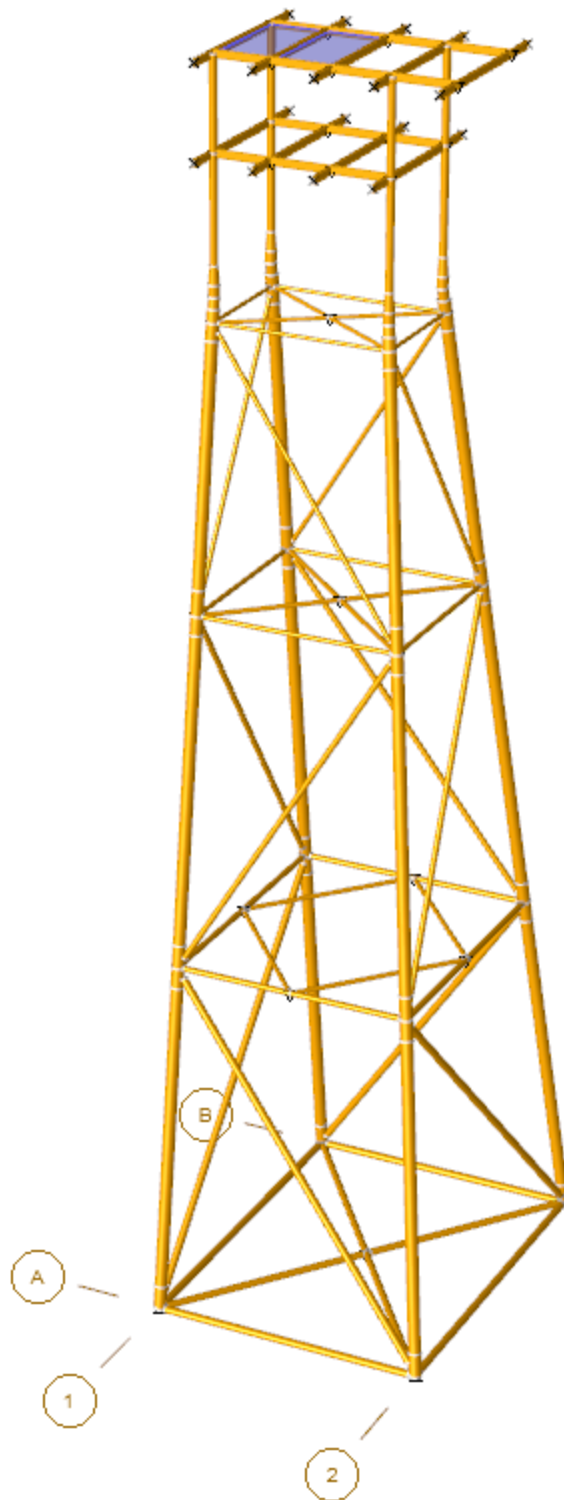
The forces due to dynamic wind loading are calculated as:

$$F_{maxj} = DAF_j F_{Mj}$$

4 SAMPLE PROBLEMS

The structures shown in Figure below was used to illustrate various capabilities of the Dynamic Response program. Seven separate response analyses are illustrated:

1. The dynamic response of the structure due to ground motion was determined using the response spectrum approach. The responses of each mode were added using the Complete Quadratic Combination (CQC) method.
2. The response of the structure due to ground motion was determined using the Time History approach.
3. This problem is an engine vibration problem where the motion of the deck due to excitation from reciprocating machinery was analyzed.
4. The dynamic response due to a step load defined using a force time history was analyzed.
5. The structural response due to extreme wind gusts from four directions was determined.
6. Fatigue damage due to wind fatigue was performed using the spectral wind capabilities.
7. Fatigue damage due to vibration caused by ice floes was calculated by the program.



4.1 RESPONSE SPECTRUM SEISMIC ANALYSIS

The following is an example of a response analysis for a base driven system using the Response Spectrum approach.

The structure in Figure 1 stands in 261 feet of water. It is located in a seismic zone with stiff soil where the ratio of effective ground acceleration to gravitational acceleration is 0.20.

The response spectrum was applied equally along both principal orthogonal horizontal axes along with an acceleration spectrum equal to one-half of that applied in the vertical direction. All three spectra were applied simultaneously and the responses were combined using the complete quadratic combination (CQC) method. Five percent critical damping was assumed.

The static analysis included gravity and buoyancy loads. The stresses induced by earthquake loads were combined with the static stresses for the purpose of member strength check. For tubular joint check, the seismic stresses were doubled then combined with the static stresses. In either case, the allowable stresses were increased by 70 percent.

Note: For any dynamic analysis, a foundation super element, dummy pile or equivalent pile stub used to simulate the soil/pile interaction must be developed using the PSI or PILE program.

For this sample problem, Load Case EQKS containing the gravity and buoyancy of the structure was generated by Seastate then solved. Dynpac was used to generate the dynamic characteristics of the structure. The mass of the structure includes the mass associated with gravity, enclosed fluid and added mass.

The Dynamic Response program was used to predict the response of the structure due to the ground motion caused by an earthquake. The structure was analyzed using the response spectrum approach with an effective horizontal ground acceleration of $0.20 \times G$.

The following is the Dynamic Response input file used in conjunction with the generalized mass file and solution file from Dynpac.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	DROPT SPEC	15	+Z	-261.				
2	* USE	5.0%	OVERALL	DAMPING				
3	SDAMP	5.0						
4	* CREATES	STATIC +	SEISMIC	COMBINATIONS	USING	STATIC	LOAD	CASE EQKS
5	* USE	1.0	X	SEISMIC	STRESS	FOR	ELEMENT	CHECK
6	* USE	2.0	X	SEISMIC	STRESS	FOR	CONNECTION	CHECK
7	STCMB	1.0	2.0	EQKS	1.0			
8	LOAD							
9	* CREATE	1	SEISMIC	LOAD	CASE	WITH	0.2G	RESPONSE
10	* WITH	1.0	X	AND	Y	DIRECTION	FACTOR	AND
11	SPLAPI	.2	1.0B	1.0B	0.5C	CQC		
12	RSFUNC	803DX	5.0	803DY	5.0	803DZ	5.0	
13	END							

The following is a detailed description of the Dynamic Response input file, the results of the analysis ensue:

Line 1. The DROPT line specifies analysis options, namely:

-
- a. A base driven spectral earthquake analysis is to be performed ('SPEC' in columns 7-10).
 - b. A mudline elevation of -261. is specified in columns 19-24.

Line 3. The SDAMP line specifies that the overall structural damping is 5.0 percent.

Line 7. The STCMB line is used to have seismic and static results combined automatically. The STCMB line designates the following:

- a. Seismic loading is to be factored by 1.0 when combined with static loading for element check load cases.
- b. Seismic loading is to be factored by 2.0 when combined with static loading for joint check load cases as designated by 2.0 in columns 13-17.
- c. Load case EQKS from the static solution file is to be combined with the seismic load cases.

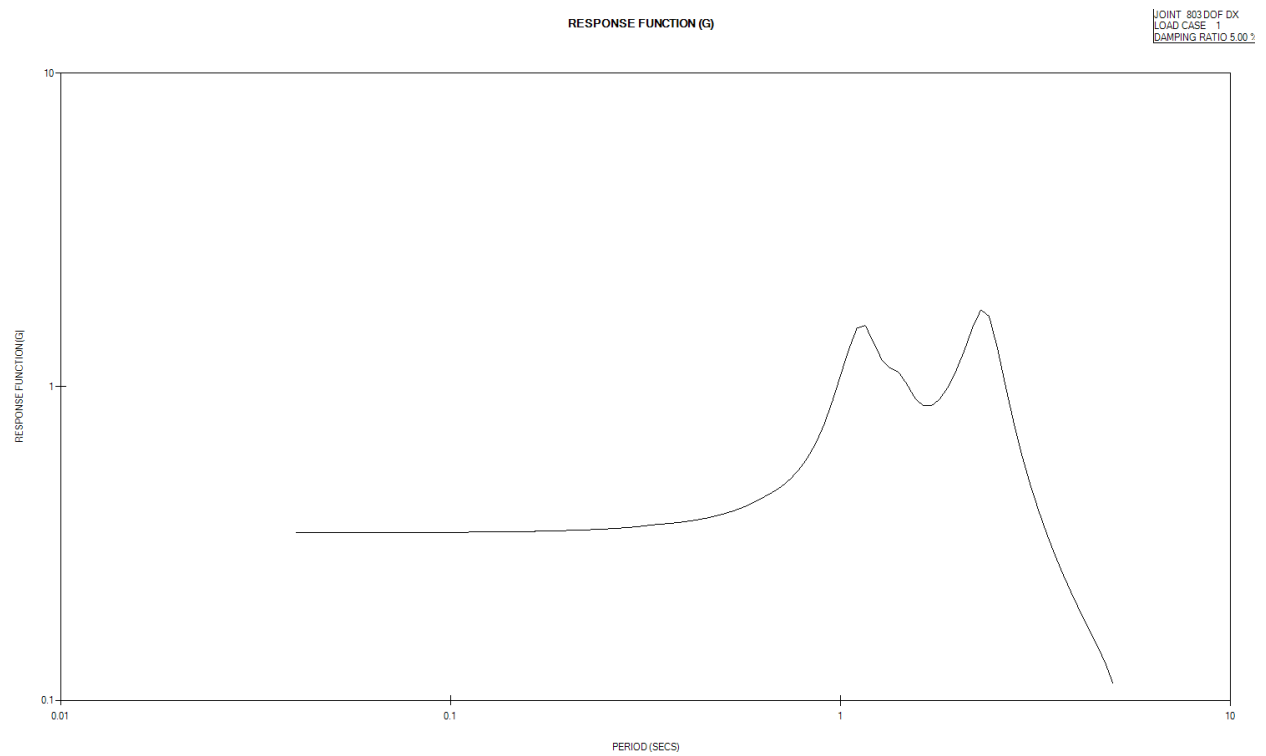
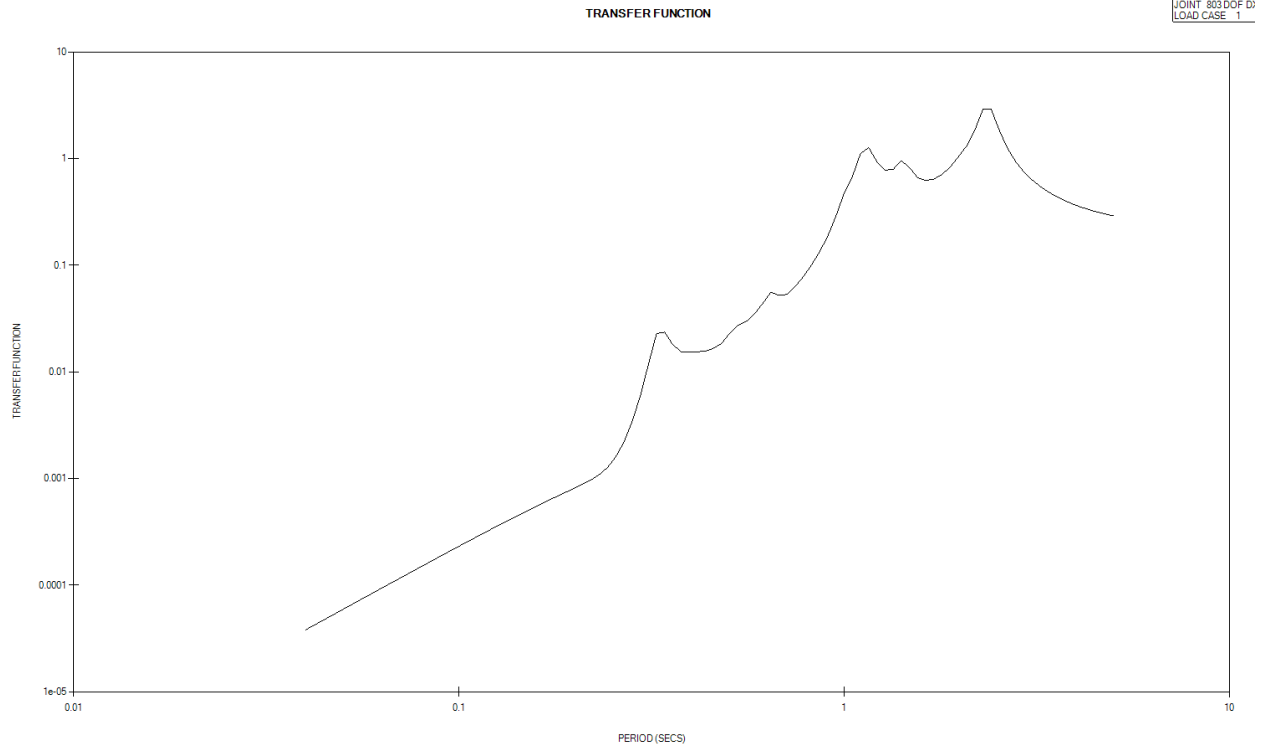
Line 8. The LOAD line specifies that loading data is to follow.

Line 11. The SPLAPI line defines the spectral analysis parameters as follows:

- a. The API RP2A spectrum for soil type 'B' is to be used for directions X and Y as specified by 'B' in columns 22, 29 and soil type 'C' is used for the Z direction as specified by 'C' in column 36.
- b. Modal responses will be combined using the CQC method as designated in columns 38-41.
- c. A response or ground acceleration factor of $0.20 * G$ is specified in columns 11-15.
- d. The percent of the ground acceleration factor to be applied in each of the global directions is 100.0, 100.0 and 50.0 for the X, Y and Z global directions respectively as specified by 1.0, 1.0 and 0.50 in columns 16-21, 23-28 and 30-35, respectively.

Line 12. The response function for the global X, Y and Z directions for joint 803 is requested on the RSFUNC line. The response function and power spectral density for 5 percent damping plotted verse time will be generated (PG1 O R in columns 70-76).

Below are the response function and spectral density plots created in Dynamic Response followed by a portion of the output containing the directional responses.



	1	2	3	4	5	6	7	8
1	1234567890123456789012345678901234567890123456789012345678901234567890							
2	OPTIONS	EN	UCJO	PTPT	PT			
3	ENGLISH TEST MODEL							
4	3 12 0 20		3					
5	1 15CQC	DYNAMIC RESPONSE						
6	P 1	0.91753E+01		0.42591E+00	0.50000E-01			
7	P 2	0.12882E+00		0.48890E+00	0.50000E-01			
8	P 3	0.30649E+00		0.70236E+00	0.50000E-01			
9	P 4	-0.17174E+01		0.88473E+00	0.50000E-01			
10	P 5	0.70270E-01		0.99501E+00	0.50000E-01			
11	P 6	0.46359E-02		0.15313E+01	0.50000E-01			
12	P 7	-0.50549E-01		0.15626E+01	0.50000E-01			
13	P 8	0.88794E-02		0.18498E+01	0.50000E-01			
14	P 9	-0.38424E-02		0.19746E+01	0.50000E-01			
15	P 10	0.28896E-01		0.24008E+01	0.50000E-01			
16	P 11	-0.26591E-02		0.24840E+01	0.50000E-01			
17	P 12	0.21402E-01		0.25291E+01	0.50000E-01			
18	P 13	-0.97135E-01		0.26450E+01	0.50000E-01			
19	P 14	-0.11110E+00		0.28225E+01	0.50000E-01			
20	P 15	-0.40150E+00		0.30421E+01	0.50000E-01			
21	2 15CQC	DYNAMIC RESPONSE						
22	P 1	0.16728E-01		0.42591E+00	0.50000E-01			
23	P 2	0.93210E+01		0.48890E+00	0.50000E-01			
24	P 3	-0.27678E+01		0.70236E+00	0.50000E-01			
25	P 4	-0.80927E-01		0.88473E+00	0.50000E-01			
26	P 5	-0.20806E+01		0.99501E+00	0.50000E-01			
27	P 6	-0.13201E+00		0.15313E+01	0.50000E-01			
28	P 7	0.51310E-02		0.15626E+01	0.50000E-01			
29	P 8	0.10128E-02		0.18498E+01	0.50000E-01			
30	P 9	-0.13799E+00		0.19746E+01	0.50000E-01			
31	P 10	-0.18181E-02		0.24008E+01	0.50000E-01			
32	P 11	-0.13417E-01		0.24840E+01	0.50000E-01			
33	P 12	0.65640E-03		0.25291E+01	0.50000E-01			
34	P 13	0.46190E-04		0.26450E+01	0.50000E-01			
35	P 14	-0.14306E-01		0.28225E+01	0.50000E-01			
36	P 15	0.21750E+00		0.30421E+01	0.50000E-01			
37	3 15CQC	DYNAMIC RESPONSE						
38	P 1	0.24800E+00		0.42591E+00	0.50000E-01			
39	P 2	0.19615E-02		0.48890E+00	0.50000E-01			
40	P 3	-0.10070E-01		0.70236E+00	0.50000E-01			
41	P 4	-0.14552E+00		0.88473E+00	0.50000E-01			
42	P 5	0.97356E-02		0.99501E+00	0.50000E-01			
43	P 6	0.43640E-02		0.15313E+01	0.50000E-01			
44	P 7	0.15387E+01		0.15626E+01	0.50000E-01			
45	P 8	-0.19477E-01		0.18498E+01	0.50000E-01			
46	P 9	-0.93781E-02		0.19746E+01	0.50000E-01			
47	P 10	-0.64096E-01		0.24008E+01	0.50000E-01			
48	P 11	0.11796E-01		0.24840E+01	0.50000E-01			
49	P 12	-0.11803E+00		0.25291E+01	0.50000E-01			
50	P 13	0.11713E+01		0.26450E+01	0.50000E-01			
51	P 14	0.28148E+00		0.28225E+01	0.50000E-01			
52	P 15	0.59587E-01		0.30421E+01	0.50000E-01			

Note: This file created by the Dynamic Response program, contains the responses for each of the three directions. The modal responses are combined using the CQC method as specified to obtain directional responses which are then combined using the RMS or SRSS method. The combine steps are executed automatically as part of the Dynamic Response analysis.

When using the STCMB line, the program creates four seismic+static load combinations, two for element check and two for joint can check, for each seismic load case as follows:

LC	Combine Type	Description
----	--------------	-------------

1	PRST	Element code check case, seismic axial tension
2	PRSC	Element code check case, seismic axial compression
3	PRST	Joint can check case, seismic axial tension
4	PRSC	Joint can check case, seismic axial compression

When using the STCMB feature, the Combine program is executed automatically to combine the seismic results and the static results. Two load cases for each member check and joint check will be created to account for the cyclic nature of the seismic loading. The following is the combine input file created by the program and used to create the combined load cases.

```

1234567890123456789012345678901234567890123456789012345678901234567890
1  ENGLISH TEST MODEL
2  CMBOPT
3  LCOND      PRST      EQK + STAT MEMB
4  COMP PEQKS  1.0000
5  COMP S    1    1.0000
6  LCOND      PRSC      EQK + STAT MEMB
7  COMP PEQKS  1.0000
8  COMP S    1    1.0000
9  LCOND      PRST      EQK + STAT JOIN
10 COMP PEQKS  1.0000
11 COMP S    1    2.0000
12 LCOND      PRSC      EQK + STAT JOIN
13 COMP PEQKS  1.0000
14 COMP S    1    2.0000
15 END

```

Note: The STCMB feature requires that the static solution file exist prior to executing the earthquake analysis. If the STCMB feature is not used, the user must create the Combine input file and execute Combine as a separate analysis step.

The following are the Post and Joint Can input files, respectively, used for post processing:

```

1234567890123456789012345678901234567890123456789012345678901234567890
1  OPTIONS      EN      SDUCJO 1 1      PTPT      PT
2  LCSEL IN      1      2
3  AMOD
4  AMOD      1 1.7      2 1.7
5  END

```

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	JCNOPT EQK EN	2.0	C NID	M FLMX			1.75	
2	LCSEL IN	3	4					
3	AMOD							
4	AMOD	3 1.7	4 1.7					
5	END							

Note: Only load cases 1 and 2 are selected in the Post input file, while only load cases 3 and 4 are selected in the Joint Can input file. Notice also that the 'JO' option which designates that stresses are to be checked only at the member ends is specified in the Post input file.

4.2 TIME HISTORY SEISMIC ANALYSIS

Sample Problem 2 is similar to Sample Problem 1 except that the dynamic response due to ground motion was calculated using the time history approach.

The structure in Figure 1 is located in a seismic zone where site specific studies have been performed. The studies have yielded ground acceleration time history functions for each direction.

For this analysis, the time histories were applied in the respective directions separately. Five percent critical damping was assumed.

Equivalent static loading for each time point is generated by the analysis which can then be combined with gravity loading for structural code checks.

The following contains selected portions of the input and results of the analysis:

The Dynamic Response program was used to predict the response of the structure due to the ground motion caused by an earthquake. The structure was analyzed using a ground acceleration time history for each direction. The following is the Dynamic Response input file used in conjunction with the generalized mass file and solution file from Dynpac.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	DROPT TIME 10 +Z							
2	SDAMP 5.0							
3	LOAD							
4	THLOAD FILE SDO		ALLESPLTPLMPLS					
5	TIME		35.0	0.02	1E-10			
6	END							

The following is a detailed description of the Dynamic Response input file:

Line 1. The DROPT line specifies analysis options, namely:

- a. A base driven time history analysis is to be performed ('TIME' in columns 7-10).

Line 2. The SDAMP line specifies that the overall structural damping is 5.0 percent.

Line 3. The LOAD line specifies that loading data is to follow.

Line 4. The THLOAD line defines the spectral analysis parameters as follows:

- a. The time history functions are defined in a separate time history input file as specified by 'FILE' in columns 9-12.
- b. Only structural damping is to be used ('SDO' in columns 21-23).

- c. All time points are to be saved as a load case as designated in columns 33-35 by 'ALL'.
- d. Equivalent static load cases are to be saved as specified by 'ESL' in columns 36-38.
- e. Plots showing the modal responses, overturning moment and base shear versus time are requested by 'PLT', 'PLM' and 'PLS' respectively in columns 39-47.

Line 5. The TIME line designates the time history start and end times as 0.0 seconds and 35.0 seconds respectively with an output time interval of 0.2 seconds and a minimum time integration step of 1E-10 seconds.

The time history input file contains the ground motion information for each time step. The following is an excerpt of the time history

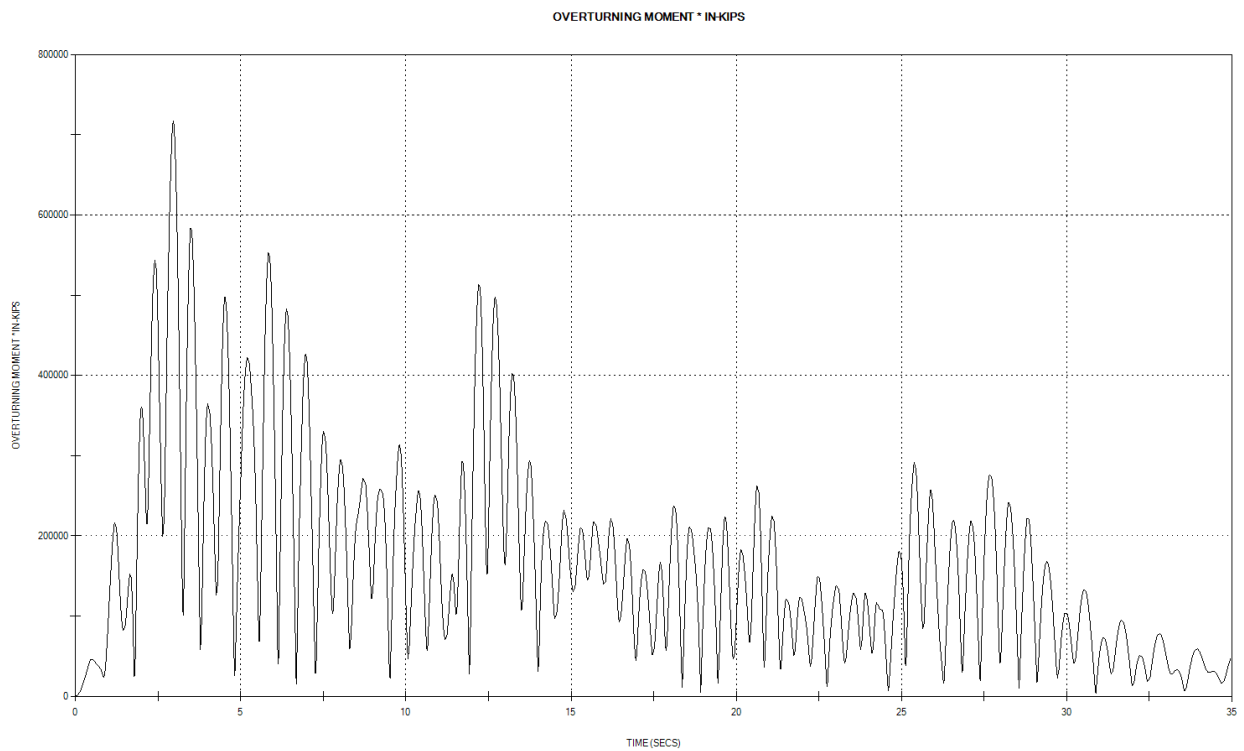
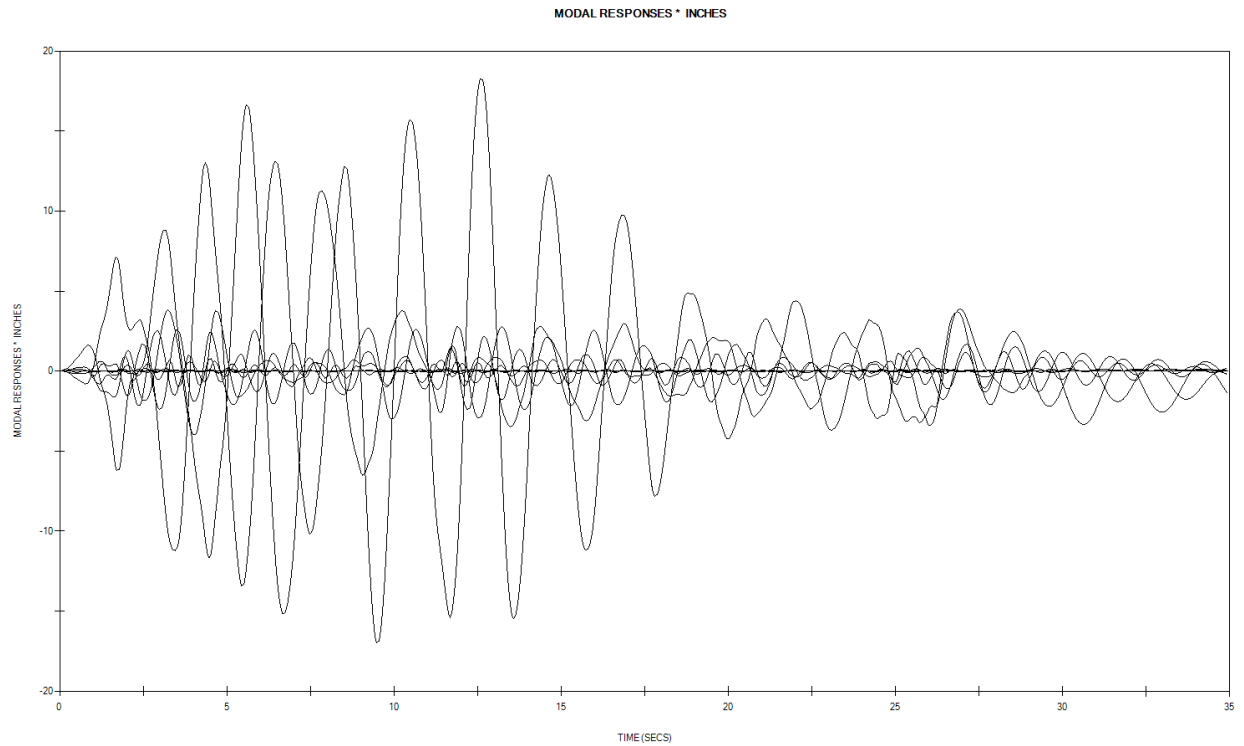
	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	ELCN	3A						
2	0.000000E+00-	.551180E+000.	118110E+000.	944880E+00				
3	0.200000E-01-	.425196E+010.	748030E+00-	.905510E+01				
4	0.400000E-01-	.397637E+010.	267716E+01-	.108267E+02				
5	0.600000E-01-	.346456E+010.	114173E+01-	.156299E+02				
6	0.800000E-01-	.374015E+010.	114173E+01-	.153543E+02				
7	0.100000E+00-	.472440E+010.	212598E+01-	.236220E+01				
8	0.120000E+00-	.559054E+010.	326771E+010.	.161023E+02				
9	0.140000E+00-	.503936E+010.	208661E+010.	.822833E+01				
10	0.160000E+00-	.433070E+010.	551180E+00-	.268897E+02				
11	0.180000E+00-	.334645E+010.	208661E+01-	.254724E+02				
12	0.200000E+00-	.334645E+010.	527558E+010.	.358267E+01				

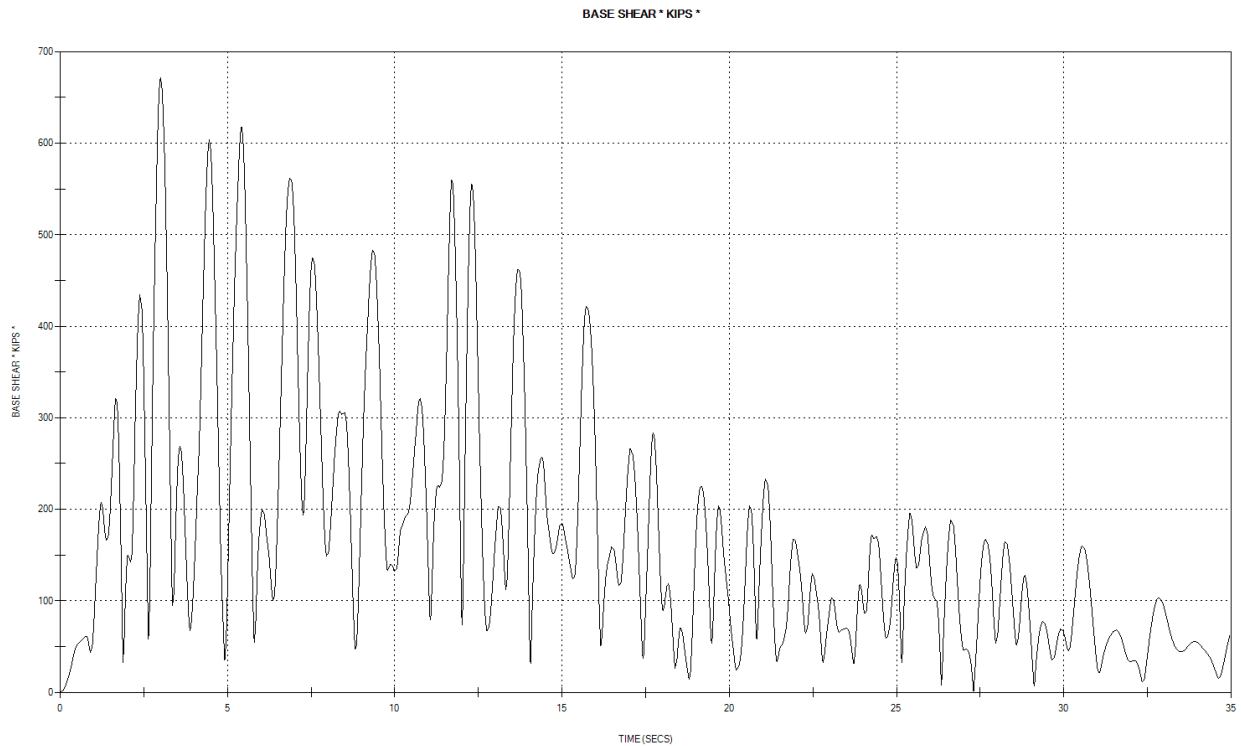
The following is a detailed description of the Time History Earthquake input file, the results of the analysis ensue:

Line 1. The name of the time history 'ELCN' is entered in columns 1-4. Three time history input functions are specified with '3' in columns 5-8, and acceleration is specified as the dependent variable with 'A' in column 9.

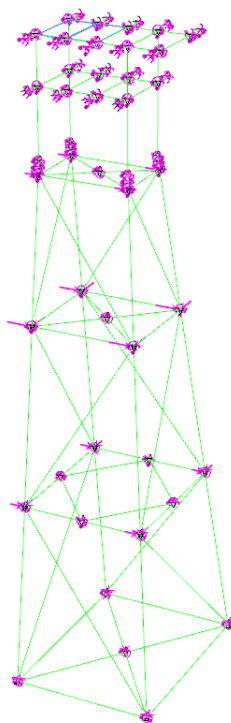
Line 2. The time increment is entered in columns 1-12, the acceleration functions in the X, Y, and Z direction is entered in columns 13-24, 25-36, and 37-48 respectively.

The plots created by the analysis followed by a portion of the output containing the responses are on the following pages.





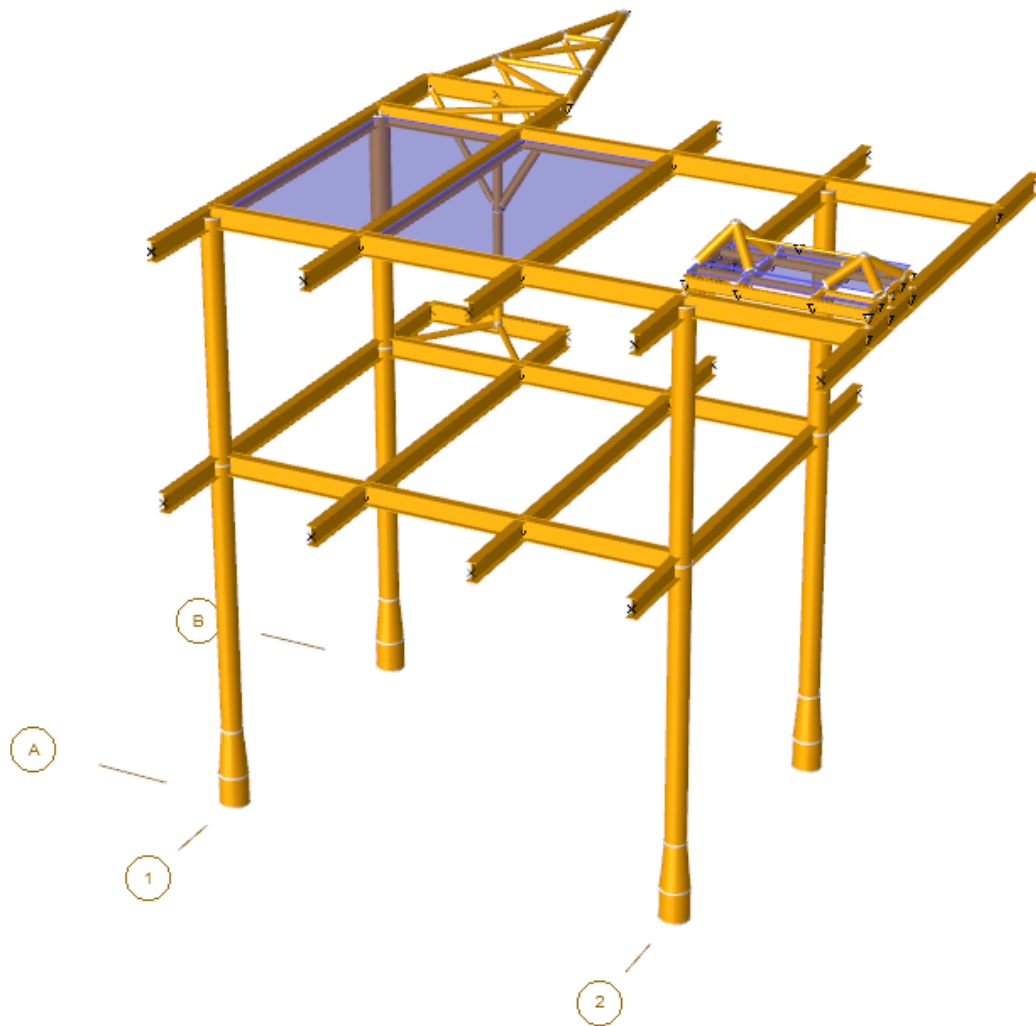
The result of the response analysis is a model file (dyroci) with equivalent static load conditions for all 1750 time points (35 seconds/0.02 seconds). The following figure depicts the equivalent static loading for load condition 149 (2.98s) which represents the maximum base shear load condition.



4.3 ENGINE/COMPRESSOR VIBRATION

Sample Problem 3 is an engine vibration problem where the motion of a deck structure due to reciprocating machinery is analyzed.

The structure in figure below contains the deck structure in conjunction with a skid containing representation of the engine and compressor.



For this analysis, the Dynpac program was used to extract 80 modes. The Dynamic Response program was used to predict the response of the structure due to the machinery. Running speeds of 600 rpm and 1200 rpm were analyzed. Gas torque and unbalanced forces were input with 2% structural damping considered for modes through the top running speed and 5% damping for higher frequency modes.

The following is the Dynamic Response input file used in conjunction with the generalized mass file and solution file from Dynpac.

```

1      1      2      3      4      5      6      7      8
1234567890123456789012345678901234567890123456789012345678901234567890
1      DROPT ENGV 80EC 1
2      * DAMPING OF 2.0% FOR MODES THROUGH THE TOP RUNNING SPEED
3      SDAMP      2.0 2.0 2.0 2.0 2.0 2.0 2.0 2.0 2.0 2.0
4      SDAMP      2.0 2.0 2.0 2.0 2.0 2.0 2.0 2.0 2.0 2.0
5      SDAMP      2.0 2.0 2.0 2.0 2.0 2.0 2.0 2.0 2.0 2.0
6      SDAMP      2.0 2.0 2.0 2.0 2.0 5.0 5.0 5.0 5.0 5.0
7      * DAMPING OF 5.0% FOR MODES HIGHER THAN TOP RUNNING SPEED
8      SDAMP      5.0 5.0 5.0 5.0 5.0 5.0 5.0 5.0 5.0 5.0
9      SDAMP      5.0 5.0 5.0 5.0 5.0 5.0 5.0 5.0 5.0 5.0
10     SDAMP      5.0 5.0 5.0 5.0 5.0 5.0 5.0 5.0 5.0 5.0
11     SDAMP      5.0 5.0 5.0 5.0 5.0 5.0 5.0 5.0 5.0 5.0
12     LOAD
13     * THE EXPECTED RUNNING SPEED IS BETWEEN 600 RPM AND 1200 RPM.
14     ENGVIB480.0 1440.0 MOD 602.0 20 10DL PT PT
15     PLTOPT SNMLDL GR
16     JNTPLT 803 836 845 846 847 848 1000 3000 835 836 807
17     * RUNNING SPEED = 600 RPM
18     * 100% GAS TORQUE LOAD (LOAD CASE WITH 597 PSI SUCTION / 1043 PSI DISCHARGE)
19     * 100% COMPRESSOR MECHANICAL UNBALANCED LOAD - NO ENGINE UNBALANCE GIVEN.
20     * ALL UNBALANCES ARE MULTIPLIED BT 1000, THEREFORE RESPONSES ARE IN "MILS"
21     * RESPONSES ARE CALCULATED WITH FOLLOWING ADJUSTMENTS:
22     * 1) MECHANICAL UNBALANCE FORCES AND MOMENTS ARE HELD CONSTANT FROM 480 TO 600
23     * RPM AND FROM 1200 RPM TO 1440 RPM.
24     * 2) GAS TORQUES ARE GIVEN FOR ONE SPEED BUT WILL BE ASSUMED CONSTANT
25     * THROUGHOUT THE SPEED RANGE.
26     *
27     * LOAD GROUP#1 - COMPRESSOR "A" GAS TORQUE CURVE (MX)
28     *
29     RSPEED 600.
30     UNBAL 1000TIM 0.010 0.0 LN 0 1 5
31     LDFACT 0.0000-323.78-756.32-1327.9-2277.6-3448.4-4491.4-6573.3-8446.4-10145.
32     LDFACT-11215.-12287.-13271.-14261.-15237.-16214.-17220.-18287.-17655.-16316.
33     LDFACT-14745.-12988.-11104.-9157.9-7216.7-5349.5-3621.6-2091.1-805.68 200.56
34     LDFACT 909.30 1319.4 1447.0 1325.4 1002.2 537.58 0.00 -587.25-1382.5-2522.9
35     LDFACT-4037.0-5866.2-7883.4-9899.9-10432.-10744.-10575.-10878.-10816.-10756.
36     LDFACT-10770.-10926.-10049.-8386.8-6761.4-5221.5-3806.1-2543.8-1452.5-540.94
37     LDFACT 190.80 749.43 1147.0 1399.4 1524.3 1540.7 1467.0 1320.4 1118.0 872.27
38     LDFACT 597.66 303.47
39     *
40     * LOAD GROUP#1 - OPPOSITE GAS TORQUE APPLIED AT THE ENGINE
41     *
42     UNBAL 3000TIM -0.010 0.0 LN 0 1 5
43     LDFACT 0.0000-323.78-756.32-1327.9-2277.6-3448.4-4491.4-6573.3-8446.4-10145.
44     LDFACT-11215.-12287.-13271.-14261.-15237.-16214.-17220.-18287.-17655.-16316.
45     LDFACT-14745.-12988.-11104.-9157.9-7216.7-5349.5-3621.6-2091.1-805.68 200.56
46     LDFACT 909.30 1319.4 1447.0 1325.4 1002.2 537.58 0.00 -587.25-1382.5-2522.9
47     LDFACT-4037.0-5866.2-7883.4-9899.9-10432.-10744.-10575.-10878.-10816.-10756.
48     LDFACT-10770.-10926.-10049.-8386.8-6761.4-5221.5-3806.1-2543.8-1452.5-540.94
49     LDFACT 190.80 749.43 1147.0 1399.4 1524.3 1540.7 1467.0 1320.4 1118.0 872.27
50     LDFACT 597.66 303.47
51     *
52     * LOAD GROUP#2 - COMPRESSOR "A" MECHANICAL UNBALANCED
53     * (MY - PRIMARY VERTICAL COUPLE)
54     *
55     UNBAL 1000 4.44 LN 0 2
56     *
57     * LOAD GROUP#3 - COMPRESSOR "A" MECHANICAL UNBALANCED
58     * (MZ - PRIMARY HORIZONTAL COUPLE)
59     *
60     UNBAL 1000 24.6 LN 0 3
61     *
62     * LOAD GROUP#4 - COMPRESSOR "A" MECHANICAL UNBALANCED
63     * (MZ - SECONDARY HORIZONTAL COUPLE)
64     *

```

```

65 UNBAL 1000 0.0 LN 0 4
66 UNBAL 1000 3.6 LN 1 4
67 * RUNNING SPEED = 1200 RPM
68 * 100% GAS TORQUE LOAD (LOAD CASE WITH 597 PSI SUCTION / 1043 PSI DISCHARGE)
69 * 100% COMPRESSOR MECHANICAL UNBALANCED LOAD - NO ENGINE UNBALANCE GIVEN.
70 * ALL UNBALANCES ARE MULTIPLIED BT 1000, THEREFORE RESPONSES ARE IN "MILS"
71 * RESPONSES ARE CALCULATED WITH FOLLOWING ADJUSTMENTS:
72 * 1) MECHANICAL UNBALANCE FORCES AND MOMENTS ARE HELD CONSTANT FROM 480 TO 600
73 * RPM AND FROM 1200 RPM TO 1440 RPM.
74 * 2) GAS TORQUES ARE GIVEN FOR ONE SPEED BUT WILL BE ASSUMED CONSTANT
75 * THROUGHOUT THE SPEED RANGE.
76 *
77 * LOAD GROUP#1 - COMPRESSOR "A" GAS TORQUE UNBALANCED (MX)
78 *
79 RSPEED 1200.
80 UNBAL 1000TIM 0.010 0.0 LN 0 1 5
81 LDFACT 0.0000-323.78-756.32-1327.9-2277.6-3448.4-4491.4-6573.3-8446.4-10145.
82 LDFACT-11215.-12287.-13271.-14261.-15237.-16214.-17220.-18287.-17655.-16316.
83 LDFACT-14745.-12988.-11104.-9157.9-7216.7-5349.5-3621.6-2091.1-805.68 200.56
84 LDFACT 909.30 1319.4 1447.0 1325.4 1002.2 537.58 0.00 -587.25-1382.5-2522.9
85 LDFACT-4037.0-5866.2-7883.4-9899.9-10432.-10744.-10575.-10878.-10816.-10756.
86 LDFACT-10770.-10926.-10049.-8386.8-6761.4-5221.5-3806.1-2543.8-1452.5-540.94
87 LDFACT 190.80 749.43 1147.0 1399.4 1524.3 1540.7 1467.0 1320.4 1118.0 872.27
88 LDFACT 597.66 303.47
89 *
90 * LOAD GROUP#1 - OPPOSITE GAS TORQUE APPLIED AT ENGINE
91 *
92 UNBAL 3000TIM -0.010 0.0 LN 0 1 5
93 LDFACT 0.0000-323.78-756.32-1327.9-2277.6-3448.4-4491.4-6573.3-8446.4-10145.
94 LDFACT-11215.-12287.-13271.-14261.-15237.-16214.-17220.-18287.-17655.-16316.
95 LDFACT-14745.-12988.-11104.-9157.9-7216.7-5349.5-3621.6-2091.1-805.68 200.56
96 LDFACT 909.30 1319.4 1447.0 1325.4 1002.2 537.58 0.00 -587.25-1382.5-2522.9
97 LDFACT-4037.0-5866.2-7883.4-9899.9-10432.-10744.-10575.-10878.-10816.-10756.
98 LDFACT-10770.-10926.-10049.-8386.8-6761.4-5221.5-3806.1-2543.8-1452.5-540.94
99 LDFACT 190.80 749.43 1147.0 1399.4 1524.3 1540.7 1467.0 1320.4 1118.0 872.27
100 LDFACT 597.66 303.47
101 *
102 * LOAD GROUP#2 - COMPRESSOR "A" MECHANICAL UNBALANCED
103 * (MY - PRIMARY VERTICAL COUPLE)
104 *
105 UNBAL 1000 17.76 LN 0 2
106 *
107 * LOAD GROUP#3 - COMPRESSOR "A" MECHANICAL UNBALANCED
108 * (MZ - PRIMARY HORIZONTAL COUPLE)
109 *
110 UNBAL 1000 98.4 LN 0 3
111 *
112 * LOAD GROUP#4 - COMPRESSOR "A" MECHANICAL UNBALANCED
113 * (MZ - SECONDARY HORIZONTAL COUPLE)
114 *
115 UNBAL 1000 0.0 LN 0 4
116 UNBAL 1000 14.4 LN 1 4
117 END
118

```

The following is a detailed description of the Dynamic Response input file, the results of the analysis ensue:

Line 1. The DROPT line specifies analysis options, namely:

- a. A engine vibration analysis is to be performed ('ENGV' in columns 7-10).
- b. 80 modes are to be used as specified in columns 13-14.
- c. The input file is to be echoed as specified by 'EC' in columns 15-16.

Lines 3-6. The first four SDAMP lines specify that the structural damping is 2.0 percent for modes 1 through 34.

Lines 6-11. The next five SDAMP lines specify that the structural damping is 5.0 percent for modes 35 through 80.

Line 12. The LOAD line specifies that loading data is to follow.

Line 14. The ENGVIB line defines the vibration analysis parameters as follows:

- a. The analysis will be performed for a lowest running speed of 480 rpm and a highest speed of 1440 rpm as defined in columns 7-11 and 14-19, respectively.
- b. Running speeds corresponding to each modal frequency will be analyzed ('MOD' in columns 21-23).
- c. 60 speed increments will be used as designated in columns 27-28.
- d. The nonlinear interpolation power is 2.0
- e. A minimum of 20 points are to be determined in one cycle of the fundamental frequency while a 10 points are to be calculated in the highest harmonic.
- f. A normalized force summary along with a joint exceedance print are requested by 'PT' in columns 53-54 and 57-58, respectively.

Line 15. The SNAME, D-Line and Military allowable curves are to be plotted as indicated on the PLTOPT line.

Line 16. The joints to be plotted are input on the JNTPLT line.

Line 29. The first input set of input data corresponds to a running speed of 600 rpm as designated by '600.' in columns 12-15 on the RSPEED line.

Line 30. The first unbalanced force is a gas torque curve about the global X axis applied at the compressor joint 1000 as defined by the UNBAL line and the subsequent LDFACT lines.

- a. A time history curve to be is to be applied at joint 1000 as specified by '1000' in columns 8-11 and 'TIM' in columns 12-14.
- b. The moment factor about the X axis is 0.01. To obtain the moment at any point, this value will be factored by the values defined using subsequent LDFACT lines.
- c. The harmonic number for this force is 0 and 5 harmonics are to be used from the Fourier series.
- d. This unbalanced load is assigned to load group 1 as specified in column 71.

Lines 31-38. The LDFACT lines designate load factors for each point on the curve. Seventy-two points representing every 5 degrees are used to define the curve.

Line 42. An equal but opposite gas torque is applied at the engine joint 3000 as defined by the UNBAL line and the subsequent LDFACT lines. Notice that this unbalanced load is applied to load group 1.

Lines 43-50. The LDFACT lines designate load factors for each point on the curve for joint 3000.

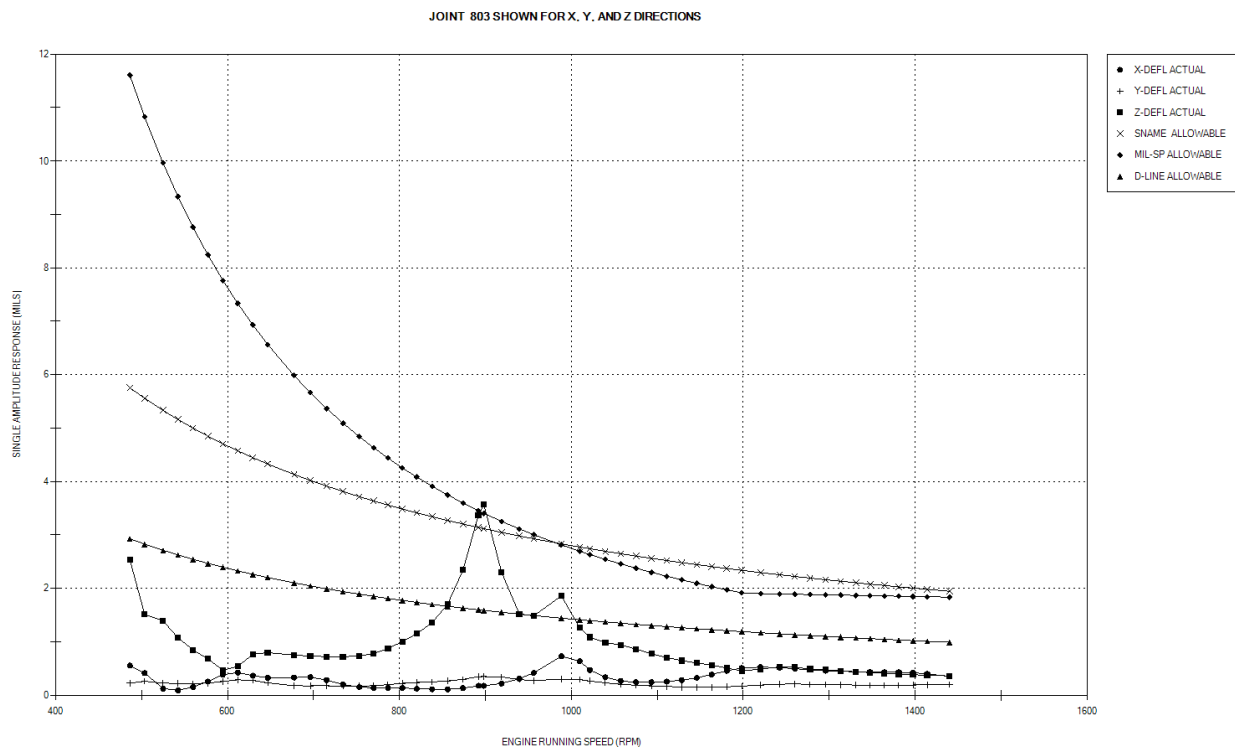
Line 55. The primary unbalanced force is defined as a moment about the Y axis as follows:

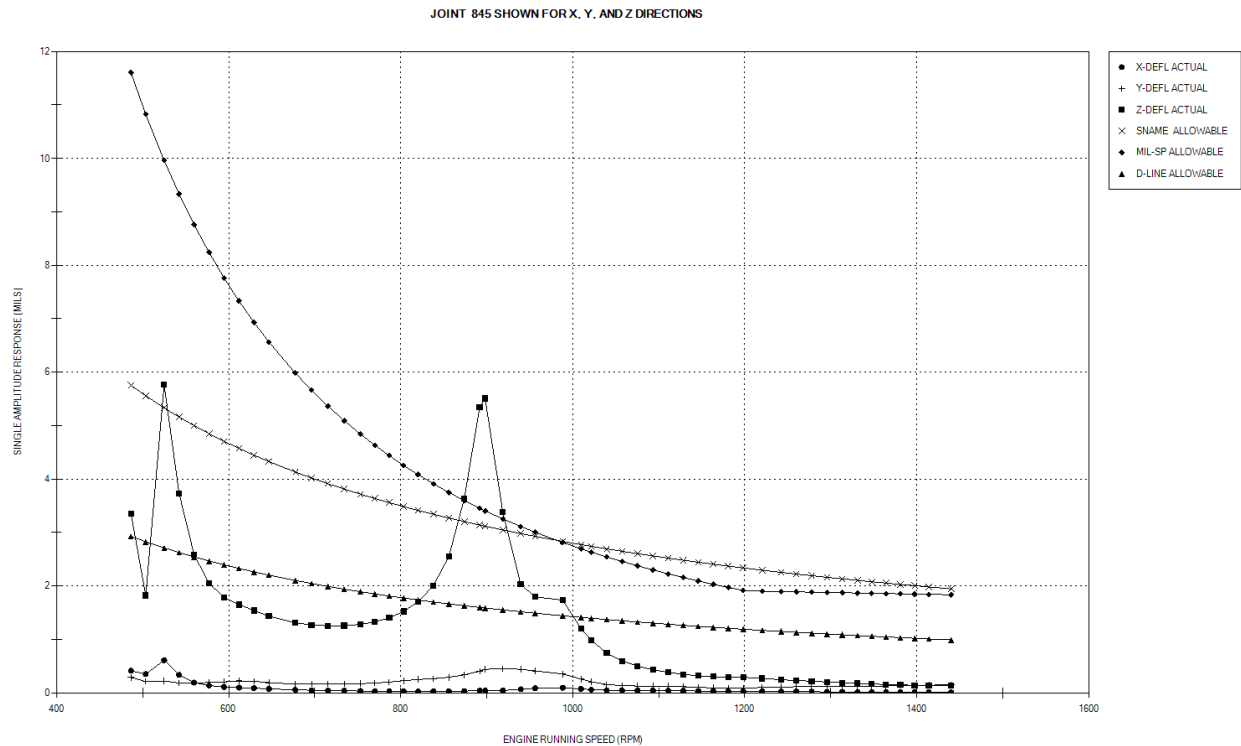
- d. A moment with magnitude 4.44 is applied about the Y axis as designated in columns 41-44.
- e. Because the phase relationship between this force and the gas torque is unknown, the force is assigned to load group 2.
- f. The harmonic number for this force is 0.

Line 60. The secondary unbalanced forces are defined as a moment about the Z axis.

Line 79. The second running for which data is to be input is defined as 1200 rpm on the RSPEED line.

Some plots created by the analysis followed by a portion of the output are on the following pages.

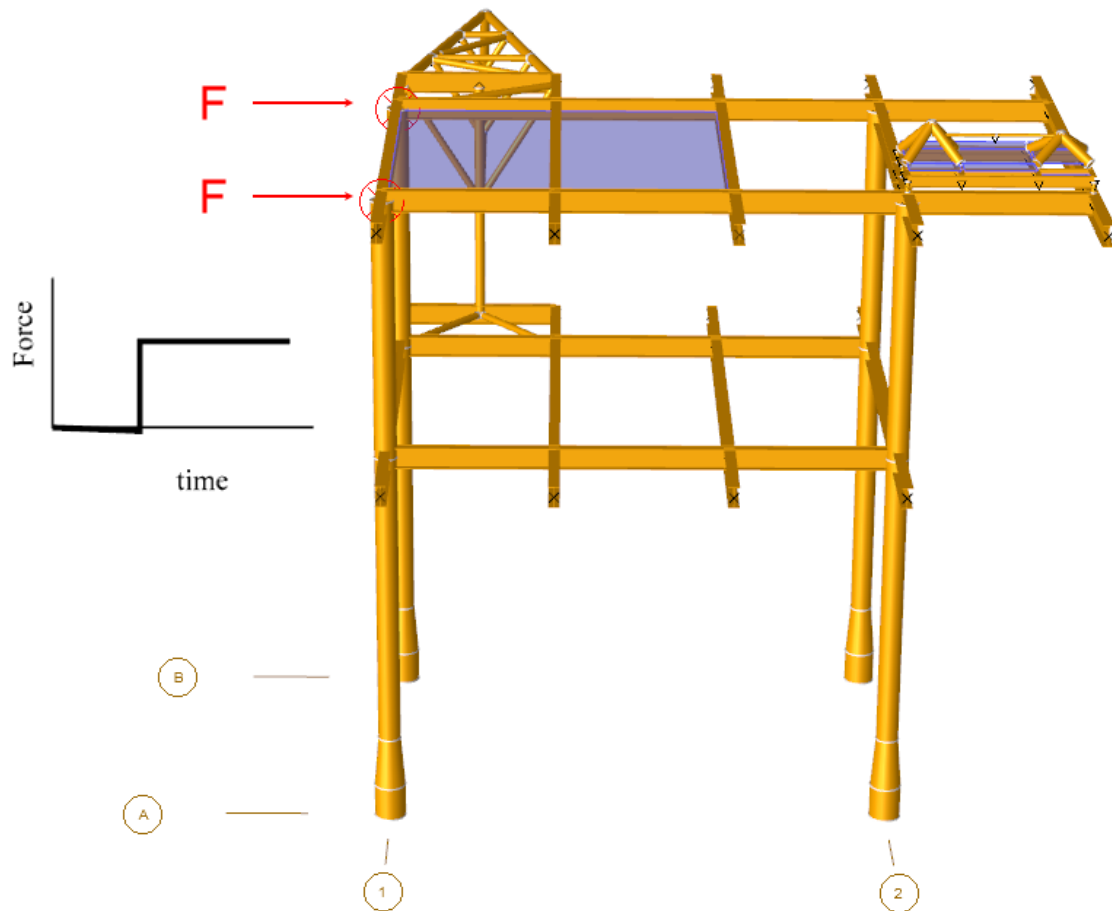




4.4 FORCE TIME HISTORY ANALYSIS

Sample Problem 4 is a force time history analysis where a step function is applied to the structure.

The step function F was applied at joints 803 and 805 as shown in the figure below:



The Dynamic Response program was used to predict the response of the structure. The following is the Dynamic Response input file used in conjunction with the generalized mass file and solution file from Dynpac.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	DROPT VIBR	20EC+Z						
2	* 2 PERCENT STRUCTURAL DAMPING							
3	SDAMP	2.						
4	LOAD							
5	* STRUCTURAL DAMPING ONLY							
6	* PLOT BASE SHEAR, OTM, ACCELERATION AND DISPLACEMENT							
7	FVIB THIS LINE SDO	LN	PLSPLM	MMS	JPA	JPD		
8	* MONITOR JOINTS 801 AND 805							
9	JTNUM	801 805						
10	* ANALYSIS TIME IS 20 SECONDS							
11	TIME	20.	0.02	1.0E-8	1.0			
12	* APPLY FORCE AT JOINTS 801 AND 805 AS STEP FUNCTION							
13	THFORCE	2NON	ABC					
14	LOAD	801	0.	0.				

15	LOAD	805	0.	0.
16	LOAD	801	.5	0.
17	LOAD	805	.5	0.
18	LOAD	801	.51	15.
19	LOAD	805	.51	15.
20	LOAD	801	20.	15.
21	LOAD	805	20.	15.
22	END			

The following is a detailed description of the Dynamic Response input file, the results of the analysis ensue:

Line 1. The DROPT line specifies analysis options, namely:

- a. A periodic vibration analysis is to be performed ('VIBR' in columns 7-10).
- b. 20 modes are to be used as specified in columns 13-14.
- c. The vertical axis is +Z..

Line 3. The SDAMP line specifies that the overall structural damping is 2.0 percent for all modes.

Line 4. The LOAD line specifies that loading data is to follow.

Line 7. The FVIB line defines the analysis parameters as follows:

- d. The analysis type is a time history analysis as designated by 'THIS' with time history input specified on input lines to follow ('LINE' in columns 12-15).
- e. Only structural damping is to be used ('SDO' in columns 17-19).
- f. Linear interpolation is to be used ('LN' in columns 28-29).
- g. Base shear and overturning moments are to be plotted versus time as indicated by 'PLS' and 'PLM'.
- h. 'MMS' in columns 41-43 specifies that load cases are to be created for the time points corresponding to time of maximum base shear and time of maximum overturning moment.
- i. Time history plots of joint acceleration and displacement are requested ('JPA' in columns 47-49 and 'JPD' in columns 53-55).

Line 9. Joints 801 and 805 are to be plotted as specified on the JNTNUM line.

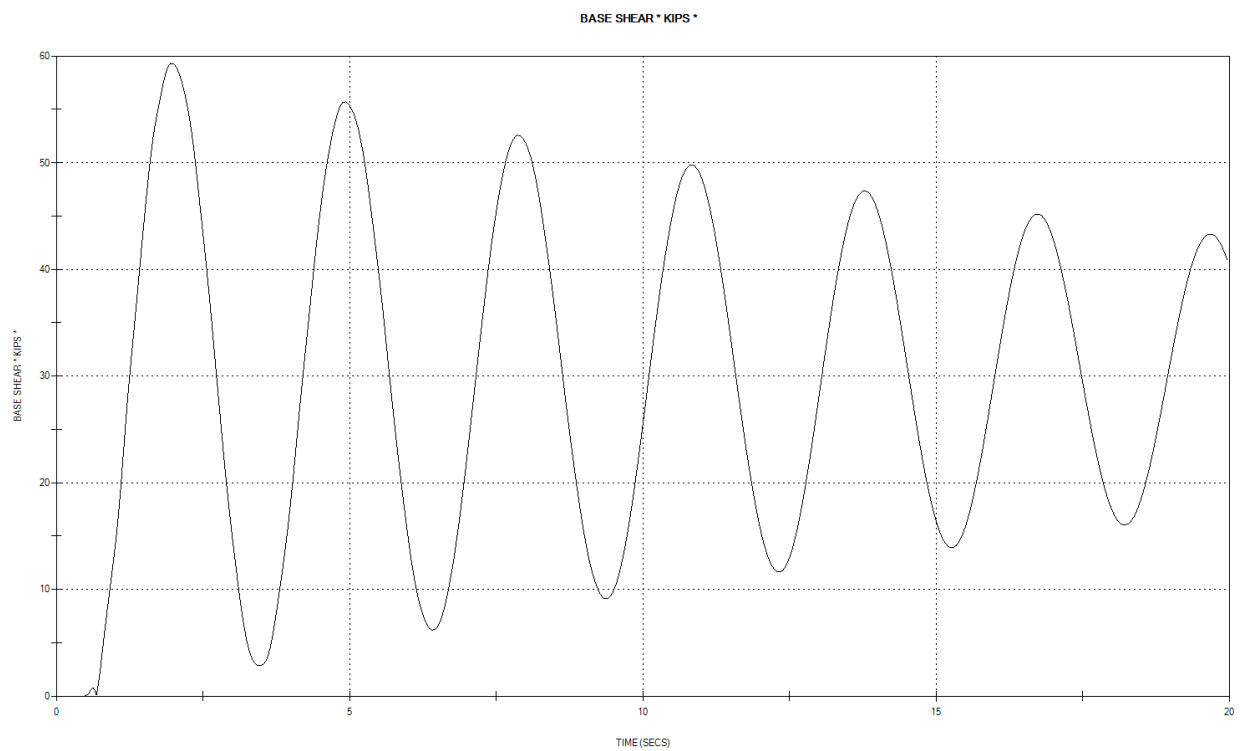
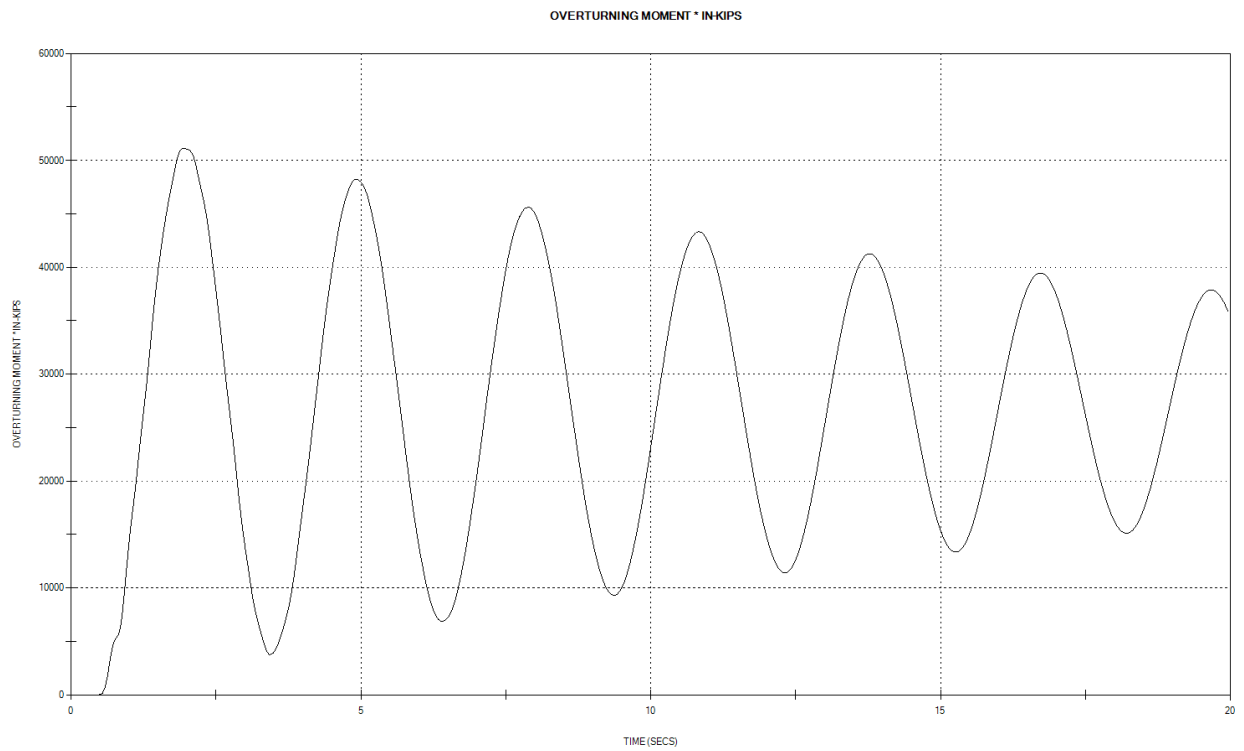
Line 11. The analysis is to run from 0.0 to 20.0 seconds as designated on the TIME line.

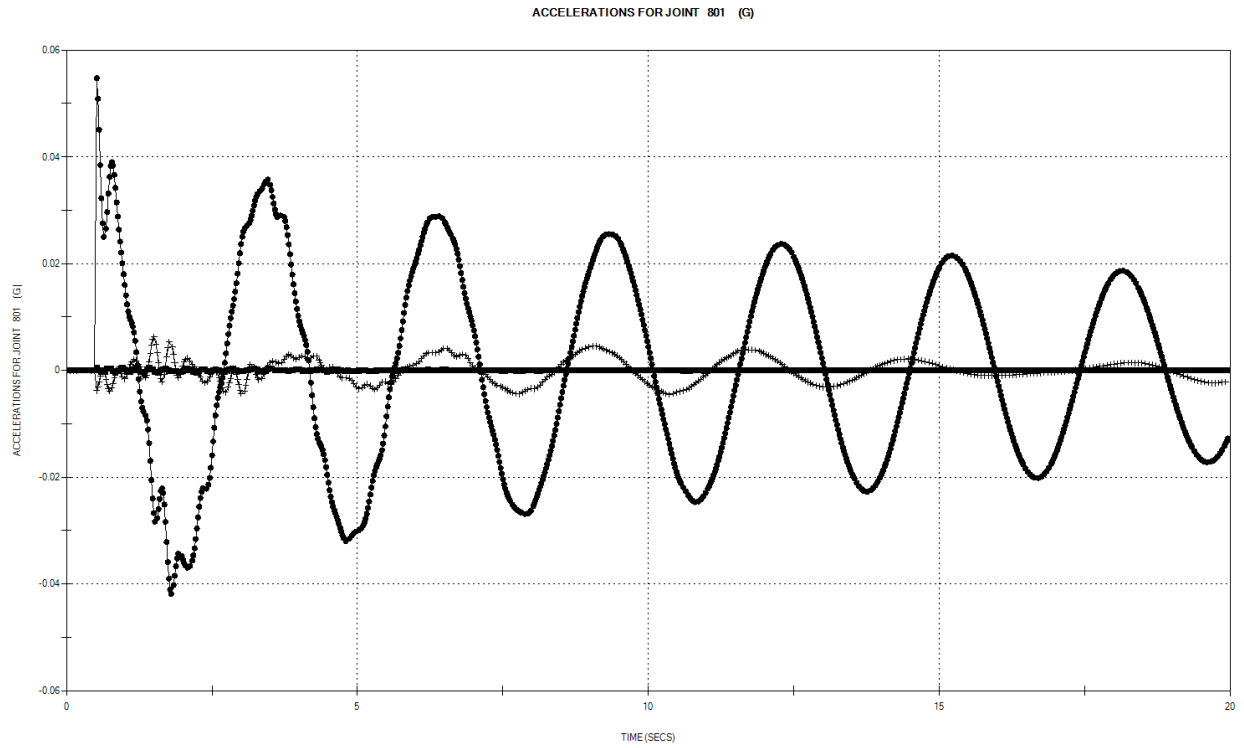
Line 13. The time history force options are specified on the THFORCE line. Force is to be applied to one joint at non-uniform time increments.

Lines 14-21. The step function is defined using the LOAD lines as follows:

- j. From time 0.0 to 0.5 seconds, no force is applied.
- k. At time 0.51 seconds, a 15.0 kip force in the X direction is applied. This force remains constant through 20.0 seconds.

Some plots created by the analysis followed by a portion of the output are on the following pages.





4.5 EXTREME SPECTRAL WIND ANALYSIS

Sample Problem 5 is an extreme spectral wind analysis.

An extreme 150 knot wind was applied to the structure along both the global X and Y directions. The wind loading was defined using the Seastate input file below:

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	LDOPT	+Z	64.20	490.00		GLOBEN WIN	NPNP	K
2	FILE S							
3	LOAD							
4	LOADCNW000							
5	WIND							
6	WIND M	100.00	33.00	0.0	AP08			
7	LOADCNW045							
8	WIND							
9	WIND M	100.00	33.00	45.00	AP08			
10	LOADCNW090							
11	WIND							
12	WIND M	100.00	33.00	90.00	AP08			
13	END							

The Dynamic Response program was used to predict the response of the structure. The following is the Dynamic Response input file used in conjunction with the generalized mass file and solution file from Dynpac.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	DROPT WIND	15	+Z					
2	SDAMP	2.0						
3	STCMB	1.	1.	DEAD1.05	MISC1.05	AREA 0.5	MACH 1.0	
4	LOAD							
5	SPCWIN EX MN PL PL				600.0			
6	END							

The following is a detailed description of the Dynamic Response input file, the results of the analysis ensue:

Line 1. The DROPT line specifies analysis options, namely:

- A spectral wind analysis is to be performed ('WIND' in columns 7-10).
- 15 modes are to be used as specified in columns 13-14.
- The vertical axis is +Z..

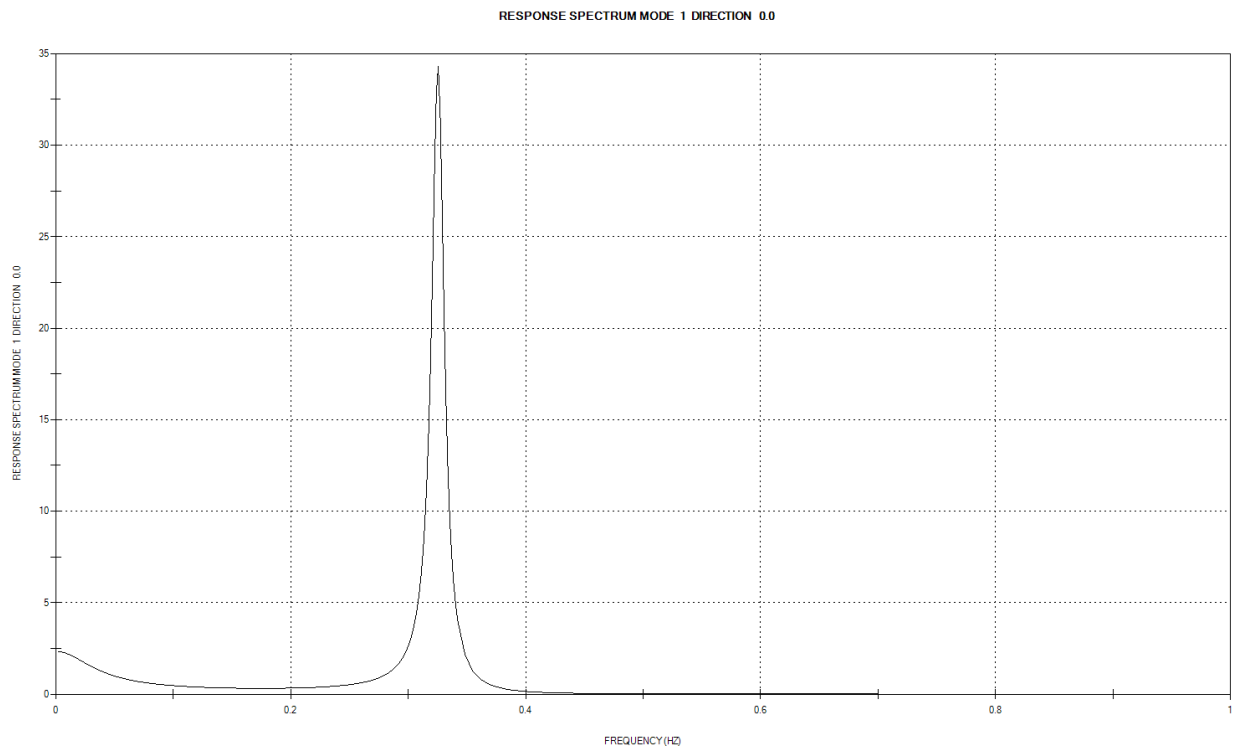
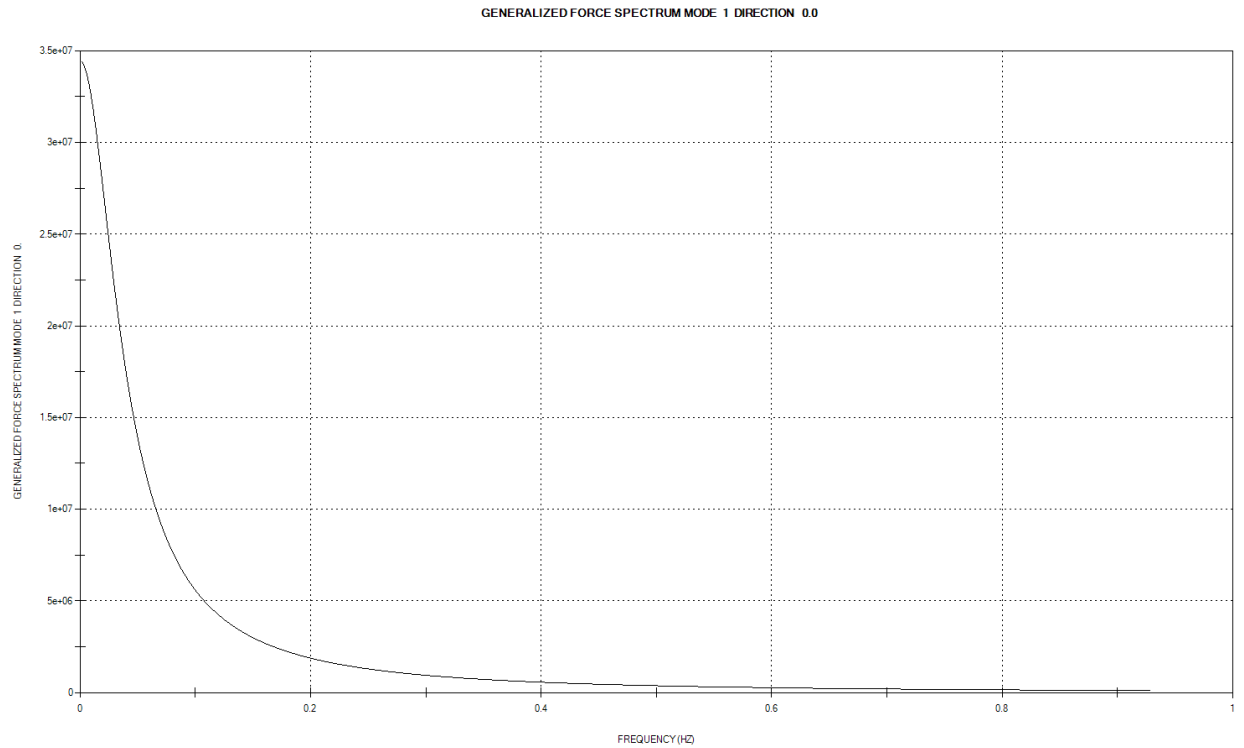
Line 2. The SDAMP line specifies that the overall structural damping is 2.0 percent for all modes.

Line 4. The LOAD line specifies that loading data is to follow.

Line 5. The SPCWIN line defines the analysis parameters as follows:

- The analysis type is an extreme wind analysis as designated by 'EX' in columns 8-9.
- Minimum print is requested along with a generalized force plot and response spectrum plot ('MN', 'PL', and 'PL' in columns 11-12, 14-15, and 17-18, respectively).
- The wind averaging time is 600 seconds.

Some plots created by the analysis followed by a portion of the output are on the following pages.



The Dynamic Response program was used to predict the response of the structure. The following is the Dynamic Response input file used in conjunction with the generalized mass file and modal solution file from Dynpac.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	DROPT	WIND	15	+Z				
2	SDAMP	2.0						
3	LOAD							
4	SPCWIN	FT	MN	PL	PL	CADR	600.0	
5	FTOPT	20.0	2.0	SMWJT	SK		EFT	
6	FTOPT2	PT						
7	JSLC	VB07VB06VB05VB04VB03VB02VB01	849	841840	VB08			
8	WINSPC	.230	WEI	2.1	10.3HR			
9	WINSPC	45.	.350	WEI	2.1	10.3HR		
10	WINSPC	90.	.420	WEI	2.1	10.3HR		
11	END							

The following is a detailed description of the Dynamic Response input file, the results of the analysis ensue:

Line 1. The DROPT line specifies analysis options, namely:

- A spectral wind analysis is to be performed ('WIND' in columns 7-10).
- 15 modes are to be used as specified in columns 13-14.
- The vertical axis is +Z..

Line 2. The SDAMP line specifies that the overall structural damping is 2.0 percent for all modes.

Line 3. The LOAD line specifies that loading data is to follow.

Line 4. The SPCWIN line defines the analysis parameters as follows:

- The analysis type is a fatigue wind analysis as designated by 'FT' in columns 8-9.
- Minimum print is requested along with a generalized force plot and response spectrum plot ('MN', 'PL', and 'PL' in columns 11-12, 14-15, and 17-18, respectively).

Line 5. Fatigue input data is specified in the file including options, joint selection, scf overrides etc.

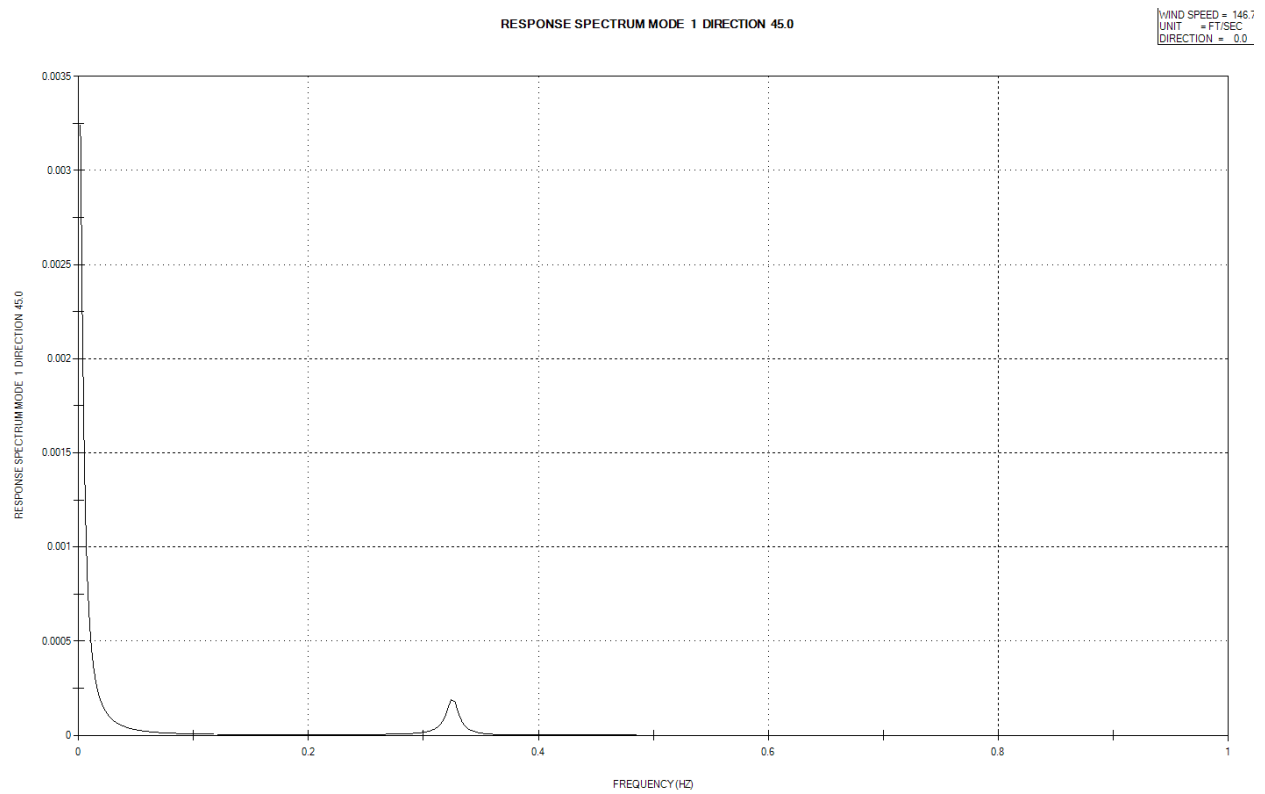
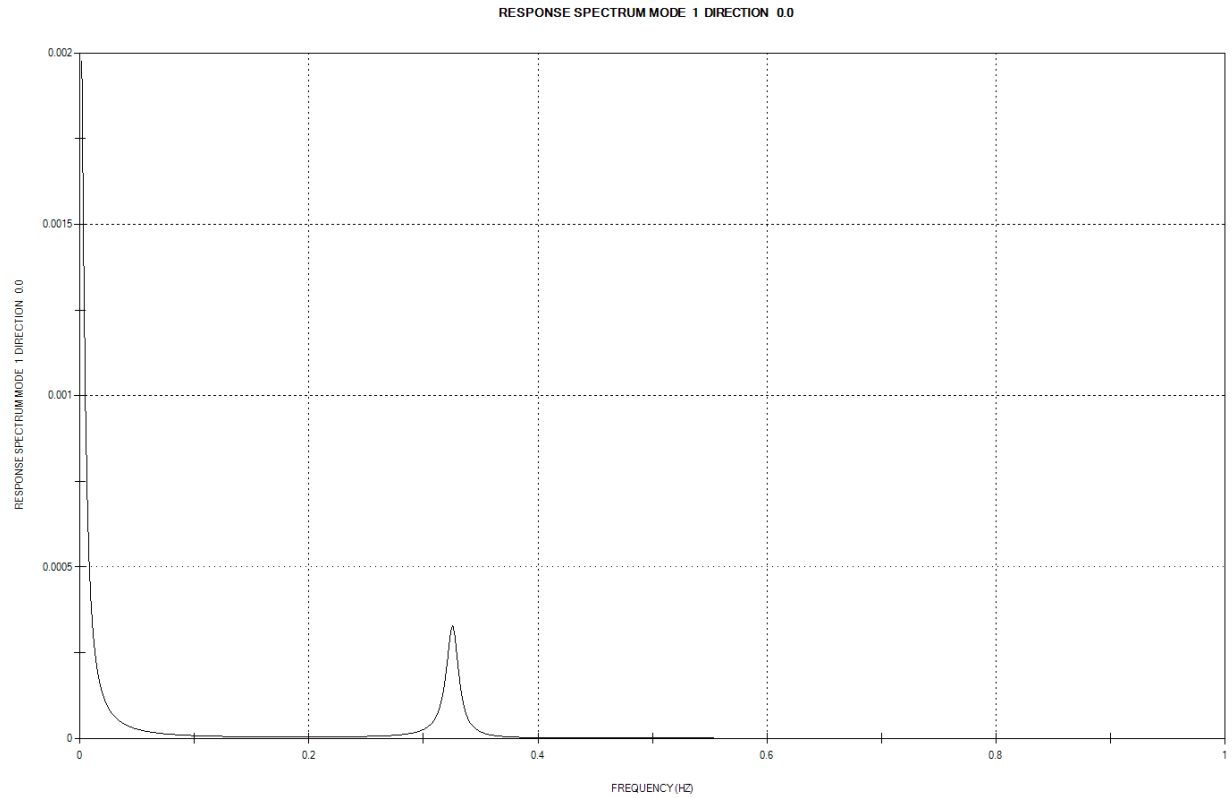
Line 8. Wind Harris spectrum data is specified for the 0 degree direction on the second WINSPC line as follows:

- The approach direction is designated as 0 degrees in columns 9-12.
- The fraction of occurrence for winds of this direction is 23% as indicated by 0.23 in columns 15-20.
- The mean wind distribution type is designated as a Weibull by 'WEI' in columns 22-24.
- The Weibull K and A parameters are specified in columns 26-32 and 33-39, respectively.

Line 9. Wind Harris spectrum data is specified for the 45 degree direction on the second WINSPC line as follows:

- e. The approach direction is designated as 45 degrees in columns 9-12.
 - f. The fraction of occurrence for winds of this direction is 35% as indicated by 0.35 in columns 15-20.
 - g. The mean wind distribution type is designated as a Weibull by 'WEI' in columns 22-24.
 - h. The Weibull K and A parameters are specified in columns 26-32 and 33-39, respectively.
- B. Wind Harris spectrum data is specified for the 90 degree direction on the third WINSPC line as follows:
- a. The approach direction is designated as 90 degrees in columns 9-12.
 - b. The fraction of occurrence for winds of this direction is 42% as indicated by 0.42 in columns 15-20.
 - c. The mean wind distribution type is designated as a Weibull by 'WEI' in columns 22-24.
 - d. The Weibull K and A parameters are specified in columns 26-32 and 33-39, respectively.

Some plots created by the analysis followed by a portion of the output are on the following pages.



4.7 ICE VIBRATION FATIGUE ANALYSIS

Sample Problem 7 calculates the fatigue damage due to ice vibration.

When ice builds up and cracks, the structure tends to vibrate thus causing cyclic stresses. The fatigue damage caused by a ten inch thick ice sheet flowing along the global X axis at the water surface was determined.

The Dynamic Response program was used to predict the response of the structure. The following is the Dynamic Response input file used in conjunction with the generalized mass file and modal solution file from Dynpac.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	DROPT	VIBR	15					+Z-261.0
2	SDAMP		5.					
3	LOAD							
4	FVIB	ICE	LINE	SDO	LN	PLSPLMIFD	PLT	
5	JTNUM		401	403	405	407		
6	TIME				200.	0.025		
7	FTOPT		20.	1.0	2.0	SMAPI	NE	KAW
8	GRPMD		PL10.00001	PL20.00001	PL30.00001	W.B0.00001		
9	ICE		10.	0.5	0.4	5.0	0.02 50.0 0.0315	0.05 240.
10	END							

The following is a detailed description of the Dynamic Response input file, the results of the analysis ensue:

Line 1. The DROPT line specifies analysis options, namely:

- A vibration study is to be performed ('VIBR' in columns 7-10).
- 15 modes are to be used as specified in columns 13-14.
- The mudline elevation is -261.0 ft.

Line 2. The SDAMP line specifies that the overall structural damping is 5.0 percent for all modes.

Line 3. The LOAD line specifies that loading data is to follow.

Line 4. The FVIB line defines the analysis parameters as follows:

- The analysis type is an ice fatigue analysis as designated by 'ICE' in columns 7-9.
- Structural damping only is used.
- Base shear and overturning moment are to be plotted versus time.
- Ice forces and displacements plots are requested as specified by 'IFD' in columns 38-40.

Line 5. Joints 401, 403, 405 and 407 are to be monitored as designated on the JNTNUM line.

Line 6. The analysis start and end time are input as 0.0 and 20.0 on the TIME line.

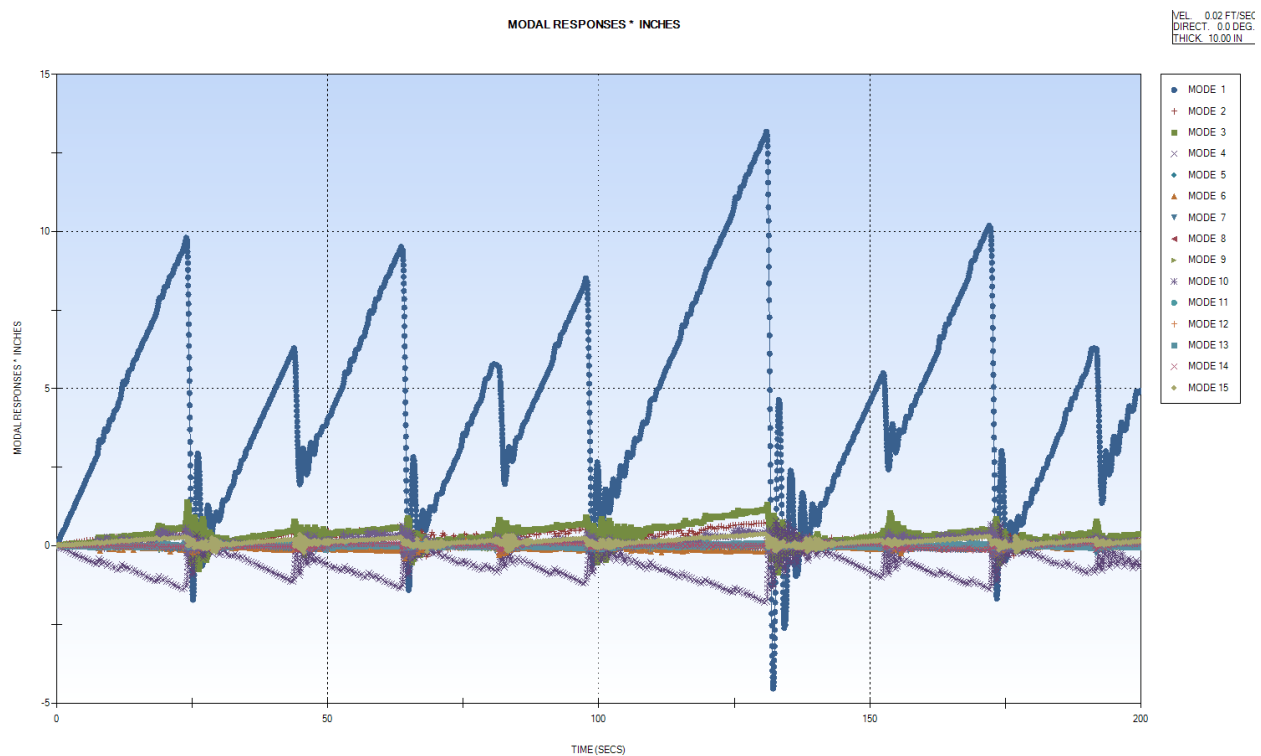
Line 7. Fatigue option data is specified in the file using the FTOPT line.

Line 8. The effective diameter for groups PL1, PL2, PL3 and W.B is set to 0.0 using the GRPMD line.

Line 9. Ice floe data in the X direction is specified using the ICE line as follows:

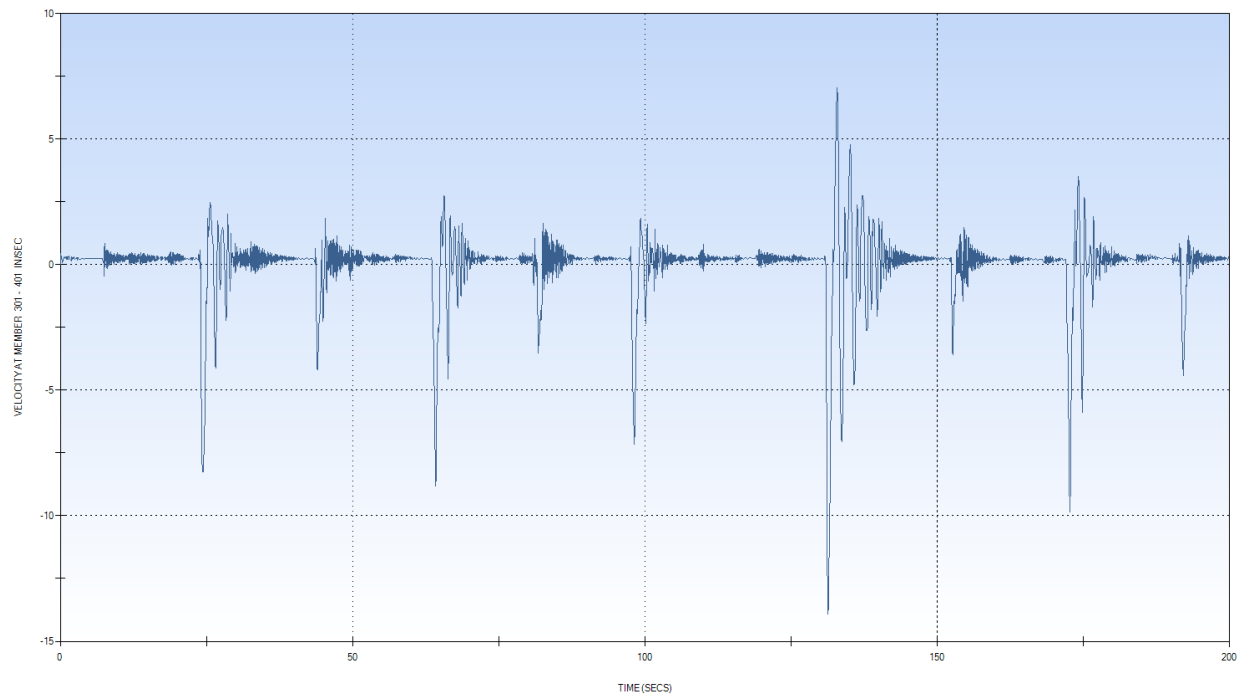
- h. The ice thickness is 10.0 as specified in columns 8-13.
- i. The elastic modulus, static crushing strength and breaking length ratio are specified as 0.50, 0.4 and 5.0, respectively.
- j. The elevation of the ice floe is 0.0 and the flow direction is along the global X axis as indicated by 0.0 in columns 32-37 and 38-43, respectively.
- k. The floe velocity is 0.02 and the analysis velocity step increment is 0.05.
- l. The duration of the ice movement is 240 seconds as input in columns 71-76.

Some plots created by the analysis followed by a portion of the output follow.



VELOCITY AT MEMBER 301 - 401 IN/SEC

VEL 0.02 FT/SEC
DIRECT 0.0 DEG
THICK 10.00 IN



5 INPUT LINES

DYNAMIC RESPONSE OPTIONS LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IN ANY DYNAMIC RESPONSE RUN. IT SPECIFIES OVERALL ANALYSIS PARAMETERS AND TYPE OF ANALYSIS.
(7-10)	ENTER THE TYPE OF ANALYSIS TO BE PERFORMED FROM THE FOLLOWING OPTIONS: 'SPEC' - BASE DRIVEN SPECTRAL ANALYSIS (SPECTRAL EARTHQUAKE) 'TIME' - BASE DRIVEN TIME HISTORY ANALYSIS (TIME HISTORY EARTHQUAKE) 'VIBR' - FORCE DRIVEN ANALYSIS (PERIODIC OR TIME HISTORY) 'WIND' - WIND SPECTRAL ANALYSIS (EXTREME OR FATIGUE ANALYSIS) 'ENGV' - ENGINE VIBRATION ANALYSIS 'TRNF' - REACTION DRIVEN TRANSFER FUNCTION ANALYSIS 'TCLP' - GENERATE INCREMENTAL LOADS FOR COLLAPSE ANALYSIS 'TESL' - GENERATE EQUIVALENT STATIC LOADS FOR LINEAR ANALYSIS 'SHIP' - DYNAMIC SHIP IMPACT ANALYSIS 'DROP' - DYNAMIC DROPPED OBJECT ANALYSIS
(11-14)	ENTER THE NUMBER OF MODES TO BE USED IN THE ANALYSIS. THE DEFAULT IS THE NUMBER OF MODES COMPUTED BY THE PREVIOUS DYNPAC RUN.
(15-16)	ENTER 'EC' IF A LISTING OF THE INPUT DATA IS TO BE INCLUDED IN THE OUTPUT LISTING FILE.
(17-18)	ENTER THE VERTICAL COORDINATE (-X, Y, OR Z, OR +X, Y, OR Z)
(19-24)	ENTER THE MUDLINE ELEVATION IN THE VERTICAL COORDINATE.
(25-27)	ENTER THE SELECTIONS FOR PRINTING THE JOINT ACCELERATIONS, VELOCITIES, AND/OR DISPLACEMENTS. FOR SPECTRAL ANALYSIS ONLY. 'A' - ACCELERATIONS 'V' - VELOCITIES 'D' - DISPLACEMENTS

LINE LABEL	ANALYSIS SELECTION OPTION	NUMBER OF MODES	INPUT ECHO OPTION	VERTICAL COORDINATE	MUDLINE ELEVATION	OUTPUT SELECTION (SPECTRAL ONLY)		
						FIRST SELECTION	SECOND SELECTION	THIRD SELECTION
DROPT								
1-- 5	7--10	11-->14	15--16	17--18	19<--24	25	26	27
DEFAULT				' +Z'	0			
ENGLISH					FT			
METRIC					M			

END LINE

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS REQUIRED AND SIGNALS THE END OF THE RESPONSE CALCULATIONS AND DIRECTS THE PROGRAM TO GENERATE THE OUTPUT FOR THE 'COMBINE' PROGRAM.

LINE LABEL	LEAVE BLANK
END	
1-- 3	4-----80

SIMULATED EARTHQUAKE OUTPUT LOADS DATA

COLUMNS	COMMENTARY
GENERAL	THIS DATA IS USED TO REQUEST EQUIVALENT STATIC LOADS TO BE GENERATED. THIS OPTION IS ESPECIALLY USEFUL TO SIMULATE EARTHQUAKE LOADS FOR SUBSEQUENT COLLAPSE ANALYSES.
(8)	ENTER THE TYPE OF LOAD TO BASE THE EQUIVALENT MODAL FACTORS. ENTER 'M' FOR OVERTURNING MOMENT OR 'S' FOR BASE SHEAR. THE RESULTING EQUIVALENT STATIC LOAD SHOULD REPRESENT THE LOAD SELECTION.
(9)	ENTER 'V' IF THE MAXIMUM VERTICAL LOADING IS TO BE INCLUDED IN THE EQUIVALENT STATIC LOADING.
(10)	SINCE THE DYNAMIC PORTION OF THE EARTHQUAKE ANALYSIS CAN REVERSE SIGN, THIS OPTION CAN BE USED TO SELECT THE OPPOSITE SIGN OR CREATE TWO LOAD CASES FOR BOTH SIGNS. ENTER 'S' FOR THE STANDARD SIGN. 'R' FOR THE REVERSED SIGN. 'B' FOR BOTH. 'A' FOR ALL DIRECTIONS.
(11-13)	THE LOAD CASES GENERATED CAN BE ADDED ON TO AN EXISTING FILE REPLACING ALL THE LOADS ON THAT FILE OR SKIPPING A NUMBER OF LOAD CASES BEFORE ADDING. IF THE DEAD LOAD CASE IS ON THE FILE, THEN IT IS USEFUL TO SKIP OVER THAT LOAD CASE BEFORE ADDING THE EARTHQUAKE LOADS. ENTER THE NUMBER OF LOAD CASES TO BE SKIPPED HERE. IF ALL THE EXISTING LOAD CASES ARE TO BE INCLUDED ON THE OUTPUT FILE, ENTER 'APP' TO APPEND THE GENERATED LOAD CASES TO THE LAST LOAD CASE ON THE EXISTING FILE.
(14-16)	ENTER THE NUMBER OF SHEAR OR MOMENT DIRECTIONS TO BE CHECKED FOR MAXIMUM SHEAR OR OVERTURNING MOMENT. MAXIMUM ALLOWED IS 100.
(17)	ENTER 'M' IF EACH MODE IS OUTPUT AS A SEPARATE LOAD CASE.

LINE LABEL	LOAD TYPE OPTION	INCLUDE MAXIMUM VERTICAL LOADS OPTION	REVERSE OPTION	NUMBER SKIPPED LOAD CASES	NUMBER OF DIRECTIONS	OPTION FOR SEPARATE MODES	LEAVE BLANK
EQKLOD							
1-- 6	8	9	10	11-->13	14-->16	17	18-----80
DEFAULT	'M'		'S'		20		

FLUID MODAL DAMPING

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE FLUID MODAL DAMPING VALUES.
(7- 8)	ENTER 'PC' IF THE PROGRAM IS TO CALCULATE THE FLUID DAMPING. THE DAMPING VALUES WILL BE CALCULATED BASED ON THE NONLINEAR FORCES ON THE STRUCTURE. IN THE CASE OF THE SPECTRAL ANALYSIS, AN EQUIVALENT DAMPING RATIO IS CALCULATED BASED ON A PARTICULAR AMPLITUDE.
(9-10)	ENTER 'NL' IF THE NONLINEAR DAMPING IS TO BE USED FOR THE TIME HISTORY ANALYSIS. OTHERWISE, A LINEAR VALUE WILL BE USED BASED ON A SPECIFIED AMPLITUDE.
(11-15)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL IF ALL MODES ARE TO HAVE THE SAME VALUE. OTHERWISE LEAVE THIS FIELD BLANK. IF A VALUE IS ENTERED HERE, LEAVE THE REMAINING FIELDS ON THIS LINE BLANK.
(16-20)	IF A SPECIFIED AMPLITUDE IS TO BE USED IN THE FLUID DAMPING CALCULATIONS, ENTER THAT VALUE HERE. USED FOR TIME HISTORY ONLY.
(21-25)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL FOR THE FIRST MODE.
(26-70)	ENTER MODAL DAMPING VALUES FOR THE REMAINING MODES. IF THE NUMBER OF MODES IS GREATER THAN TEN, USE ADDITIONAL LINES WITH THE REMAINING DAMPING VALUES ENTERED IN COLUMNS 21-70.

LINE LABEL	FLUID DAMPING SOURCE	NONLINEAR FLUID DAMPING OPTION	OVERALL MODAL DAMPING	FLUID DAMPING AMPLITUDE	*2FLUID MODAL DAMPING VALUES (PERCENT OF CRITICAL)									
					MODE 1	MODE 2	MODE 3	MODE 4	MODE 5	MODE 6	MODE 7	MODE 8	MODE 9	MODE 10
FDAMP														
1-- 5	7-- 8	9--10	11<--15	16<--20	21<--25	26<--30	31<--35	36<--40	41<--45	46<--50	51<--55	56<--60	61<--65	66<--70

LOAD HEADER

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS REQUIRED AND SPECIFIES THAT LOADING DATA (SPECTRAL, TIME HISTORY, OR VIBRATION) FOLLOWS AS DESIGNATED ON THE DROPT LINE.

LINE LABEL	LEAVE BLANK
LOAD	
1-- 4	5-----80

POWER SPECTRAL DENSITY DATA

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED ONLY IF 'LINE' IS SELECTED AS THE SOURCE OF THE SPECTRAL DATA AND 'PSD' IS SELECTED AS THE TYPE. THIS LINE FOLLOWS THE POWER SPECTRAL DENSITY HEADER LINE. REPEAT THIS LINE TO ENTER THE SPECIFIED NUMBER OF TABLE ENTRY POINTS.
(21-30)	ENTER THE FREQUENCY FOR THE FIRST TABLE ENTRY. FREQUENCIES ARE ENTERED IN ASCENDING ORDER.
(31-40)	ENTER THE FIRST PSD VALUE. THE UNITS FOR THIS DATA DEPENDS ON THE TYPE OF POWER SPECTRAL DENSITY BEING ENTERED AS FOLLOWS: PSD TYPE ENGLISH METRIC ACCELERATION G**2/CPS G**2/CPS VELOCITY (IN/SEC) **2/CPS (CM/SEC) **2/CPS DISPLACEMENT IN**2/CPS CM**2/CPS
(41-60)	ENTER THE SECOND FREQUENCY AND PSD FOR THE SECOND ENTRY.

LINE LABEL	FIRST TABLE ENTRY		SECOND TABLE ENTRY		LEAVE BLANK
	FREQUENCY	PSD VALUE	FREQUENCY	PSD VALUE	
PSDSPC					
1-- 6	21<--30	31<--40	41<--50	51<--60	61-----80
DEFAULT					
ENGLISH	HZ	SEE COMMENTS	HZ	SEE COMMENTS	
METRIC	HZ	SEE COMMENTS	HZ	SEE COMMENTS	

POWER SPECTRAL DENSITY HEADER

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED ONLY IF 'LINE' IS SELECTED AS THE SOURCE OF THE SPECTRAL DATA AND 'PSD' IS SELECTED AS THE TYPE.
(7-10)	INPUT THE NUMBER OF FREQUENCY VALUES TO BE USED IN CONSTRUCTING THE TABLE OF POWER SPECTRAL DENSITY VERSUS FREQUENCY. THIS TABLE IS USED WITH LINEAR LOG-LOG INTERPOLATION AND IT IS NECESSARY TO INPUT AS LEAST TWO TABLE ENTRIES.

LINE LABEL	NUMBER OF FREQUENCY VALUES	LEAVE BLANK
PSDSPC		
1-- 6	7-->10	11-----80

RESPONSE FUNCTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS OPTIONAL AND IS USED TO SPECIFY DATA TO GENERATE RESPONSE FUNCTIONS AT SPECIFIED LOCATIONS ON THE STRUCTURE. THIS LINE IS ONLY USED FOR SPECTRAL EARTHQUAKE ANALYSIS AND FOLLOWS THE INPUT SPECTRUM DATA DEFINITION, IF ANY, AFTER THE SPLOAD LINE. THIS LINE CAN BE REPEATED AS MANY TIMES AS DESIRED TO GET AS MANY RESPONSE FUNCTIONS AS NEEDED. THE LAST RSFUNC LINE WILL REMAIN IN EFFECT FOR ALL SUCCEEDING SPLOAD CASES.
(7-10)	ENTER THE JOINT NAME THAT THIS RESPONSE FUNCTION IS TO BE DEFINED.
(11-12)	ENTER THE DEGREE OF FREEDOM FOR THIS RESPONSE FUNCTION. THE CHOICES ARE: 'DX' - X DIRECTION. 'DY' - Y DIRECTION. 'DZ' - Z DIRECTION.
(13-16)	ENTER THE DAMPING DESIRED FOR THE RESPONSE FUNCTION IN PERCENT OF CRITICAL.
(17-66)	ENTER THE DATA FOR UP TO 6 RESPONSE FUNCTIONS. ADDITIONAL RESPONSE FUNCTIONS CAN BE CREATED USING ADDITIONAL RESPONSE FUNCTION LINES.
(67-69)	ENTER THE NUMBER OF POINTS USED IN DETERMINING THE RESPONSE FUNCTION. THE NUMBER OF POINTS NEEDED DEPENDS ON THE PERIOD SPAN OF THE INPUT SPECTRUM AND THE REQUIRED ACCURACY OF THE OUTPUT SPECTRUM. THE POINTS ARE EQUALLY SPACED ON A FREQUENCY LOG BASIS.
(70)	IF THE PLOTS ARE TO BE CREATED VERSUS FREQUENCY, ENTER 'F', IF THE PLOTS ARE VERSUS PERIOD, ENTER 'P'.
(71-72)	IF A FULL GRID IS TO DRAWN INSTEAD OF TIC MARKS ON THE AXES, THEN ENTER 'G1' FOR GRID LINES ON THE MAJOR DIVISIONS AND 'G2' FOR GRID LINES ON ALL DIVISIONS. THESE GRID LINES ARE USEFUL WHEN READING NUMERICAL INFORMATION FROM THE PLOTS.
(73-77)	SELECT THE DESIRED PLOTS FROM THE FOLLOWING CHOICES: 'I' - INPUT POWER SPECTRAL DENSITY. 'O' - OUTPUT POWER SPECTRAL DENSITY. 'T' - TRANSFER FUNCTION. 'R' - RESPONSE FUNCTION. 'W' - RESPONSE FUNCTION BASED ON WHITE NOISE ASSUMPTION.

LINE LABEL	RESPONSE FUNCTION SELECTIONS																		NUMBER POINTS	INDEP. VAR.	GRID OPTION	PLOT OPTIONS					LEAVE BLANK
	1ST FUNCTION			2ND FUNCTION			3RD FUNCTION			4TH FUNCTION			5TH FUNCTION			6TH FUNCTION						O P T 1	O P T 2	O P T 3	O P T 4	O P T 5	
	JOINT NAME	DOF	DAMP. RATIO	JOINT NAME	DOF	DAMP. RATIO	JOINT NAME	DOF	DAMP. RATIO	JOINT NAME	DOF	DAMP. RATIO	JOINT NAME	DOF	DAMP. RATIO	JOINT NAME	DOF	DAMP. RATIO									
RSFUNC																											
1-- 6	7-->10	11--12	13<--16	17-->20	21--22	23<--26	27-->30	31--32	33<--36	37-->40	41--42	43<--46	47-->50	51--52	53<--56	57-->60	61--62	63<--66	67-->69	70	71--72	73	74	75	76	77	78--80
DEFAULT																			100	'F'							

INPUT SPECTRAL RESPONSE DATA

COLUMNS	COMMENTARY												
GENERAL	THIS LINE IS REQUIRED ONLY IF 'LINE' IS SELECTED AS THE SOURCE OF THE SPECTRAL DATA AND 'RSP' IS SELECTED AS THE TYPE. THIS LINE FOLLOWS THE INPUT SPECTRAL HEADER LINE.												
(7-10)	INPUT THE NUMBER OF PERIOD VALUES TO BE USED IN CONSTRUCTING THE TWO-DIMENSIONAL TABLE OF SPECTRAL RESPONSES VERSUS DAMPING AND PERIOD. EACH SET OF RESPONSE VERSUS PERIOD VALUES NEED NOT HAVE THE SAME NUMBER OF ENTRIES FOR EACH DAMPING VALUE. THIS LINE IS REPEATED AS MANY TIMES AS REQUIRED TO INPUT THIS NUMBER OF RESPONSE VERSUS PERIOD TABLE ENTRIES WITH TWO ENTRIES PER LINE. ON THE SUCCEEDING LINES, IT IS NOT NECESSARY TO ENTER THE NUMBER OF PERIOD VALUES OR THE DAMPING RATIO.												
(11-16)	ENTER THE DAMPING RATIO IN PERCENT OF CRITICAL FOR THIS SET OF RESPONSE VERSUS PERIOD VALUES. THIS TWO-DIMENSIONAL TABLE IS READ IN AS FOLLOWS: ALL RESPONSES FOR THE LOWEST DAMPING VALUE IN ASCENDING PERIOD ORDER, FOLLOWED BY (BEGINNING ON A SEPARATE LINE) ALL RESPONSES FOR THE NEXT LOWEST DAMPING VALUE, ETC.												
(21-30)	ENTER THE PERIOD FOR THE FIRST TABLE ENTRY. PERIODS ARE TO BE ENTERED IN ASCENDING ORDER.												
(31-40)	ENTER THE FIRST RESPONSE VALUE. THE UNITS FOR THIS DATA DEPENDS ON THE TYPE OF RESPONSE SPECTRUM BEING ENTERED AS FOLLOWS: <table><tr><td>RESPONSE TYPE</td><td>ENGLISH</td><td>METRIC</td></tr><tr><td>ACCELERATION</td><td>G'S</td><td>G'S</td></tr><tr><td>VELOCITY</td><td>IN/SEC</td><td>CM/SEC</td></tr><tr><td>DISPLACEMENT</td><td>IN</td><td>CM</td></tr></table>	RESPONSE TYPE	ENGLISH	METRIC	ACCELERATION	G'S	G'S	VELOCITY	IN/SEC	CM/SEC	DISPLACEMENT	IN	CM
RESPONSE TYPE	ENGLISH	METRIC											
ACCELERATION	G'S	G'S											
VELOCITY	IN/SEC	CM/SEC											
DISPLACEMENT	IN	CM											
(41-60)	ENTER THE SECOND PERIOD AND RESPONSE FOR THE SECOND ENTRY.												

LINE LABEL	NUMBER OF PERIODS	DAMPING RATIO (PERCENT CRITICAL)	FIRST TABLE ENTRY		SECOND TABLE ENTRY		LEAVE BLANK
			PERIOD	RESPONSE VALUE	PERIOD	RESPONSE VALUE	
RSPSPC							
1-- 6	7-->10	11<--16	21<--30	31<--40	41<--50	51<--60	61--80
DEFAULT							
ENGLISH			SEC	SEE COMMENTS	SEC	SEE COMMENTS	
METRIC			SEC	SEE COMMENTS	SEC	SEE COMMENTS	

INPUT SPECTRAL RESPONSE HEADER

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED ONLY IF 'LINE' IS SELECTED AS THE SOURCE OF THE SPECTRAL DATA AND 'RSP' IS SELECTED AS THE TYPE.
(7-10)	INPUT THE NUMBER OF DAMPING VALUES TO BE USED IN CONSTRUCTING THE TWO-DIMENSIONAL TABLE OF SPECTRAL RESPONSES VERSUS DAMPING AND PERIOD. IF THIS VALUE IS ENTERED AS '1', THE DAMPING VALUE SPECIFIED SHOULD BE THE VALUE SPECIFIED AS THE DAMPING OF THE MODEL BEING ANALYZED.

LINE LABEL	NUMBER OF DAMPING VALUES	LEAVE BLANK
RSPSPC		
1-- 6	7-->10	11-----80

USER-DEFINED RESPONSE SPECTRUM DATA

COLUMNS	COMMENTARY
GENERAL	THIS DATA DEFINES THE RESPONSE SPECTRUM DATA FOR A SPECTRAL EARTHQUAKE ANALYSIS. THIS LINE CAN BE USED TO DEFINE NEW SOIL TYPES FOR API SPECTRAL ANALYSES USING THE SPLAPI LINE.
(8)	ENTER THE IDENTIFIER USED TO REFERENCE THIS SPECTRUM.
(10)	SPECIFY THE NUMBER OF DAMPING VALUES USED TO DEFINE THIS RESPONSE SPECTRUM. MAXIMUM NUMBER OF DAMPING VALUES IS THREE.

LINE LABEL	RESPONSE SPECTRUM ID	NUMBER OF DAMPING VALUES	LEAVE BLANK
RSPU1			
1-- 5	8	10	11-----80

USER-DEFINED RESPONSE SPECTRUM DATA

COLUMNS	COMMENTARY
GENERAL	THIS DATA DEFINES THE NORMALIZED RESPONSE SPECTRUM DATA FOR API SPECTRAL EARTHQUAKE ANALYSIS. THIS LINE IS REPEATED FOR NUMBER OF DAMPING VALUES SPECIFIED ON THE 'RSPU1' INPUT.
(9-14)	ENTER THE DAMPING VALUE FOR THIS RESPONSE SPECTRUM.
(21-26)	ENTER THE PERIOD FOR THE FIRST POINT IN THE RESPONSE SPECTRUM VERSUS PERIOD INPUT. THE POINTS MUST BE ENTERED IN ORDER OF ASCENDING PERIODS.
(27-32)	ENTER THE NORMALIZED RESPONSE SPECTRUM VALUE. THIS VALUE IS THE SPECTRAL ACCELERATION DIVIDED BY GRAVITY.
(33-80)	ENTER THE REMAINING POINTS. THE MAXIMUM NUMBER OF POINTS IS FIFTEEN. TO SPECIFY MORE THAN FIVE POINTS REPEAT THE 'RSPU2' LINE WITH THE SAME DAMPING RATIO.

LINE LABEL	DAMPING RATIO (PERCENT CRITICAL)	FIRST POINT		SECOND POINT		THIRD POINT		FOURTH POINT		FIFTH POINT	
		PERIOD	NORMALIZED RESPONSE SPECTRUM VALUE	PERIOD	NORMALIZED RESPONSE SPECTRUM VALUE	PERIOD	NORMALIZED RESPONSE SPECTRUM VALUE	PERIOD	NORMALIZED RESPONSE SPECTRUM VALUE	PERIOD	NORMALIZED RESPONSE SPECTRUM VALUE
RSPU2											
1-- 5	9<--14	21<--26	27<--32	33<--38	39<--44	45<--50	51<--56	57<--62	63<--68	69<--74	75<--80
DEFAULT											
ENGLISH		SEC		SEC		SEC		SEC		SEC	
METRIC		SEC		SEC		SEC		SEC		SEC	

STRUCTURAL MODAL DAMPING

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE MODAL DAMPING VALUES. THESE VALUES SHOULD INCLUDE ALL SOURCES OF DAMPING OTHER THAN THE FLUID DAMPING IF AN ITERATIVE SOLUTION IS BEING USED. IN THAT CASE THE FLUID DAMPING IS ACCOUNTED FOR BY CALCULATING THE ACTUAL FLUID FORCES DUE TO MOTION OF THE STRUCTURE AND CALCULATING AN EQUIVALENT LINEAR MODAL DAMPING VALUE WHICH IS ADDED TO THESE VALUES TO OBTAIN THE TOTAL DAMPING. IF THE FLUID DAMPING IS NOT BEING CALCULATED BY THE PROGRAM, THEN THE USER SHOULD INCLUDE THE FLUID DAMPING IN THIS DATA OR ENTER THE FLUID DAMPING ON THE FDAMP LINE.
(11-15)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL IF ALL MODES ARE TO HAVE THE SAME VALUE, OTHERWISE LEAVE THIS FIELD BLANK. IF A VALUE IS ENTERED HERE, LEAVE THE REMAINING FIELDS ON THIS LINE BLANK.
(21-25)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL FOR THE FIRST MODE.
(26-70)	ENTER MODAL DAMPING VALUES FOR THE REMAINING MODES. IF THE NUMBER OF MODES IS GREATER THAN TEN, USE ADDITIONAL LINES WITH THE REMAINING DAMPING VALUES ENTERED IN COLUMNS 21-70.

LINE LABEL	MODAL DAMPING VALUES (PERCENT OF CRITICAL)										
	OVERALL MODAL DAMPING	MODE 1	MODE 2	MODE 3	MODE 4	MODE 5	MODE 6	MODE 7	MODE 8	MODE 9	MODE 10
SDAMP											
1-- 5	11<--15	21<--25	26<--30	31<--35	36<--40	41<--45	46<--50	51<--55	56<--60	61<--65	66<--70

API SPECTRAL ANALYSIS LOAD

COLUMNS	COMMENTARY
GENERAL	THIS DATA OR THE SPLOAD DATA IS REQUIRED FOR A SPECTRAL ANALYSIS. IT DEFINES THE RESPONSE FACTORS, SOIL TYPES AND OUTPUT FORM.
(11-15)	ENTER THE RESPONSE FACTOR TO BE APPLIED ON ALL THE DIRECTIONALITY FACTORS. THIS CAN BE USED TO SPECIFY THE "G" LEVEL FOR PARTICULAR EARTHQUAKE ZONES.
(16-21)	SPECIFY THE FACTOR FOR THE X-DIRECTION FOR THE INPUT MOTIONS AT THE BASE OF THE STRUCTURE.
(22)	ENTER THE SOIL TYPE FOR THE X-DIRECTION, EITHER 'A', 'B', OR 'C' FOR API OR THE ONE LETTER IDENTIFIER FROM THE USER-DEFINED RESPONSE SPECTRUM DEFINED ON THE 'RSPU1' LINE.
(23-28)	SPECIFY THE FACTOR FOR THE Y-DIRECTION FOR THE INPUT MOTIONS AT THE BASE OF THE STRUCTURE.
(29)	ENTER THE SOIL TYPE FOR THE Y-DIRECTION, EITHER 'A', 'B', OR 'C' FOR API OR THE ONE LETTER IDENTIFIER FROM THE USER-DEFINED RESPONSE SPECTRUM DEFINED ON THE 'RSPU1' LINE.

COLUMNS	COMMENTARY
(30-35)	SPECIFY THE FACTOR FOR THE Z-DIRECTION FOR THE INPUT MOTIONS AT THE BASE OF THE STRUCTURE.
(36)	ENTER THE SOIL TYPE FOR THE Z-DIRECTION, EITHER 'A', 'B', OR 'C' FOR API OR THE ONE LETTER IDENTIFIER FROM THE USER-DEFINED RESPONSE SPECTRUM DEFINED ON THE 'RSPU1' LINE.
(38-41)	TYPE OF MODAL COMBINATIONS: 'SRSS' - SQUARE ROOT OF SUM OF THE SQUARES. 'PRMS' - PEAK PLUS THE SQUARE ROOT OF THE SUM OF THE SQUARES. 'PEAK' - LINEAR ADDITION OF ABSOLUTE VALUES. 'CQC ' - COMPLETE QUADRATIC COMBINATION.
(42-44)	ENTER 'A', 'V', OR 'D' FOR JOINT ACCELERATIONS, VELOCITIES, OR DISPLACEMENTS TO BE PRINTED.
(45-47)	ENTER 'PRS' TO CREATE SPECTRUM PLOTS
(48)	ENTER 'P' IF THE CQC MATRIX IS TO BE LISTED.
(70)	ENTER 'N' IF THE FACTOR 'D' IS NOT TO BE USED FOR API SPECTRUM ENTER 'D' OR LEAVE BLANK IF TO BE USED

LINE LABEL	OVERALL RESPONSE FACTOR	DIRECTIONALITY FACTORS AND SOIL TYPES						OUTPUT MODAL COMBINATIONS	JOINT DATA PRINT			PLOT SPECTRUM	CQC PRINT OPTION	LEAVE BLANK	FACTOR D	LEAVE BLANK
		X-DIRECTION		Y-DIRECTION		Z-DIRECTION			FIRST SELECTION	SECOND SELECTION	THIRD SELECTION					
		FACTOR	SOIL TYPE	FACTOR	SOIL TYPE	FACTOR	SOIL TYPE									
SPLAPI																
1-- 6	11<--15	16<--21	22	23<--28	29	30<--35	36	38--41	42	43	44	45--47	48	49--69	70	71--80
DEFAULT								'CQC '							'D'	

SPECTRAL ANALYSIS LOAD

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED FOR A SPECTRAL ANALYSIS. IT DEFINES THE INPUT SPECTRUM, DAMPING CONSIDERATIONS, OUTPUT FORM, AND RESPONSE FACTORS.
(9-12)	ENTER THE SPECTRUM SOURCE AS FOLLOWS: 'API ' - AS SPECIFIED IN THE API RP2A 7TH EDITION. 'APIA' - AS SPECIFIED IN THE API RP2A 15TH EDITION FOR SOIL "A". 'APIB' - AS SPECIFIED IN THE API RP2A 15TH EDITION FOR SOIL "B". 'APIC' - AS SPECIFIED IN THE API RP2A 15TH EDITION FOR SOIL "C". 'LINE' - SPECTRUM IS INPUT ON SUBSEQUENT LINES. 'PREV' - USE SPECTRUM FROM PREVIOUS LOAD CASE.
(15-17)	ENTER THE SPECTRUM TYPE AS FOLLOWS: 'RSP' - RESPONSE SPECTRUM. 'PSD' - INPUT POWER SPECTRAL DENSITY. ' ' - LEAVE BLANK FOR API SPECTRUMS.
(18)	ENTER THE SPECTRUM FORM: 'A' - ACCELERATION. 'V' - VELOCITY. 'D' - DISPLACEMENT. ' ' - LEAVE BLANK FOR API SPECTRUMS.
(21-23)	SPECIFY THE DAMPING TYPE: 'SDO' - STRUCTURAL DAMPING ONLY. 'FDS' - EQUIVALENT FLUID DAMPING AT A SPECIFIED AMPLITUDE. 'FDA' - EQUIVALENT FLUID DAMPING AT THE ACTUAL AMPLITUDE.

COLUMNS	COMMENTARY
(25-28)	SPECIFY THE TYPE OF MODAL COMBINATIONS: 'RMS ' - SQUARE ROOT OF SUM OF THE SQUARES. 'PRMS' - PEAK PLUS THE SQUARE ROOT OF THE SUM OF THE SQUARES. 'PEAK' - LINEAR ADDITION OF ABSOLUTE VALUES. 'CQC ' - COMPLETE QUADRATIC COMBINATION. 'CQC2' - CQC WITH TWO INDEPENDENT SPECTRA. REQUIRES TWO SPLOAD SETS. (USED WITH EQUIVALENT STATIC LOAD GENERATION) 'CQC3' - CQC WITH THREE INDEPENDENT SPECTRA. REQUIRES THREE SPLOAD SETS. (USED WITH EQUIVALENT STATIC LOAD GENERATION)
(29-31)	ENTER 'A', 'V', OR 'D' FOR JOINT ACCELERATIONS, VELOCITIES, OR DISPLACEMENTS TO BE PRINTED.
(39-44)	ENTER THE VALUE FOR THE DAMPING AMPLITUDE TO BE USED WITH THE 'FDS' OPTION.
(45-50)	ENTER THE RESPONSE FACTOR TO BE APPLIED ON ALL THE DIRECTIONALITY FACTORS. THIS CAN BE USED TO SPECIFY THE "G" LEVEL FOR PARTICULAR EARTHQUAKE ZONES.
(51-68)	SPECIFY THE FACTORS IN EACH DIRECTION FOR THE INPUT MOTIONS AT THE BASE OF THE STRUCTURE.
(69)	ENTER 'P' IF THE CQC MATRIX IS TO BE LISTED.
(70)	ENTER 'N' IF THE FACTOR 'D' IS NOT TO BE USED FOR API SPECTRUM ENTER 'D' OR LEAVE BLANK IF TO BE USED

LINE LABEL	INPUT SPECTRUM SOURCE	INPUT SPECTRUM TYPE	INPUT SPECTRUM FORM	DAMPING TYPE	OUTPUT MODAL COMBINATIONS	JOINT DATA PRINT			DAMPING AMPLITUDE	RESPONSE FACTOR	DIRECTIONALITY FACTORS			CQC PRINT OPTION	FACTOR D
						FIRST SELECTION	SECOND SELECTION	THIRD SELECTION			X DIRECTION	Y DIRECTION	Z DIRECTION		
SPLOAD															
1-- 6	9--12	15--17	18	21--23	25--28	29	30	31	39<--44	45<--50	51<--56	57<--62	63<--68	69	70
DEFAULT															'D'
ENGLISH									IN						
METRIC									CM						

STATIC + DYNAMIC COMBINATION DATA

COLUMNS	COMMENTARY
GENERAL	THIS DATA IS USED TO SPECIFY HOW THE STATIC LOADS ARE TO BE COMBINED WITH THE DYNAMIC LOADING.
(7)	ENTER 'N' IF THE LOAD COMBINATIONS OF THE STATIC LOADS AND EQUIVALENT STATIC DYNAMIC LOADS ARE NOT TO BE INCLUDED IN THE GENERATED LOAD OUTPUT FILE.
(8-12)	ENTER THE FACTOR TO BE USED ON THE EARTHQUAKE LOADS TO BE COMBINED WITH THE STATIC LOAD CASES FOR BEAM AND PLATE ELEMENT CODE CHECKS.
(13-17)	ENTER THE FACTOR TO BE USED ON THE EARTHQUAKE LOADS TO BE COMBINED WITH THE STATIC LOAD CASES FOR PUNCHING SHEAR CODE CHECKS.
(18-21)	ENTER THE LOAD CASE NAME OF THE FIRST STATIC LOAD CASE TO BE COMBINED WITH THE EARTHQUAKE LOADS. THIS LOAD CASE IS TO BE CREATED IN A SEPARATE STATIC EXECUTION PRIOR TO THE EARTHQUAKE ANALYSIS.
(22-26)	ENTER THE LOAD CASE FACTOR FOR THE FIRST STATIC LOAD CASE.
(27-80)	ENTER THE REMAINING STATIC LOAD CASES AS REQUIRED. UP TO 50 'STCMB' LINES MAY BE USED TO CREATE STATIC COMBINATION CASES.

[illegible]

DYNAMIC RESPONSE OPTIONS LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IN ANY DYNAMIC RESPONSE RUN. IT SPECIFIES OVERALL ANALYSIS PARAMETERS AND TYPE OF ANALYSIS.
(7-10)	ENTER THE TYPE OF ANALYSIS TO BE PERFORMED FROM THE FOLLOWING OPTIONS: 'SPEC' - BASE DRIVEN SPECTRAL ANALYSIS (SPECTRAL EARTHQUAKE) 'TIME' - BASE DRIVEN TIME HISTORY ANALYSIS (TIME HISTORY EARTHQUAKE) 'VIBR' - FORCE DRIVEN ANALYSIS (PERIODIC OR TIME HISTORY) 'WIND' - WIND SPECTRAL ANALYSIS (EXTREME OR FATIGUE ANALYSIS) 'ENGV' - ENGINE VIBRATION ANALYSI 'TRNF' - REACTION DRIVEN TRANSFER FUNCTION ANALYSIS 'TCLP' - GENERATE INCREMENTAL LOADS FOR COLLAPSE ANALYSIS 'TESL' - GENERATE EQUIVALENT STATIC LOADS FOR LINEAR ANALYSIS 'SHIP' - DYNAMIC SHIP IMPACT ANALYSIS 'DROP' - DYNAMIC DROPPED OBJECT ANALYSIS
(11-14)	ENTER THE NUMBER OF MODES TO BE USED IN THE ANALYSIS. THE DEFAULT IS THE NUMBER OF MODES COMPUTED BY THE PREVIOUS DYNPAC RUN.
(15-16)	ENTER 'EC' IF A LISTING OF THE INPUT DATA IS TO BE INCLUDED IN THE OUTPUT LISTING FILE.
(17-18)	ENTER THE VERTICAL COORDINATE (-X, Y, OR Z, OR +X, Y, OR Z)
(19-24)	ENTER THE MUDLINE ELEVATION IN THE VERTICAL COORDINATE.
(25-27)	ENTER THE SELECTIONS FOR PRINTING THE JOINT ACCELERATIONS, VELOCITIES, AND/OR DISPLACEMENTS. FOR SPECTRAL ANALYSIS ONLY. 'A' - ACCELERATIONS 'V' - VELOCITIES 'D' - DISPLACEMENTS

COLUMNS	COMMENTARY
(79-80)	ENTER 'NR' TO EXCLUDE ROTATIONAL DEGREES OF FREEDOM OF MASS MATRIX FROM EQUIVALENT STATIC LOADS OF THE BASE ACCELERATION. DEFAULT: THE ROTATIONAL DOFS ARE INCLUDED.

LINE LABEL	ANALYSIS SELECTION OPTION	NUMBER OF MODES	INPUT ECHO OPTION	VERTICAL COORDINATE	MUDLINE ELEVATION	OUTPUT SELECTION (SPECTRAL ONLY)			BLANK	EXCLUDE ROTATIONAL DOFS FOR ESL?
						FIRST SELECTION	SECOND SELECTION	THIRD SELECTION		
DROPT										
1-- 5	7--10	11-->14	15--16	17--18	19<--24	25	26	27	28--78	79--80
DEFAULT				'+Z'	0					
ENGLISH					FT					
METRIC					M					

END LINE

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS REQUIRED AND SIGNALS THE END OF THE RESPONSE CALCULATIONS AND DIRECTS THE PROGRAM TO GENERATE THE OUTPUT FOR THE 'COMBINE' PROGRAM.

LINE LABEL	LEAVE BLANK
END	
1-- 3	4-----80

FLUID MODAL DAMPING

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE FLUID MODAL DAMPING VALUES.
(7- 8)	ENTER 'PC' IF THE PROGRAM IS TO CALCULATE THE FLUID DAMPING. THE DAMPING VALUES WILL BE CALCULATED BASED ON THE NONLINEAR FORCES ON THE STRUCTURE. IN THE CASE OF THE SPECTRAL ANALYSIS, AN EQUIVALENT DAMPING RATIO IS CALCULATED BASED ON A PARTICULAR AMPLITUDE.
(9-10)	ENTER 'NL' IF THE NONLINEAR DAMPING IS TO BE USED FOR THE TIME HISTORY ANALYSIS. OTHERWISE, A LINEAR VALUE WILL BE USED BASED ON A SPECIFIED AMPLITUDE.
(11-15)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL IF ALL MODES ARE TO HAVE THE SAME VALUE. OTHERWISE LEAVE THIS FIELD BLANK. IF A VALUE IS ENTERED HERE, LEAVE THE REMAINING FIELDS ON THIS LINE BLANK.
(16-20)	IF A SPECIFIED AMPLITUDE IS TO BE USED IN THE FLUID DAMPING CALCULATIONS, ENTER THAT VALUE HERE. USED FOR TIME HISTORY ONLY.
(21-25)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL FOR THE FIRST MODE.
(26-70)	ENTER MODAL DAMPING VALUES FOR THE REMAINING MODES. IF THE NUMBER OF MODES IS GREATER THAN TEN, USE ADDITIONAL LINES WITH THE REMAINING DAMPING VALUES ENTERED IN COLUMNS 21-70.

LINE LABEL	FLUID DAMPING SOURCE	NONLINEAR FLUID DAMPING OPTION	OVERALL MODAL DAMPING	FLUID DAMPING AMPLITUDE	*2FLUID MODAL DAMPING VALUES (PERCENT OF CRITICAL)									
					MODE 1	MODE 2	MODE 3	MODE 4	MODE 5	MODE 6	MODE 7	MODE 8	MODE 9	MODE 10
FDAMP														
1-- 5	7-- 8	9--10	11<--15	16<--20	21<--25	26<--30	31<--35	36<--40	41<--45	46<--50	51<--55	56<--60	61<--65	66<--70

JOINT SELECTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IF ANY OUTPUT SELECTED ON THE THLOAD LINE IS FOR OUTPUT FOR SELECTED JOINTS SUCH AS 'JTA', ETC. THIS LINE FOLLOWS THE THLOAD LINE.
(11-74)	ENTER THE JOINT NAMES FOR JOINTS SELECTED FOR OUTPUT. ANY NUMBER OF JOINTS MAY BE SELECTED AND ENTERED IN ANY ORDER. MULTIPLE 'JTNUM' LINES MAY BE USED FOR INPUT WITHOUT RESTRICTIONS.

LINE LABEL	SELECTED JOINTS FOR OUTPUT															
	JOINT 1	JOINT 2	JOINT 3	JOINT 4	JOINT 5	JOINT 6	JOINT 7	JOINT 8	JOINT 9	JOINT 10	JOINT 11	JOINT 12	JOINT 13	JOINT 14	JOINT 15	JOINT 16
JTNUM																
1-- 5	11-->14	15-->18	19-->22	23-->26	27-->30	31-->34	35-->38	39-->42	43-->46	47-->50	51-->54	55-->58	59-->62	63-->66	67-->70	71-->74

LOAD HEADER

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS REQUIRED AND SPECIFIES THAT LOADING DATA (SPECTRAL, TIME HISTORY, OR VIBRATION) FOLLOWS AS DESIGNATED ON THE DROPT LINE.

LINE LABEL	LEAVE BLANK
LOAD	
1-- 4	5-----80

STRUCTURAL MODAL DAMPING

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE MODAL DAMPING VALUES. THESE VALUES SHOULD INCLUDE ALL SOURCES OF DAMPING OTHER THAN THE FLUID DAMPING IF AN ITERATIVE SOLUTION IS BEING USED. IN THAT CASE THE FLUID DAMPING IS ACCOUNTED FOR BY CALCULATING THE ACTUAL FLUID FORCES DUE TO MOTION OF THE STRUCTURE AND CALCULATING AN EQUIVALENT LINEAR MODAL DAMPING VALUE WHICH IS ADDED TO THESE VALUES TO OBTAIN THE TOTAL DAMPING. IF THE FLUID DAMPING IS NOT BEING CALCULATED BY THE PROGRAM, THEN THE USER SHOULD INCLUDE THE FLUID DAMPING IN THIS DATA OR ENTER THE FLUID DAMPING ON THE FDAMP LINE.
(11-15)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL IF ALL MODES ARE TO HAVE THE SAME VALUE, OTHERWISE LEAVE THIS FIELD BLANK. IF A VALUE IS ENTERED HERE, LEAVE THE REMAINING FIELDS ON THIS LINE BLANK.
(21-25)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL FOR THE FIRST MODE.
(26-70)	ENTER MODAL DAMPING VALUES FOR THE REMAINING MODES. IF THE NUMBER OF MODES IS GREATER THAN TEN, USE ADDITIONAL LINES WITH THE REMAINING DAMPING VALUES ENTERED IN COLUMNS 21-70.

LINE LABEL	MODAL DAMPING VALUES (PERCENT OF CRITICAL)										
	OVERALL MODAL DAMPING	MODE 1	MODE 2	MODE 3	MODE 4	MODE 5	MODE 6	MODE 7	MODE 8	MODE 9	MODE 10
SDAMP											
1-- 5	11<--15	21<--25	26<--30	31<--35	36<--40	41<--45	46<--50	51<--55	56<--60	61<--65	66<--70

TIME HISTORY DATA HEADER

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IF THE TIME HISTORY SOURCE IS SPECIFIED AS 'LINE' ON THE 'THLOAD' LINE. THE OVERALL PARAMETERS OF THE TIME HISTORY INPUT ARE DESCRIBED HERE.
(8-10)	ENTER THE NUMBER OF TIME HISTORY FUNCTIONS TO INPUT. THIS VALUE CAN BE FROM 1 TO 3. TYPICALLY AN EARTHQUAKE INPUT WILL HAVE THE THREE VALUES FROM THE THREE DIRECTIONS. THESE DIRECTIONS CAN BE CHANGED WITH THE USE OF THE THFACT LINE TO ELIMINATE THE NECESSITY OF MODIFYING THE INPUT TIME HISTORY.
(11-13)	ENTER THE INPUT FORMAT FOR THE SUBSEQUENT DATA INPUT. THERE ARE TWO OPTIONS, 'STD' FOR THE STANDARD INPUT USING 'THDATA' LINES AND 'CMP' FOR THE COMPRESSED INPUT USING 'THCOMP' LINES.
(14-20)	ENTER THE TIME INTERVAL BETWEEN INPUT TIME POINTS FOR THE COMPRESSED INPUT FORMAT. LEAVE BLANK FOR THE STANDARD INPUT.
(22-25)	ENTER ANY FOUR CHARACTER NAME TO IDENTIFY THE TIME HISTORY INPUT IN THE OUTPUT LISTING.
(30)	ENTER THE TIME HISTORY INPUT TYPE. ***** UNITS ***** TYPE ENGLISH METRIC 'A' - ACCELERATION IN/SEC**2 CM/SEC**2 'V' - VELOCITY IN/SEC CM/SEC 'D' - DISPLACEMENT IN CM 'G' - ACCELERATION G'S G'S

LINE LABEL	NUMBER OF TIME HISTORY FUNCTIONS	INPUT FORMAT	TIME INTERVAL (SEC) - 'CMP' ONLY -	TIME HISTORY INPUT NAME	INPUT TIME HISTORY TYPE	LEAVE BLANK
THBEGIN						
1-- 7	8-->10	11--13	14<--20	22--25	30	31-----80
DEFAULT		'STD'			'A'	

TIME HISTORY DATA (COMPRESSED)

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IF THE TIME HISTORY SOURCE IS SPECIFIED AS 'LINE' ON THE THLOAD LINE AND 'CMP' IS SPECIFIED ON THE THBEGIN LINE.
(8-10)	ENTER 'END' ON THE LAST LINE IN THIS INPUT.
(11-70)	ENTER THE TIME HISTORY VALUES FOR THE NUMBER OF FUNCTIONS AS ENTERED ON THE THBEGIN LINE. THE UNITS FOR THESE FUNCTIONS DEPEND ON THE TIME HISTORY TYPE AS SELECTED ON THE THBEGIN LINE.
	* ENGLISH * * METRIC * ACCELERATION VELOCITY DISPLACEMENT G IN/SEC IN G CM/SEC CM
	THE ORDER OF INPUT OF THE TIME HISTORY VARIABLES IS THAT ALL FUNCTIONS ARE ENTERED FOR EACH TIME POINT FOLLOWED BY THE NEXT TIME POINT. FOR EXAMPLE, IF THERE ARE TWO FUNCTIONS BEING INPUT, THEN THE FIRST VALUE OF THE FIRST FUNCTION IS INPUT FIRST FOLLOWED BY THE FIRST VALUE OF THE SECOND FUNCTION. THE NEXT VALUE WOULD BE THE SECOND VALUE OF THE FIRST FUNCTION, ETC.

LINE LABEL	END OF DATA	TIME HISTORY DATA						LEAVE BLANK
		FIRST VALUE	SECOND VALUE	THIRD VALUE	FOURTH VALUE	FIFTH VALUE	SIXTH VALUE	
THCOMP								
1-- 6	8--10	11<--20	21<--30	31<--40	41<--50	51<--60	61<--70	71--80

TIME HISTORY DATA

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IF THE TIME HISTORY SOURCE IS SPECIFIED AS 'LINE' ON THE THLOAD LINE AND 'STD' IS SPECIFIED ON THE THBEGIN LINE.
(11-20)	ENTER THE TIME IN SECONDS. THE FIRST TIME POINT IS ASSUMED TO BE ZERO AND THE ASSOCIATED TIME HISTORY FUNCTIONS ARE ALSO ASSUMED TO BE ZERO. DO NOT ENTER THE FIRST TIME POINT AS LESS THAN OR EQUAL TO ZERO. TO TERMINATE THE INPUT TIME HISTORY, THE LAST LINE WILL BE A THDATA LINE WITH ALL FIELDS LEFT BLANK.
(21-50)	ENTER THE TIME HISTORY VALUES FOR THE NUMBER OF FUNCTIONS AS ENTERED ON THE THBEGIN LINE. THE UNITS FOR THESE FUNCTIONS DEPEND ON THE TIME HISTORY TYPE AS SELECTED ON THE THBEGIN LINE.

LINE LABEL	TIME (SEC)	TIME HISTORY DATA			LEAVE BLANK
		FIRST FUNCTION	SECOND FUNCTION	THIRD FUNCTION	
THDATA					
1-- 6	11<--20	21<--30	31<--40	41<--50	51--80

TIME HISTORY DIRECTIONALITY FACTOR

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS OPTIONAL AND IS USED TO MODIFY THE INPUT TIME HISTORY BY APPLYING FACTORS IN EACH OF THREE DIRECTIONS FOR EACH OF THREE INPUT FUNCTIONS. IF THIS LINE IS OMITTED, THE DEFAULT FACTORS ARE 1.0 FOR THE X-DIRECTION FOR THE FIRST FUNCTION, 1.0 FOR THE Y-DIRECTION FOR THE SECOND FUNCTION, AND 1.0 FOR THE Z-DIRECTION FOR THE THIRD FUNCTION WITH THE REMAINING FACTORS = 0.0. THIS ESSENTIALLY APPLIES THE FIRST FUNCTION TO THE X-DIRECTION, SECOND TO THE Y, AND THIRD TO THE Z. BY APPLYING DIFFERENT FACTORS TO DIFFERENT FUNCTIONS, THE USER CAN USE ANY FUNCTION FOR ANY DIRECTION OR ANY COMBINATION TO ACHIEVE THE DESIRED INPUT TIME HISTORY.
(11-15)	ENTER THE FACTOR TO BE APPLIED TO THE FIRST INPUT TIME HISTORY FUNCTION FOR THE X-DIRECTION.
(16-55)	ENTER THE REMAINING FACTORS FOR THE OTHER DIRECTIONS AND FUNCTIONS.

LINE LABEL	DIRECTIONAL TIME HISTORY FACTORS								
	FIRST TIME HISTORY			SECOND TIME HISTORY			THIRD TIME HISTORY		
	X FACTOR	Y FACTOR	Z FACTOR	X FACTOR	Y FACTOR	Z FACTOR	X FACTOR	Y FACTOR	Z FACTOR
THFACT									
1-- 6	11<--15	16<--20	21<--25	26<--30	31<--35	36<--40	41<--45	46<--50	51<--55

TIME HISTORY LOAD

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED FOR A TIME HISTORY ANALYSIS. IT DEFINES THE INPUT TIME HISTORY, DAMPING, AND OUTPUT OPTIONS.
(9-12)	ENTER THE TIME HISTORY INPUT SOURCE AS FOLLOWS: 'LINE' - TIME HISTORY IS INPUT ON SUBSEQUENT LINES. 'PREV' - USE TIME HISTORY FROM PREVIOUS LOAD CASE. 'FILE' - TIME HISTORY CONTAINED ON DATA FILE. 'SHIP' - DYNAMIC SHIP IMPACT ANALYSIS. 'DROP' - DYNAMIC DROPPED OBJECT ANALYSIS. ENTER A FOUR CHARACTER NAME OF A SPECIFIC TIME HISTORY CONTAINED IN A FILE.
(18-20)	SPECIFY THE DAMPING TYPE: 'SDO' - STRUCTURAL DAMPING ONLY. 'LFD' - LINEARIZED FLUID DAMPING AT A SPECIFIED AMPLITUDE (REQUIRES 'PC' ON FDAMP LINE). 'NFD' - NONLINEAR FLUID DAMPING WITH FLUID FORCES CALCULATED AT EVERY TIME STEP DURING INTEGRATION (REQUIRES 'PC' ON FDAMP LINE).
(21-28)	IF THE LINEARIZED FLUID DAMPING IS SELECTED, ENTER THE AMPLITUDE TO BE USED TO COMPUTE THE EQUIVALENT LINEAR FLUID DAMPING.
(29-30)	SELECT THE INPUT INTERPOLATION SCHEME. INTERMEDIATE VALUES BETWEEN THE INPUT VALUES ARE CALCULATED USING THE FOLLOWING TECHNIQUES: 'LN' - LINEAR DEFAULT FOR ACCELERATION INPUT. 'QD' - QUADRATIC DEFAULT FOR VELOCITY INPUT. 'CU' - CUBIC DEFAULT FOR DISPLACEMENT INPUT.
(31-32)	ENTER 'GP' IF GAP ELEMENTS ARE INCLUDED IN THIS ANALYSIS.

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COLUMNS                                COMMENTARY
(33-59)
SELECT ANY OF THE FOLLOWING OUTPUT OPTIONS:
(A) LOAD CASE OPTIONS
'MXM' - CREATES LOAD CASE CORRESPONDING TO TIME POINT HAVING
      THE MAXIMUM OVERTURNING MOMENT.
'MXS' - CREATES LOAD CASE CORRESPONDING TO TIME POINT HAVING
      THE MAXIMUM BASE SHEAR.
'ALL' - CREATES LOAD CASE FOR ALL TIME POINTS.
'ESL' - CREATES EQUIVALENT STATIC LOADS
'CLP' - CREATE LOADS AND INPUT FILE FOR COLLAPSE
'WVR' - CREATE LOADS AND TIME-HISTORY INPUT FILE FOR THE WAVE
      RESPONSE ANALYSIS
(B) MODAL RESPONSE OPTIONS
'PRT' - PRINTS MODAL RESPONSES FOR ALL TIME POINTS.
'PLT' - PLOTS MODAL RESPONSES VERSUS TIME.
(C) JOINT RESULTS OPTIONS
'JMA' - OUTPUTS MAXIMUM AND MINIMUM TIME HISTORY ACCELERATIONS
      OF JOINTS SELECTED ON THE JTNUM LINES IN ADDITION TO
      THE MAXIMUM ABSOLUTE VALUE.
'JPA' - SAME AS 'JMA' PLUS CREATES TIME HISTORY PLOTS OF JOINT
      ACCELERATION.
'JTA' - SAME AS 'JMA' PLUS CREATES PRINT OF ACCELERATION TIME
      HISTORY.
'JTV' - SAME AS 'JTA' EXCEPT FOR VELOCITIES.
'JPV' - SAME AS 'JPA' EXCEPT FOR VELOCITIES.
'JMV' - SAME AS 'JMA' EXCEPT FOR VELOCITIES.
'JTD' - SAME AS 'JTA' EXCEPT FOR DISPLACEMENTS.
'JPD' - SAME AS 'JPA' EXCEPT FOR DISPLACEMENTS.
'JMD' - SAME AS 'JMA' EXCEPT FOR DISPLACEMENTS.
(D) MISCELLANEOUS PLOT OPTIONS
'PLM' - PLOTS OVERTURNING MOMENT.
'PLS' - PLOTS BASE SHEAR.
(E) LOAD DEVIATION/COMPARE OPTION
'Cxx' - COMPARES DEVIATION OF LOAD SUMMATION WHEN 'xx'
      NUMBER OF OUTPUT POINTS ARE SKIPPED.

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[illegible]

INTEGRATION CONTROL

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE INTEGRATION PARAMETERS FOR A TIME HISTORY ANALYSIS. IF THIS LINE IS OMITTED, THE START TIME IS SET EQUAL TO THE START TIME OF THE TIME HISTORY INPUT AND THE STOP TIME TO THE END TIME OF THE INPUT TIME HISTORY. OTHER PARAMETERS WILL HAVE THEIR NORMAL DEFAULT VALUES.
(11-20)	ENTER THE START TIME FOR THE BEGINNING OF THE TIME HISTORY INTEGRATION.
(21-30)	ENTER THE END TIME FOR THE INTEGRATION. IF LEFT BLANK, THE PROGRAM WILL USE THE END OF THE INPUT TIME HISTORY.
(31-40)	OUTPUT VALUES INCLUDING OVERTURNING MOMENT AND BASE SHEAR WILL BE CALCULATED AT THIS TIME INTERVAL. IT IS RECOMMENDED THAT THIS VALUE BE NO GREATER THAN ONE FOURTH OF THE SMALLEST NATURAL STRUCTURAL PERIOD OF INTEREST. THE DEFAULT WILL BE SELECTED BY THE PROGRAM BASED ON THE MODE HAVING THE SMALLEST PERIOD.
(41-50)	ENTER THE MINIMUM INTEGRATION TIME STEP ALLOWED DURING THE INTEGRATION PROCESS. SINCE A VARIABLE TIME STEP INTEGRATION PROCEDURE IS BEING USED, A MINIMUM TIME STEP IS NECESSARY TO PREVENT THE PROGRAM FROM REDUCING THE TIME STEP TO AN INFINITESIMAL VALUE. HOWEVER, THIS VALUE SHOULD BE SUFFICIENTLY SMALL TO ALLOW THE INTEGRATION TO PROCEED PAST ANY RAPIDLY CHANGING CONDITIONS DURING THE TIME HISTORY ANALYSIS.
(51-60)	ENTER THE TOLERANCE FACTOR ALLOWED FOR CONTROLLING THE BUILDUP OF ERRORS DURING THE INTEGRATION. THIS FACTOR IS APPLIED TO THE BUILT-IN ERROR CONTROLS SUCH THAT A FACTOR OF 1.0 (DEFAULT) NORMALLY KEEPS THE ERROR BUILDUP WITHIN SATISFACTORY LIMITS. IF IT SHOULD OCCUR THAT THE MINIMUM INTEGRATION TIME STEP LIMIT IS REACHED, IT MAY BECOME NECESSARY TO INCREASE THIS TOLERANCE FACTOR OR TO DECREASE THE MINIMUM ALLOWABLE INTEGRATION TIME STEP.

LINE LABEL	START TIME	END TIME	OUTPUT TIME INTERVAL	MINIMUM INTEGRATION TIME STEP	TOLERANCE FACTOR
TIME					
1-- 4	11<--20	21<--30	31<--40	41<--50	51<--60
DEFAULT				1.00E-08	1
ENGLISH	SEC	SEC	SEC	SEC	
METRIC	SEC	SEC	SEC	SEC	

DROPPED OBJECT IMPACT LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE PARAMETERS ASSOCIATED WITH A DROPPED OBJECT IMPACTING THE STRUCTURE AT A DESIGNATED LOCATION.
(9-18)	ENTER THE OBJECT'S WEIGHT.
(19)	SELECT OPTION FOR OBJECT AND STRUCTURE INTERACTION AFTER ' ' - OBJECT ALLOWED TO BOUNCE (COMPRESSION FORCE ONLY). 'A' - OBJECT ADHERES AFTER IMPACT (COMPRESSION AND TENSILE FORCE). 'S' - SINGLE IMPACT (USE OBJECT MASS AT JOINT IN DYNPAC FOR ACCURATE POST IMPACT VIBRATION).
(20)	ENTER 'O' TO OMIT THE FREE FALL DURATION OF THE OBJECT.
(21-27)	ENTER THE OBJECT'S INITIAL VELOCITY.
(42-48)	OPTIONALLY ENTER THE DISTANCE THE OBJECT TRAVELS BEFORE IMPACT. THIS IS THE DISTANCE THE OBJECT FALLS FROM TIME POINT ZERO WHERE THE INITIAL VELOCITY IS DEFINED TO THE POINT OF IMPACT.
(63-66)	ENTER THE JOINT NUMBER WHERE THIS IMPACT OCCURS.

LINE LABEL	OBJECT WEIGHT	ADHERE OPTION	FREE FALL OPTION	INITIAL VELOCITY	DISTANCE BEFORE IMPACT	IMPACT JOINT NUMBER	LEAVE BLANK
DROP							
1-- 4	9---18	19	20	21<--27	42<--48	63-->66	67--80
DEFAULT		0	0		0		
ENGLISH	KIPS			FPS	FT		
METRIC	TONNES			MPS	M		

DYNAMIC RESPONSE OPTIONS LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IN ANY DYNAMIC RESPONSE RUN. IT SPECIFIES OVERALL ANALYSIS PARAMETERS AND TYPE OF ANALYSIS.
(7-10)	ENTER THE TYPE OF ANALYSIS TO BE PERFORMED FROM THE FOLLOWING OPTIONS: 'SPEC' - BASE DRIVEN SPECTRAL ANALYSIS (SPECTRAL EARTHQUAKE) 'TIME' - BASE DRIVEN TIME HISTORY ANALYSIS (TIME HISTORY EARTHQUAKE) 'VIBR' - FORCE DRIVEN ANALYSIS (PERIODIC OR TIME HISTORY) 'WIND' - WIND SPECTRAL ANALYSIS (EXTREME OR FATIGUE ANALYSIS) 'ENGV' - ENGINE VIBRATION ANALYSIS 'TRNF' - REACTION DRIVEN TRANSFER FUNCTION ANALYSIS 'TCLP' - GENERATE INCREMENTAL LOADS FOR COLLAPSE ANALYSIS 'TESL' - GENERATE EQUIVALENT STATIC LOADS FOR LINEAR ANALYSIS 'SHIP' - DYNAMIC SHIP IMPACT ANALYSIS 'DROP' - DYNAMIC DROPPED OBJECT ANALYSIS
(11-14)	ENTER THE NUMBER OF MODES TO BE USED IN THE ANALYSIS. THE DEFAULT IS THE NUMBER OF MODES COMPUTED BY THE PREVIOUS DYNPAC RUN.
(15-16)	ENTER 'EC' IF A LISTING OF THE INPUT DATA IS TO BE INCLUDED IN THE OUTPUT LISTING FILE.
(17-18)	ENTER THE VERTICAL COORDINATE (-X, Y, OR Z, OR +X, Y, OR Z)
(19-24)	ENTER THE MUDLINE ELEVATION IN THE VERTICAL COORDINATE.
(25-27)	ENTER THE SELECTIONS FOR PRINTING THE JOINT ACCELERATIONS, VELOCITIES, AND/OR DISPLACEMENTS. FOR SPECTRAL ANALYSIS ONLY. 'A' - ACCELERATIONS 'V' - VELOCITIES 'D' - DISPLACEMENTS

LINE LABEL	ANALYSIS SELECTION OPTION	NUMBER OF MODES	INPUT ECHO OPTION	VERTICAL COORDINATE	MUDLINE ELEVATION	OUTPUT SELECTION (SPECTRAL ONLY)		
						FIRST SELECTION	SECOND SELECTION	THIRD SELECTION
DROPT								
1-- 5	7--10	11-->14	15--16	17--18	19<--24	25	26	27
DEFAULT				' +Z'	0			
ENGLISH					FT			
METRIC					M			

END LINE

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS REQUIRED AND SIGNALS THE END OF THE RESPONSE CALCULATIONS AND DIRECTS THE PROGRAM TO GENERATE THE OUTPUT FOR THE 'COMBINE' PROGRAM.

LINE LABEL	LEAVE BLANK
END	
1-- 3	4-----80

FLUID MODAL DAMPING

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE FLUID MODAL DAMPING VALUES.
(7- 8)	ENTER 'PC' IF THE PROGRAM IS TO CALCULATE THE FLUID DAMPING. THE DAMPING VALUES WILL BE CALCULATED BASED ON THE NONLINEAR FORCES ON THE STRUCTURE. IN THE CASE OF THE SPECTRAL ANALYSIS, AN EQUIVALENT DAMPING RATIO IS CALCULATED BASED ON A PARTICULAR AMPLITUDE.
(9-10)	ENTER 'NL' IF THE NONLINEAR DAMPING IS TO BE USED FOR THE TIME HISTORY ANALYSIS. OTHERWISE, A LINEAR VALUE WILL BE USED BASED ON A SPECIFIED AMPLITUDE.
(11-15)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL IF ALL MODES ARE TO HAVE THE SAME VALUE. OTHERWISE LEAVE THIS FIELD BLANK. IF A VALUE IS ENTERED HERE, LEAVE THE REMAINING FIELDS ON THIS LINE BLANK.
(16-20)	IF A SPECIFIED AMPLITUDE IS TO BE USED IN THE FLUID DAMPING CALCULATIONS, ENTER THAT VALUE HERE. USED FOR TIME HISTORY ONLY.
(21-25)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL FOR THE FIRST MODE.
(26-70)	ENTER MODAL DAMPING VALUES FOR THE REMAINING MODES. IF THE NUMBER OF MODES IS GREATER THAN TEN, USE ADDITIONAL LINES WITH THE REMAINING DAMPING VALUES ENTERED IN COLUMNS 21-70.

LINE LABEL	FLUID DAMPING SOURCE	NONLINEAR FLUID DAMPING OPTION	OVERALL MODAL DAMPING	FLUID DAMPING AMPLITUDE	*2FLUID MODAL DAMPING VALUES (PERCENT OF CRITICAL)									
					MODE 1	MODE 2	MODE 3	MODE 4	MODE 5	MODE 6	MODE 7	MODE 8	MODE 9	MODE 10
FDAMP														
1-- 5	7-- 8	9--10	11<--15	16<--20	21<--25	26<--30	31<--35	36<--40	41<--45	46<--50	51<--55	56<--60	61<--65	66<--70

JOINT SELECTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IF ANY OUTPUT SELECTED ON THE THLOAD LINE IS FOR OUTPUT FOR SELECTED JOINTS SUCH AS 'JTA', ETC. THIS LINE FOLLOWS THE THLOAD LINE.
(11-74)	ENTER THE JOINT NAMES FOR JOINTS SELECTED FOR OUTPUT. ANY NUMBER OF JOINTS MAY BE SELECTED AND ENTERED IN ANY ORDER. MULTIPLE 'JTNUM' LINES MAY BE USED FOR INPUT WITHOUT RESTRICTIONS.

LINE LABEL	SELECTED JOINTS FOR OUTPUT															
	JOINT 1	JOINT 2	JOINT 3	JOINT 4	JOINT 5	JOINT 6	JOINT 7	JOINT 8	JOINT 9	JOINT 10	JOINT 11	JOINT 12	JOINT 13	JOINT 14	JOINT 15	JOINT 16
JTNUM																
1-- 5	11-->14	15-->18	19-->22	23-->26	27-->30	31-->34	35-->38	39-->42	43-->46	47-->50	51-->54	55-->58	59-->62	63-->66	67-->70	71-->74

LOAD HEADER

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED AND SPECIFIES THAT LOADING DATA (SPECTRAL, TIME HISTORY, OR VIBRATION) FOLLOWS AS DESIGNATED ON THE DROPT LINE.

LINE LABEL	LEAVE BLANK
LOAD	
1-- 4	5-----80

STRUCTURAL MODAL DAMPING

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE MODAL DAMPING VALUES. THESE VALUES SHOULD INCLUDE ALL SOURCES OF DAMPING OTHER THAN THE FLUID DAMPING IF AN ITERATIVE SOLUTION IS BEING USED. IN THAT CASE THE FLUID DAMPING IS ACCOUNTED FOR BY CALCULATING THE ACTUAL FLUID FORCES DUE TO MOTION OF THE STRUCTURE AND CALCULATING AN EQUIVALENT LINEAR MODAL DAMPING VALUE WHICH IS ADDED TO THESE VALUES TO OBTAIN THE TOTAL DAMPING. IF THE FLUID DAMPING IS NOT BEING CALCULATED BY THE PROGRAM, THEN THE USER SHOULD INCLUDE THE FLUID DAMPING IN THIS DATA OR ENTER THE FLUID DAMPING ON THE FDAMP LINE.
(11-15)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL IF ALL MODES ARE TO HAVE THE SAME VALUE, OTHERWISE LEAVE THIS FIELD BLANK. IF A VALUE IS ENTERED HERE, LEAVE THE REMAINING FIELDS ON THIS LINE BLANK.
(21-25)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL FOR THE FIRST MODE.
(26-70)	ENTER MODAL DAMPING VALUES FOR THE REMAINING MODES. IF THE NUMBER OF MODES IS GREATER THAN TEN, USE ADDITIONAL LINES WITH THE REMAINING DAMPING VALUES ENTERED IN COLUMNS 21-70.

LINE LABEL	MODAL DAMPING VALUES (PERCENT OF CRITICAL)										
	OVERALL MODAL DAMPING	MODE 1	MODE 2	MODE 3	MODE 4	MODE 5	MODE 6	MODE 7	MODE 8	MODE 9	MODE 10
SDAMP											
1-- 5	11<--15	21<--25	26<--30	31<--35	36<--40	41<--45	46<--50	51<--55	56<--60	61<--65	66<--70

SHIP IMPACT LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE PARAMETERS ASSOCIATED WITH A SHIP IMPACTING THE STRUCTURE AT A DESIGNATED
(9-18)	ENTER THE SHIP WEIGHT.
(19-20)	ENTER 'KN' IF THE SHIP'S INITIAL VELOCITY IS TO ENTERED IN
(21-27)	ENTER THE SHIP'S INITIAL VELOCITY.
(28-34)	ENTER THE TRAVEL DIRECTION, MEASURED FROM THE 'X' AXIS COUNTERCLOCKWISE TOWARD THE GLOBAL 'Y' AXIS.
(42-48)	OPTIONALLY ENTER THE DISTANCE THE SHIP TRAVELS BEFORE IMPACT. THIS IS THE DISTANCE THE SHIP TRAVELS FROM TIME POINT ZERO WHERE THE INITIAL VELOCITY IS DEFINED TO THE POINT OF
(49-55)	ENTER THE IMPACT ANGLE WHERE ZERO CORRESPONDS TO A HEAD ON IMPACT, 90 DEGREES IS A PORT BROADSIDE IMPACT AND -90 DEGREES IS A STARBOARD BROADSIDE IMPACT.
(56-62)	ENTER THE FRICTION COEFFICIENT. THIS IS USED TO CALCULATE THE TANGENTIAL FORCE BASED ON THE NORMAL IMPACT LOADING.
(63-66)	ENTER THE JOINT NUMBER WHERE THIS IMPACT OCCURS.

COLUMNS	COMMENTARY
(67-70)	ENTER THE ADDED MASS FACTOR TO ACCOUNT FOR THE FLUID ADDED MASS. THIS FACTOR IS USED TO MULTIPLY THE SHIP'S WEIGHT TO OBTAIN THE COMBINED EFFECTIVE MASS.
(71)	IF THE IMPACT MEMBER INDENTATION IS TO BE INCLUDED PER API RP2A-WSD SECTION C18.9.2C, ENTER 'F' FOR THE FURNES OPTION EQ. C18.9.2-2 OR 'E' FOR THE ELLINAS OPTION EQ. C18.9.2-3. THESE OPTIONS DEPEND ON A TUBULAR MEMBER BEING CONNECTED TO THE IMPACT JOINT. LEAVE BLANK IF MEMBER INDENTATION IS NOT TO BE INCLUDED.
(72-75)	ENTER THE "B" LIMIT (MEMBER DIAMETER/DENT DEPTH). IF ENTERED, A WARNING MESSAGE WILL ISSUED IF THE RATIO IS BELOW THE SPECIFIED LIMIT.
(76-79)	ENTER THE RATIO OF DENT ENERGY TO INITIAL SHIP'S ENERGY. IF ENTERED, A WARNING MESSAGE WILL BE ENTERED IF THIS VALUE IS EXCEEDED.
(80)	ENTER SHIP INDENTATION CURVE LEAVE BLANK FOR NO SHIP INDENTATION. 1 - BOW IMPACT FROM DNV TN 202 2 - BROADSIDE IMPACT (D=1.5 M) FROM DNV TN 202 3 - BROADSIDE IMPACT (D=10 M) FROM DNV TN 202 4 - STERN IMPACT (D=1.5 M) FROM DNV TN 202 5 - STERN IMPACT (D=10 M) FROM DNV TN 202

LINE LABEL	SHIP WEIGHT	SHIP VELOCITY IN KNOTS OPTION	INITIAL VELOCITY	DIRECTION	DISTANCE BEFORE IMPACT	IMPACT ANGLE	COEFF. OF FRICTION	IMPACT JOINT NUMBER	ADDED MASS FACTOR	DENT PARAMETERS			SHIP INDENTATION CURVE
										CALC. OPTION	B LIMIT	ENERGY RATIO LIMIT	
SHIP													
1-- 4	9<--18	19<--20	21<--27	28<--34	42<--48	49<--55	56<--62	63-->66	67<--70	71	72--75	76--79	80
DEFAULT					0	0	0.3		1	NONE			
ENGLISH	SHORT TONS		>FTSEC	DEG	FT	DEG							
METRIC	TONNES		M/SEC	DEG	M	DEG							

TIME HISTORY LOAD

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED FOR A TIME HISTORY ANALYSIS. IT DEFINES THE INPUT TIME HISTORY, DAMPING, AND OUTPUT OPTIONS.
(9-12)	ENTER THE TIME HISTORY INPUT SOURCE AS FOLLOWS: 'LINE' - TIME HISTORY IS INPUT ON SUBSEQUENT LINES. 'PREV' - USE TIME HISTORY FROM PREVIOUS LOAD CASE. 'FILE' - TIME HISTORY CONTAINED ON DATA FILE. 'SHIP' - DYNAMIC SHIP IMPACT ANALYSIS. 'DROP' - DYNAMIC DROPPED OBJECT ANALYSIS. ENTER A FOUR CHARACTER NAME OF A SPECIFIC TIME HISTORY CONTAINED IN A FILE.
(18-20)	SPECIFY THE DAMPING TYPE: 'SDO' - STRUCTURAL DAMPING ONLY. 'LFD' - LINEARIZED FLUID DAMPING AT A SPECIFIED AMPLITUDE (REQUIRES 'PC' ON FDAMP LINE). 'NFD' - NONLINEAR FLUID DAMPING WITH FLUID FORCES CALCULATED AT EVERY TIME STEP DURING INTEGRATION (REQUIRES 'PC' ON FDAMP LINE).
(21-28)	IF THE LINEARIZED FLUID DAMPING IS SELECTED, ENTER THE AMPLITUDE TO BE USED TO COMPUTE THE EQUIVALENT LINEAR FLUID DAMPING.
(29-30)	SELECT THE INPUT INTERPOLATION SCHEME. INTERMEDIATE VALUES BETWEEN THE INPUT VALUES ARE CALCULATED USING THE FOLLOWING TECHNIQUES: 'LN' - LINEAR DEFAULT FOR ACCELERATION INPUT. 'QD' - QUADRATIC DEFAULT FOR VELOCITY INPUT. 'CU' - CUBIC DEFAULT FOR DISPLACEMENT INPUT.
(31-32)	ENTER 'GP' IF GAP ELEMENTS ARE INCLUDED IN THIS ANALYSIS.

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COLUMNS
(33-59)
COMMENTARY
SELECT ANY OF THE FOLLOWING OUTPUT OPTIONS:
(A) LOAD CASE OPTIONS
'MXM' - CREATES LOAD CASE CORRESPONDING TO TIME POINT HAVING
        THE MAXIMUM OVERTURNING MOMENT.
'MXS' - CREATES LOAD CASE CORRESPONDING TO TIME POINT HAVING
        THE MAXIMUM BASE SHEAR.
'ALL' - CREATES LOAD CASE FOR ALL TIME POINTS.
'ESL' - CREATES EQUIVALENT STATIC LOADS
'CLP' - CREATE LOADS AND INPUT FILE FOR COLLAPSE
(B) MODAL RESPONSE OPTIONS
'PRT' - PRINTS MODAL RESPONSES FOR ALL TIME POINTS.
'PLT' - PLOTS MODAL RESPONSES VERSUS TIME.
(C) JOINT RESULTS OPTIONS
'JMA' - OUTPUTS MAXIMUM AND MINIMUM TIME HISTORY ACCELERATIONS
        OF JOINTS SELECTED ON THE JTNUM LINES IN ADDITION TO
        THE MAXIMUM ABSOLUTE VALUE.
'JPA' - SAME AS 'JMA' PLUS CREATES TIME HISTORY PLOTS OF JOINT
        ACCELERATION.
'JTA' - SAME AS 'JMA' PLUS CREATES PRINT OF ACCELERATION TIME
        HISTORY.
'JTV' - SAME AS 'JTA' EXCEPT FOR VELOCITIES.
'JPV' - SAME AS 'JPA' EXCEPT FOR VELOCITIES.
'JMV' - SAME AS 'JMA' EXCEPT FOR VELOCITIES.
'JTD' - SAME AS 'JTA' EXCEPT FOR DISPLACEMENTS.
'JPD' - SAME AS 'JPA' EXCEPT FOR DISPLACEMENTS.
'JMD' - SAME AS 'JMA' EXCEPT FOR DISPLACEMENTS.
(D) MISCELLANEOUS PLOT OPTIONS
'PLM' - PLOTS OVERTURNING MOMENT.
'PLS' - PLOTS BASE SHEAR.
(E) LOAD DEVIATION/COMPARE OPTION
'Cxx' - COMPARES DEVIATION OF LOAD SUMMATION WHEN 'xx'
        NUMBER OF OUTPUT POINTS ARE SKIPPED.

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[illegible]

INTEGRATION CONTROL

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE INTEGRATION PARAMETERS FOR A TIME HISTORY ANALYSIS. IF THIS LINE IS OMITTED, THE START TIME IS SET EQUAL TO THE START TIME OF THE TIME HISTORY INPUT AND THE STOP TIME TO THE END TIME OF THE INPUT TIME HISTORY. OTHER PARAMETERS WILL HAVE THEIR NORMAL DEFAULT VALUES.
(11-20)	ENTER THE START TIME FOR THE BEGINNING OF THE TIME HISTORY INTEGRATION.
(21-30)	ENTER THE END TIME FOR THE INTEGRATION. IF LEFT BLANK, THE PROGRAM WILL USE THE END OF THE INPUT TIME HISTORY.
(31-40)	OUTPUT VALUES INCLUDING OVERTURNING MOMENT AND BASE SHEAR WILL BE CALCULATED AT THIS TIME INTERVAL. IT IS RECOMMENDED THAT THIS VALUE BE NO GREATER THAN ONE FOURTH OF THE SMALLEST NATURAL STRUCTURAL PERIOD OF INTEREST. THE DEFAULT WILL BE SELECTED BY THE PROGRAM BASED ON THE MODE HAVING THE SMALLEST PERIOD.
(41-50)	ENTER THE MINIMUM INTEGRATION TIME STEP ALLOWED DURING THE INTEGRATION PROCESS. SINCE A VARIABLE TIME STEP INTEGRATION PROCEDURE IS BEING USED, A MINIMUM TIME STEP IS NECESSARY TO PREVENT THE PROGRAM FROM REDUCING THE TIME STEP TO AN INFINITESIMAL VALUE. HOWEVER, THIS VALUE SHOULD BE SUFFICIENTLY SMALL TO ALLOW THE INTEGRATION TO PROCEED PAST ANY RAPIDLY CHANGING CONDITIONS DURING THE TIME HISTORY ANALYSIS.
(51-60)	ENTER THE TOLERANCE FACTOR ALLOWED FOR CONTROLLING THE BUILDUP OF ERRORS DURING THE INTEGRATION. THIS FACTOR IS APPLIED TO THE BUILT-IN ERROR CONTROLS SUCH THAT A FACTOR OF 1.0 (DEFAULT) NORMALLY KEEPS THE ERROR BUILDUP WITHIN SATISFACTORY LIMITS. IF IT SHOULD OCCUR THAT THE MINIMUM INTEGRATION TIME STEP LIMIT IS REACHED, IT MAY BECOME NECESSARY TO INCREASE THIS TOLERANCE FACTOR OR TO DECREASE THE MINIMUM ALLOWABLE INTEGRATION TIME STEP.

LINE LABEL	START TIME	END TIME	OUTPUT TIME INTERVAL	MINIMUM INTEGRATION TIME STEP	TOLERANCE FACTOR
TIME					
1-- 4	11<--20	21<--30	31<--40	41<--50	51<--60
DEFAULT				1.00E-08	1
ENGLISH	SEC	SEC	SEC	SEC	
METRIC	SEC	SEC	SEC	SEC	

DYNAMIC RESPONSE OPTIONS LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IN ANY DYNAMIC RESPONSE RUN. IT SPECIFIES OVERALL ANALYSIS PARAMETERS AND TYPE OF ANALYSIS.
(7-10)	ENTER THE TYPE OF ANALYSIS TO BE PERFORMED FROM THE FOLLOWING OPTIONS: 'SPEC' - BASE DRIVEN SPECTRAL ANALYSIS (SPECTRAL EARTHQUAKE) 'TIME' - BASE DRIVEN TIME HISTORY ANALYSIS (TIME HISTORY EARTHQUAKE) 'VIBR' - FORCE DRIVEN ANALYSIS (PERIODIC OR TIME HISTORY) 'WIND' - WIND SPECTRAL ANALYSIS (EXTREME OR FATIGUE ANALYSIS) 'ENGV' - ENGINE VIBRATION ANALYSIS 'TRNF' - REACTION DRIVEN TRANSFER FUNCTION ANALYSIS 'TCLE' - GENERATE INCREMENTAL LOADS FOR COLLAPSE ANALYSIS 'TESL' - GENERATE EQUIVALENT STATIC LOADS FOR LINEAR ANALYSIS 'SHIP' - DYNAMIC SHIP IMPACT ANALYSIS 'DROP' - DYNAMIC DROPPED OBJECT ANALYSIS
(11-14)	ENTER THE NUMBER OF MODES TO BE USED IN THE ANALYSIS. THE DEFAULT IS THE NUMBER OF MODES COMPUTED BY THE PREVIOUS DYNPAC RUN.
(15-16)	ENTER 'EC' IF A LISTING OF THE INPUT DATA IS TO BE INCLUDED IN THE OUTPUT LISTING FILE.
(17-18)	ENTER THE VERTICAL COORDINATE (-X, Y, OR Z, OR +X, Y, OR Z)
(19-24)	ENTER THE MUDLINE ELEVATION IN THE VERTICAL COORDINATE.
(25-27)	ENTER THE SELECTIONS FOR PRINTING THE JOINT ACCELERATIONS, VELOCITIES, AND/OR DISPLACEMENTS. FOR SPECTRAL ANALYSIS ONLY. 'A' - ACCELERATIONS 'V' - VELOCITIES 'D' - DISPLACEMENTS

LINE LABEL	ANALYSIS SELECTION OPTION	NUMBER OF MODES	INPUT ECHO OPTION	VERTICAL COORDINATE	MUDLINE ELEVATION	OUTPUT SELECTION (SPECTRAL ONLY)		
						FIRST SELECTION	SECOND SELECTION	THIRD SELECTION
DROPT								
1-- 5	7--10	11-->14	15--16	17--18	19<--24	25	26	27
DEFAULT				' +Z'	0			
ENGLISH					FT			
METRIC					M			

END LINE

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS REQUIRED AND SIGNALS THE END OF THE RESPONSE CALCULATIONS AND DIRECTS THE PROGRAM TO GENERATE THE OUTPUT FOR THE 'COMBINE' PROGRAM.

LINE LABEL	LEAVE BLANK
END	
1-- 3	4-----80

ENGINE VIBRATION ANALYSIS OPTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED FOR AN ENGINE VIBRATION ANALYSIS. IT DEFINES THE OVERALL ANALYSIS PARAMETERS AND OUTPUT OPTIONS.
(7-13)	ENTER THE BEGINNING ENGINE SPEED. THIS DEFINES THE LOWEST SPEED THAT WILL BE ANALYZED.
(14-20)	ENTER THE END ENGINE SPEED. RUNNING SPEEDS WILL BE ANALYZED BETWEEN THE BEGINNING AND END ENGINE SPEEDS.
(21-23)	SELECT FROM THE FOLLOWING FOR THE BASIS FOR THE ENGINE SPEED INCREMENTS: 'CON' - CONSTANT INCREMENTS. 'MOD' - INCREMENTS WILL VARY TO HAVE AN ANALYSIS SPEED AT EACH MODAL FREQUENCY. THE INCREMENTS BETWEEN MODES WILL BE SUCH THAT THE INCREMENTS WILL NOT BE LESS THAN THE CONSTANT INCREMENT SIZE. 'MAH' - SAME AS 'MOD' EXCEPT THAT ALL HARMONIC FREQUENCIES WILL ALSO BE INCLUDED. 'USR' - USER DEFINED. SEE 'USRSP' DATA RECORD.
(24-28)	ENTER THE NUMBER OF ANALYSIS SPEED POINTS. THIS WILL BE THE TOTAL NUMBER OF ANALYSIS SPEED POINTS USED IN THE CASE OF THE CONSTANT SPEED INCREMENT SIZE. FOR THE MODAL INCREMENT OPTION, THIS VALUE WILL BE USED TO CALCULATE THE MINIMUM INCREMENT SIZE.
(29-33)	THE FORCES ARE CALCULATED BETWEEN INPUT RUNNING SPEEDS, USING EITHER A LINEAR OR NONLINEAR INTERPOLATION. FOR THE NONLINEAR INTERPOLATION, ENTER THE POWER TO BE USED FOR THIS INTERPOLATION.

COLUMNS	COMMENTARY
(34-36)	ENTER THE MINIMUM NUMBER OF POINTS TO BE CALCULATED IN ONE CYCLE OF THE FUNDAMENTAL FREQUENCY. IF HARMONICS ARE ENCOUNTERED, THEN THE NUMBER OF POINTS FOR A CYCLE WILL BE THE MAXIMUM OF THIS VALUE AND THAT CALCULATED FOR THE HARMONICS. A VALUE OF 20 WILL GIVE A MAXIMUM ERROR IN CALCULATING THE PEAK OF 1.3 PERCENT.
(37-39)	ENTER THE NUMBER OF POINTS TO BE CALCULATED FOR THE HIGHEST HARMONIC CALCULATED. A VALUE OF 10 WILL GIVE A MAXIMUM ERROR FOR THE HIGHEST HARMONIC OF 4.9 PERCENT.
(40-41)	SELECT THE ALLOWABLE TO BE USED FOR DEFLECTION UNITY CHECKS: 'DL' - D LINE ALLOWABLE. 'SN' - SNAME. 'ML' - MILITARY SPECIFICATION. 'VE' - MAXIMUM VELOCITY. 'AC' - MAXIMUM ACCELERATION.
(42-46)	IF 'VE' IS SELECTED FOR UNITY CHECKS, ENTER THE MAXIMUM VELOCITY ALLOWED. IF 'AC' IS SELECTED, ENTER THE MAXIMUM ACCELERATION ALLOWED.
(53-54)	SELECT THE GENERALIZED FORCE PRINT OPTION: 'PT' - NORMAL PRINT. 'FL' - FULL PRINT.
(55-56)	ENTER 'PL' IF ALL JOINT RESULTS ARE TO BE PLOTTED.
(57-58)	ENTER 'PT' IF ALL RUNNING SPEEDS FOR ALL JOINTS THAT EXCEED THE ALLOWABLE ARE TO BE PRINTED.

LINE LABEL	BEGIN ENGINE SPEED	END ENGINE SPEED	SPEED INCREMENT TYPE	NUMBER SPEED INCREMENTS	NONLINEAR INTERPOLATION POWER	NUMBER POINTS PER CYCLE (FUNDAMENTAL)	NUMBER POINTS PER CYCLE (HARMONICS)	ALLOWABLE OPTION	MAXIMUM VELOCITY/ACCEL. ALLOWED	OUTPUT OPTION SELECTIONS						
										GENL. FORCE PRINT	ALL JOINTS PLOT OPTION	JOINT EXCEED PRINT OPTION	4TH	5TH	6TH	7TH
ENGVIB																
1-- 6	7<--13	14<--20	21--23	24-->28	29<--33	34-->36	37-->39	40--41	42<--46	53--54	55--56	57--58	59--60	61--62	63--64	65--66
DEFAULT			'MOD'	2	2	20	10	'DL'	2.5 METRIC							
ENGLISH	RPM	RPM							MILS/SEC OR MILS/SEC/SEC							
METRIC	RPM	RPM							MM/SEC OR MM/SEC/SEC							

JOINT PLOT SELECTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS OPTIONAL AND IS USED TO DESIGNATE THE JOINTS TO BE INCLUDED IN THE PLOTS. IF OMITTED, ALL JOINTS WILL BE PLOTTED IF THAT OPTION WAS SELECTED ON THE ENGVIB RECORD. OTHERWISE, NO SPECIFIC JOINTS WILL BE PLOTTED. THIS DATA CAN BE REPEATED AS MANY TIMES AS NECESSARY TO PLOT AS MANY JOINTS AS DESIRED. NOTE THAT A JOINT MUST BE INCLUDED IN THE ANALYSIS TO BE PLOTTED.
(11-79)	ENTER THE JOINT NAMES FOR JOINTS SELECTED FOR PLOTTING.

LINE LABEL	SELECTED JOINTS FOR PLOTTING													
	JOINT 1	JOINT 2	JOINT 3	JOINT 4	JOINT 5	JOINT 6	JOINT 7	JOINT 8	JOINT 9	JOINT 10	JOINT 11	JOINT 12	JOINT 13	JOINT 14
JNTPLT														
1-- 6	11-->14	16-->19	21-->24	26-->29	31-->34	36-->39	41-->44	46-->49	51-->54	56-->59	61-->64	66-->69	71-->74	76-->79

JOINT SELECTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS OPTIONAL AND IS USED TO DESIGNATE THE JOINTS TO BE INCLUDED IN THE ANALYSIS. IF OMITTED, ALL JOINTS WILL BE ANALYZED. THIS DATA CAN BE REPEATED AS MANY TIMES AS NECESSARY TO INCLUDE AS MANY JOINTS AS DESIRED.
(11-79)	ENTER THE JOINT NAMES FOR JOINTS SELECTED FOR ANALYSIS.

LINE LABEL	SELECTED JOINTS FOR ANALYSIS													
	JOINT 1	JOINT 2	JOINT 3	JOINT 4	JOINT 5	JOINT 6	JOINT 7	JOINT 8	JOINT 9	JOINT 10	JOINT 11	JOINT 12	JOINT 13	JOINT 14
JNTSEL														
1-- 6	11-->14	16-->19	21-->24	26-->29	31-->34	36-->39	41-->44	46-->49	51-->54	56-->59	61-->64	66-->69	71-->74	76-->79

LOAD FACTOR DATA

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED WHEN THE 'TIM' OPTION IS SELECTED ON THE UNBAL RECORD AND IMMEDIATELY FOLLOWS THE UNBAL RECORD. THE LOAD FACTORS ARE USED TO DESCRIBE A PERIODIC FORCE VERSUS TIME FOR ONE FORCE CYCLE ASSUMING EQUALLY SPACED POINTS. THIS DATA CAN BE REPEATED UP TO 100 TIME POINTS.
(7-76)	ENTER THE LOAD FACTORS FOR ALL TIME POINTS. IF A FIELD IS LEFT BLANK, THAT FIELD WILL BE SKIPPED.

LINE LABEL	LOAD FACTOR VERSUS TIME									
	1ST FACTOR	2ND FACTOR	3RD FACTOR	4TH FACTOR	5TH FACTOR	6TH FACTOR	7TH FACTOR	8TH FACTOR	9TH FACTOR	10TH FACTOR
LDFACT										
1-- 6	7<--13	14<--20	21<--27	28<--34	35<--41	42<--48	49<--55	56<--62	63<--69	70<--76

LOAD HEADER

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS REQUIRED AND SPECIFIES THAT LOADING DATA (SPECTRAL, TIME HISTORY, OR VIBRATION) FOLLOWS AS DESIGNATED ON THE DROPT LINE.

LINE LABEL	LEAVE BLANK
LOAD	
1-- 4	5-----80

MODE SELECTION LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SELECT MODES TO BE USED IN THE ANALYSIS. THE NUMBER OF MODES SELECTED MUST AGREE WITH THE NUMBER ENTERED IN COLUMNS 11-14 OF THE 'DROPT' LINE. MODE SELECTION IS ONLY USED FOR ENGINE VIBRATION, PERIODIC VIBRATION, OR WIND SPECTRUM ANALYSES.
(7- 9)	ENTER THE FIRST MODE SELECTION. THIS VALUE MUST BE GREATER THAN ZERO AND LESS THAN OR EQUAL TO THE HIGHEST MODE SELECTED IN THE DYNPAC ANALYSIS.
(10-12)	ENTER THE SECOND MODE.
(13-78)	ENTER THE REMAINING MODES DESIRED. IF THE NUMBER OF MODES DESIRED IS GREATER THAN 24, CONTINUE ON A SECOND 'MODSEL' LINE.

LINE LABEL	MODE SELECTIONS																							
	1ST	2ND	3RD	4TH	5TH	6TH	7TH	8TH	9TH	10TH	11TH	12TH	13TH	14TH	15TH	16TH	17TH	18TH	19TH	20TH	21TH	22TH	23TH	24TH
MODSEL																								
1-- 6	7--> 9	10-->12	13-->15	16-->18	19-->21	22-->24	25-->27	28-->30	31-->33	34-->36	37-->39	40-->42	43-->45	46-->48	49-->51	52-->54	55-->57	58-->60	61-->63	64-->66	67-->69	70-->72	73-->75	76-->78

PLOT OPTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS OPTIONAL AND IS ONLY USED TO CHANGE THE PLOT PARAMETERS IF DESIRED. HOWEVER, IF OMITTED NO ALLOWABLES WILL BE INCLUDED ON THE PLOTS.
(11-16)	SELECT THE ALLOWABLES TO BE SHOWN ON EACH PLOT. UP TO THREE ALLOWABLES CAN BE INCLUDED: 'DL' - D-LINE ALLOWABLE. 'SN' - SNAME. 'ML' - MILITARY SPECIFICATION. 'VE' - MAXIMUM VELOCITY.
(17-34)	ENTER THE PLOT SIZE DESIRED IF DIFFERENT FROM THE DEFAULT.
(35-36)	IF A GRID IS DESIRED ON THE PLOTS, ENTER 'GR'. OTHERWISE, LEAVE BLANK.

LINE LABEL	ALLOWABLE SELECTIONS			PLOT SIZE			GRID OPTION	LEAVE BLANK
	1ST	2ND	3RD	X-SIZE	Y-SIZE	CHARACTER SIZE		
PLTOPT								
1-- 6	11--12	13--14	15--16	17<--22	23<--28	29<--34	35--36	37--80
DEFAULT				11.0 ENGL	8.5 ENGL	0.08 ENGL	NONE	
ENGLISH				IN	IN	IN		
METRIC				CM	CM	CM		

RUNNING SPEED HEADER

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED FOR EACH ENGINE RUNNING SPEED THAT THE FORCE UNBALANCE IS DEFINED. THE RUNNING SPEEDS MUST BE INPUT IN ASCENDING ORDER. THIS RECORD IS FOLLOWED BY THE DEFINITION OF THE UNBALANCED FORCES FOR THIS SPEED.
(9-15)	ENTER THE ENGINE RUNNING SPEED.

LINE LABEL	RUNNING SPEED	LEAVE BLANK
RSPEED		
1-- 6	9<--15	16-----80
DEFAULT		
ENGLISH	RPM	
METRIC	RPM	

STRUCTURAL MODAL DAMPING

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE MODAL DAMPING VALUES. THESE VALUES SHOULD INCLUDE ALL SOURCES OF DAMPING OTHER THAN THE FLUID DAMPING IF AN ITERATIVE SOLUTION IS BEING USED. IN THAT CASE THE FLUID DAMPING IS ACCOUNTED FOR BY CALCULATING THE ACTUAL FLUID FORCES DUE TO MOTION OF THE STRUCTURE AND CALCULATING AN EQUIVALENT LINEAR MODAL DAMPING VALUE WHICH IS ADDED TO THESE VALUES TO OBTAIN THE TOTAL DAMPING. IF THE FLUID DAMPING IS NOT BEING CALCULATED BY THE PROGRAM, THEN THE USER SHOULD INCLUDE THE FLUID DAMPING IN THIS DATA OR ENTER THE FLUID DAMPING ON THE FDAMP LINE.
(11-15)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL IF ALL MODES ARE TO HAVE THE SAME VALUE, OTHERWISE LEAVE THIS FIELD BLANK. IF A VALUE IS ENTERED HERE, LEAVE THE REMAINING FIELDS ON THIS LINE BLANK.
(21-25)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL FOR THE FIRST MODE.
(26-70)	ENTER MODAL DAMPING VALUES FOR THE REMAINING MODES. IF THE NUMBER OF MODES IS GREATER THAN TEN, USE ADDITIONAL LINES WITH THE REMAINING DAMPING VALUES ENTERED IN COLUMNS 21-70.

LINE LABEL	MODAL DAMPING VALUES (PERCENT OF CRITICAL)										
	OVERALL MODAL DAMPING	MODE 1	MODE 2	MODE 3	MODE 4	MODE 5	MODE 6	MODE 7	MODE 8	MODE 9	MODE 10
SDAMP											
1-- 5	11<--15	21<--25	26<--30	31<--35	36<--40	41<--45	46<--50	51<--55	56<--60	61<--65	66<--70

UNBALANCED FORCE DATA

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE UNBALANCED TIME HISTORY OR PERIODIC LOADING FOR THE VIBRATION ANALYSIS. A SET OF THESE LINES SPECIFIES THE TIME HISTORY LOADING FOR A PARTICULAR RUNNING SPEED. IF THE 'TIM' OPTION IS SELECTED, THE 'LDFACT' DATA IMMEDIATELY FOLLOWS THIS DATA.
(8-11)	ENTER THE JOINT TO WHICH THIS LOAD IS TO BE APPLIED.
(12-14)	IF THE LOAD IS TO BE DESCRIBED BY A SINGLE SINE WAVE (AMPLITUDE AND PHASE ANGLE), ENTER 'SIN'. IF THE LOAD IS TO BE DESCRIBED BY A SERIES OF EQUALLY SPACED POINTS IN TIME, ENTER 'TIM'.
(17-58)	ENTER THE FORCES AND MOMENTS TO ACT AT THE DEFINED JOINT.
(59-65)	FOR THE SINE INPUT, ENTER THE PHASE ANGLE IN DEGREES.
(66-67)	ENTER THE TYPE OF INTERPOLATION TO USED FOR THIS LOADING: 'LN' - LINEAR. 'NL' - NONLINEAR. IF NONLINEAR IS CHOSEN, THE NONLINEAR POWER USED IS AS ENTERED ON THE 'ENGVIB' RECORD.
(68-69)	ENTER THE HARMONIC NUMBER FOR THIS FORCE. IF THE FORCE HAS THE SAME FREQUENCY AS THE RUNNING SPEED, ENTER ZERO OR LEAVE BLANK. IF THE FREQUENCY IS TWICE THE RUNNING SPEED, ENTER ONE, ETC.
(70-71)	ENTER THE LOAD GROUP NUMBER FOR THIS FORCE. LOAD GROUPS ARE CONSIDERED TO ACT INDEPENDENTLY FROM OTHER LOAD GROUPS AND, ACCORDINGLY, THE MAXIMUM DEFLECTIONS RESULTING FROM DIFFERENT LOAD GROUPS ARE ADDED TOGETHER.
(72-73)	ENTER THE HIGHEST HARMONIC NUMBER TO BE USED FROM THE FOURIER SERIES REPRESENTATION OF THE 'TIM' FUNCTION.

LINE LABEL	JOINT NAME	FORCE TYPE	JOINT FORCES AND MOMENTS IN GLOBAL COORDINATES						PHASE ANGLE	INTERPOLATION TYPE	HARMONIC NUMBER	LOAD GROUP NUMBER	HIGHEST HARMONIC
			FORCES			MOMENTS							
			Fx	Fy	Fz	Mx	My	Mz					
UNBAL													
1-- 5	8-->11	12--14	17<--23	24<--30	31<--37	38<--44	45<--51	52<--58	59<--65	66--67	68-->69	70-->71	72-->73
DEFAULT		'SIN'								'LN'	0	1	10
ENGLISH			KIP	KIP	KIP	KIP-IN	KIP-IN	KIP-IN	DEG				
METRIC (KN)			KN	KN	KN	KN-M	KN-M	KN-M	DEG				
METRIC (KG)			KG	KG	KG	KG-M	KG-M	KG-M	DEG				

USER-DEFINED ANALYSIS SPEED DATA

COLUMNS

COMMENTARY

GENERAL THIS LINE IS REQUIRED WHEN THE 'USR' OPTION IS SELECTED ON THE 'ENGVIB' RECORD (COLUMNS 21-23). THE MAXIMUM NUMBER OF USER-DEFINED SPEEDS IS 200.

(7-76) ENTER THE USER-DEFINED ANALYSIS SPEEDS IN ASCENDING ORDER. IF
A FIELD IS LEFT BLANK, THAT FIELD WILL BE SKIPPED.

[illegible]

JOINT DISPLACEMENT PART 1

COMMENTARY	COLUMNS
	GENERAL THIS LINE SET ENABLES THE TIME HISTORY APPLICATION OF TRANSLATIONS AND ROTATIONS ON THE JOINTS.
(1- 4)	ENTER 'DISP' ON EACH LINE IN THIS SET.
(8- 11)	ENTER THE JOINT NAME TO WHICH DISPLACEMENTS/VELOCITIES / ACCELERATIONS ON THIS LINE ARE TO BE APPLIED.
(12-16)	FOR TIME HISTORY INPUT, ENTER THE TIME THAT THIS LOAD IS TO BE APPLIED. LEAVE BLANK FOR PERIODIC VIBRATION. THE INTERPOLATION BETWEEN INPUT VALUES DEPENDS ON THE OPTION SELECTED ON THE FVIB LINE.
(17-23)	TRANSLATIONAL DISPLACEMENTS IN GLOBAL X DIRECTION ON THIS JOINT. A POSITIVE DISPLACEMENT IS IN THE POSITIVE X DIRECTION. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(24-30)	TRANSLATIONAL DISPLACEMENTS IN GLOBAL Y DIRECTION ON THIS JOINT. A POSITIVE DISPLACEMENT IS IN THE POSITIVE Y DIRECTION. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(31-37)	TRANSLATIONAL DISPLACEMENTS IN GLOBAL Z DIRECTION ON THIS JOINT. A POSITIVE DISPLACEMENT IS IN THE POSITIVE Z DIRECTION. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(38-44)	ROTATIONAL DISPLACEMENTS IN GLOBAL X DIRECTION ON THIS JOINT. THE SIGN OF THE ROTATION IS DETERMINED USING THE RIGHT-HAND RULE. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(45-51)	ROTATIONAL DISPLACEMENTS IN GLOBAL Y DIRECTION ON THIS JOINT. THE SIGN OF THE ROTATION IS DETERMINED USING THE RIGHT-HAND RULE. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(52-58)	ROTATIONAL DISPLACEMENTS IN GLOBAL Z DIRECTION ON THIS JOINT. THE SIGN OF THE ROTATION IS DETERMINED USING THE RIGHT-HAND RULE. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(59-142)	SEE DISP PART 2.

LINE LABEL	JOINT NAME	TIME HISTORY TIME	JOINT TRANSLATION AND ROTATION DATA IN GLOBAL COORDINATES						SEE DISP LINE PART 2
			TRANSLATIONS			ROTATIONS			
			Dx	Dy	Dz	RDx	RDy	RDz	
DISP									
1-- 4	8<--11	12<--16	17<--23	24<--30	31<--37	38<--44	45<--51	52<--58	59<--142
DEFAULT		SEC							
ENGLISH		SEC	FT	FT	FT	RADIANS	RADIANS	RADIANS	
METRIC (KN)		SEC	M	M	M	RADIANS	RADIANS	RADIANS	
METRIC (KG)			M	M	M	RADIANS	RADIANS	RADIANS	

JOINT DISPLACEMENT PART 2

COMMENTARY	COLUMNS
	GENERAL THIS LINE SET ENABLES THE TIME HISTORY APPLICATION OF TRANSLATIONS AND ROTATIONS ON THE JOINTS.
	(1- 58) SEE DISP PART 1
(59-65)	TRANSLATIONAL VELOCITIES IN GLOBAL X DIRECTION ON THIS JOINT. A POSITIVE DISPLACEMENT IS IN THE POSITIVE X DIRECTION. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(66-72)	TRANSLATIONAL VELOCITIES IN GLOBAL Y DIRECTION ON THIS JOINT. A POSITIVE DISPLACEMENT IS IN THE POSITIVE Y DIRECTION. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(73-79)	TRANSLATIONAL VELOCITIES IN GLOBAL Z DIRECTION ON THIS JOINT. A POSITIVE DISPLACEMENT IS IN THE POSITIVE Z DIRECTION. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(80-86)	ROTATIONALVELOCITIES IN GLOBAL X DIRECTION ON THIS JOINT. THE SIGN OF THE ROTATION IS DETERMINED USING THE RIGHT-HAND RULE. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(87-93)	ROTATIONAL VELOCITIES IN GLOBAL Y DIRECTION ON THIS JOINT. THE SIGN OF THE ROTATION IS DETERMINED USING THE RIGHT-HAND RULE. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(94-100)	ROTATIONAL VELOCITIES IN GLOBAL Z DIRECTION ON THIS JOINT. THE SIGN OF THE ROTATION IS DETERMINED USING THE RIGHT-HAND RULE. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
	(101-142) SEE DISP PART 3.

LINE LABEL	SEE DISP LINE PART 1	JOINT TRANSLATION AND ROTATION DATA IN GLOBAL COORDINATES						SEE DISP LINE PART 3
		TRANSLATIONS			ROTATIONS			
		Vx	Vy	Vz	RVx	RVy	RVz	
DISP								
1-- 4	5--> 58	59<--65	66<--72	73<--79	80<--86	87<--93	94<--100	101<--142
DEFAULT								
ENGLISH		FT	FT	FT	RADIANS	RADIANS	RADIANS	
METRIC (KN)		M	M	M	RADIANS	RADIANS	RADIANS	
METRIC (KG)		M	M	M	RADIANS	RADIANS	RADIANS	

JOINT DISPLACEMENT PART 3

COLUMNS	COMMENTARY
GENERAL	THIS LINE SET ENABLES THE TIME HISTORY APPLICATION OF TRANSLATIONS AND ROTATIONS ON THE JOINTS.
(1- 100)	SEE DISP PART 2.
(101-107)	TRANSLATIONAL DISP/VEL/ACC IN GLOBAL X DIRECTION ON THIS JOINT. A POSITIVE DISPLACEMENT IS IN THE POSITIVE X DIRECTION. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(108-114)	TRANSLATIONAL DISP/VEL/ACC IN GLOBAL Y DIRECTION ON THIS JOINT. A POSITIVE DISPLACEMENT IS IN THE POSITIVE Y DIRECTION. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(115-121)	TRANSLATIONAL DISP/VEL/ACC IN GLOBAL Z DIRECTION ON THIS JOINT. A POSITIVE DISPLACEMENT IS IN THE POSITIVE Z DIRECTION. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(122-128)	ROTATIONAL DISP/VEL/ACC IN GLOBAL X DIRECTION ON THIS JOINT. THE SIGN OF THE ROTATION IS DETERMINED USING THE RIGHT-HAND RULE. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(129-135)	ROTATIONAL DISP/VEL/ACC IN GLOBAL Y DIRECTION ON THIS JOINT. THE SIGN OF THE ROTATION IS DETERMINED USING THE RIGHT-HAND RULE. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(136-142)	ROTATIONAL DISP/VEL/ACC IN GLOBAL Z DIRECTION ON THIS JOINT. THE SIGN OF THE ROTATION IS DETERMINED USING THE RIGHT-HAND RULE. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.

LINE LABEL	SEE JLOD LINE PART 2	JOINT TRANSLATION AND ROTATION DATA IN GLOBAL COORDINATES					
		TRANSLATIONS			ROTATIONS		
		Ax	Ay	Az	RAx	RAy	RAz
DISP							
1-- 4	5--> 100	101<--107	108<--114	115<--121	122<--128	129<--135	136<--142
DEFAULT							
ENGLISH		FT	FT	FT	RADIANS	RADIANS	RADIANS
METRIC (KN)		M	M	M	RADIANS	RADIANS	RADIANS
METRIC (KG)		M	M	M	RADIANS	RADIANS	RADIANS

DYNAMIC RESPONSE OPTIONS LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IN ANY DYNAMIC RESPONSE RUN. IT SPECIFIES OVERALL ANALYSIS PARAMETERS AND TYPE OF ANALYSIS.
(7-10)	ENTER THE TYPE OF ANALYSIS TO BE PERFORMED FROM THE FOLLOWING OPTIONS: 'SPEC' - BASE DRIVEN SPECTRAL ANALYSIS (SPECTRAL EARTHQUAKE) 'TIME' - BASE DRIVEN TIME HISTORY ANALYSIS (TIME HISTORY EARTHQUAKE) 'VIBR' - FORCE DRIVEN ANALYSIS (PERIODIC OR TIME HISTORY) 'WIND' - WIND SPECTRAL ANALYSIS (EXTREME OR FATIGUE ANALYSIS) 'ENGV' - ENGINE VIBRATION ANALYSIS 'TRNF' - REACTION DRIVEN TRANSFER FUNCTION ANALYSIS 'TCLP' - GENERATE INCREMENTAL LOADS FOR COLLAPSE ANALYSIS 'TESL' - GENERATE EQUIVALENT STATIC LOADS FOR LINEAR ANALYSIS 'SHIP' - DYNAMIC SHIP IMPACT ANALYSIS 'DROP' - DYNAMIC DROPPED OBJECT ANALYSIS
(11-14)	ENTER THE NUMBER OF MODES TO BE USED IN THE ANALYSIS. THE DEFAULT IS THE NUMBER OF MODES COMPUTED BY THE PREVIOUS DYNPAC RUN.
(15-16)	ENTER 'EC' IF A LISTING OF THE INPUT DATA IS TO BE INCLUDED IN THE OUTPUT LISTING FILE.
(17-18)	ENTER THE VERTICAL COORDINATE (-X, Y, OR Z, OR +X, Y, OR Z)
(19-24)	ENTER THE MUDLINE ELEVATION IN THE VERTICAL COORDINATE.
(25-27)	ENTER THE SELECTIONS FOR PRINTING THE JOINT ACCELERATIONS, VELOCITIES, AND/OR DISPLACEMENTS. FOR SPECTRAL ANALYSIS ONLY. 'A' - ACCELERATIONS 'V' - VELOCITIES 'D' - DISPLACEMENTS

LINE LABEL	ANALYSIS SELECTION OPTION	NUMBER OF MODES	INPUT ECHO OPTION	VERTICAL COORDINATE	MUDLINE ELEVATION	OUTPUT SELECTION (SPECTRAL ONLY)		
						FIRST SELECTION	SECOND SELECTION	THIRD SELECTION
DROPT								
1-- 5	7--10	11-->14	15--16	17--18	19<--24	25	26	27
DEFAULT				' +Z'	0			
ENGLISH					FT			
METRIC					M			

END LINE

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS REQUIRED AND SIGNALS THE END OF THE RESPONSE CALCULATIONS AND DIRECTS THE PROGRAM TO GENERATE THE OUTPUT FOR THE 'COMBINE' PROGRAM.

LINE LABEL	LEAVE BLANK
END	
1-- 3	4-----80

FLUID MODAL DAMPING

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE FLUID MODAL DAMPING VALUES.
(7- 8)	ENTER 'PC' IF THE PROGRAM IS TO CALCULATE THE FLUID DAMPING. THE DAMPING VALUES WILL BE CALCULATED BASED ON THE NONLINEAR FORCES ON THE STRUCTURE. IN THE CASE OF THE SPECTRAL ANALYSIS, AN EQUIVALENT DAMPING RATIO IS CALCULATED BASED ON A PARTICULAR AMPLITUDE.
(9-10)	ENTER 'NL' IF THE NONLINEAR DAMPING IS TO BE USED FOR THE TIME HISTORY ANALYSIS. OTHERWISE, A LINEAR VALUE WILL BE USED BASED ON A SPECIFIED AMPLITUDE.
(11-15)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL IF ALL MODES ARE TO HAVE THE SAME VALUE. OTHERWISE LEAVE THIS FIELD BLANK. IF A VALUE IS ENTERED HERE, LEAVE THE REMAINING FIELDS ON THIS LINE BLANK.
(16-20)	IF A SPECIFIED AMPLITUDE IS TO BE USED IN THE FLUID DAMPING CALCULATIONS, ENTER THAT VALUE HERE. USED FOR TIME HISTORY ONLY.
(21-25)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL FOR THE FIRST MODE.
(26-70)	ENTER MODAL DAMPING VALUES FOR THE REMAINING MODES. IF THE NUMBER OF MODES IS GREATER THAN TEN, USE ADDITIONAL LINES WITH THE REMAINING DAMPING VALUES ENTERED IN COLUMNS 21-70.

LINE LABEL	FLUID DAMPING SOURCE	NONLINEAR FLUID DAMPING OPTION	OVERALL MODAL DAMPING	FLUID DAMPING AMPLITUDE	*2FLUID MODAL DAMPING VALUES (PERCENT OF CRITICAL)									
					MODE 1	MODE 2	MODE 3	MODE 4	MODE 5	MODE 6	MODE 7	MODE 8	MODE 9	MODE 10
FDAMP														
1-- 5	7-- 8	9--10	11<--15	16<--20	21<--25	26<--30	31<--35	36<--40	41<--45	46<--50	51<--55	56<--60	61<--65	66<--70

FORCED VIBRATION ANALYSIS LOAD

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED FOR A FORCED VIBRATION ANALYSIS. IT DEFINES THE TYPE OF VIBRATION ANALYSIS, SOURCE OF INPUT TIME HISTORY, DAMPING TYPE, LOAD CASE SELECTION, AND OUTPUT OPTIONS. IT IS NECESSARY TO SELECT AN OPTION THAT GENERATES AN OUTPUT LOAD CASE, IF ANY POST-PROCESSING ANALYSIS IS TO BE PERFORMED.
(7-10)	ENTER THE TYPE OF FORCED RESPONSE ANALYSIS: 'THIS' - TIME HISTORY INPUT. 'PERI' - PERIODIC INPUT. 'SPEC' - SPECTRAL INPUT (NOT PRESENTLY AVAILABLE). 'ICE' - ICE VIBRATION ANALYSIS.
(12-15)	LEAVE BLANK FOR PERIODIC ANALYSIS. FOR TIME HISTORY ANALYSIS, ENTER THE SOURCE OF THE INPUT DATA AS FOLLOWS: 'LINE' - TIME HISTORY IS INPUT ON SUBSEQUENT LINES. 'PREV' - USE TIME HISTORY FROM PREVIOUS LOAD CASE. 'FILE' - USE TIME HISTORY FROM EXTERNAL LOAD FILE.
(17-19)	SPECIFY THE DAMPING TYPE: 'SDO' - STRUCTURAL DAMPING ONLY. 'LFD' - LINEARIZED FLUID DAMPING AT A SPECIFIED AMPLITUDE. 'NFD' - NONLINEAR FLUID DAMPING AT THE ACTUAL AMPLITUDE.
(20-27)	ENTER THE DAMPING AMPLITUDE VALUE TO BE USED WITH 'LFD' OPTION
(28-29)	ENTER THE TIME HISTORY INPUT INTERPOLATION SCHEME: 'LN' - LINEAR (USE FOR STEP FUNCTIONS, RAMPS ETC.) 'QD' - QUADRATIC (SMOOTHS INPUT) 'CU' - CUBIC (PROVIDES SMOOTHER INPUT)
(32-58)	OUTPUT OPTION SELECTIONS: (A) LOAD CASE CREATION OPTIONS 'MXM' - CREATE LOAD CASE AT TIME POINT WITH THE MAX OTM. 'MXS' - CREATE LOAD CASE AT TIME POINT WITH THE MAX BASE SHEAR. 'MMS' - CREATE LOAD CASE AT TIME POINT HAVING THE MAXIMUM BASE SHEAR AND AT THE TIME POINT HAVING MAXIMUM OTM. 'ALL' - CREATE LOADS AT ALL TIME POINTS. 'ESL' - GENERATE EQUIVALENT STATIC LOADS (B) MODAL RESPONSE OPTIONS 'PRT' - PRINTS MODAL RESPONSE AT ALL TIME POINTS. 'PLT' - PLOTS MODAL RESPONSE VERSUS TIME. 'PPT' - PRINTS AND PLOTS MODAL RESPONSE VERSUS TIME.

COLUMNS	COMMENTARY
	(C) TIME HISTORY ACCELERATION OPTIONS
	'JMA' - OUTPUTS MAX/MIN ACCEL. VALUES FOR EACH DIRECTION AND OUTPUTS THE MAXIMUM ABSOLUTE ACCELERATIONS.
	'JPA' - SAME AS 'JMA' PLUS PLOTS TIME HISTORY ACCELERATION OF JOINTS SPECIFIED ON JTNUM INPUT LINES.
	'JTA' - SAME AS 'JMA' PLUS PRINTS AND PLOTS TIME HISTORY ACCELERATION OF JOINTS SPECIFIED ON JTNUM INPUT LINES.
	(D) TIME HISTORY VELOCITY OPTIONS
	'JMV' - SAME AS 'JMA' EXCEPT FOR VELOCITIES INSTEAD OF ACCELERATIONS.
	'JPV' - SAME AS 'JPA' EXCEPT FOR VELOCITIES INSTEAD OF ACCELERATIONS.
	'JTV' - SAME AS 'JTA' EXCEPT FOR VELOCITIES INSTEAD OF ACCELERATIONS.
	(E) TIME HISTORY DISPLACEMENT OPTIONS
	'JMD' - SAME AS 'JMA' EXCEPT FOR DISPLACEMENTS.
	'JPD' - SAME AS 'JPA' EXCEPT FOR DISPLACEMENTS.
	'JTD' - SAME AS 'JTA' EXCEPT FOR DISPLACEMENTS.
	(F) PERIODIC VIBRATION DEFLECTION OPTIONS
	'MXD' - PRINTS MAX. ABSOLUTE X,Y, AND Z DEFLECTIONS OF EACH JOINT OF THIS PERIODIC FORCED VIBRATION ANALYSIS.
	'SMD' - SELECTS MAXIMUM DEFLECTIONS FOR ALL PERIODIC FORCED VIBRATION ANALYSES FOR EACH JOINT.
	'DSM' - ABSOLUTE SUM OF MXD VALUES FOR ALL PERIODIC VIBRATION ANALYSES TO PRODUCE MAXIMUM POSSIBLE DISPLACEMENTS AT EACH JOINT IN EACH DIRECTION.
	(G) MISCELLANEOUS PLOT OPTIONS (ENTER ANY OF THE FOLLOWING)
	'PLM' - PLOTS OVERTURNING MOMENT VERSUS TIME.
	'PLS' - PLOTS BASE SHEAR VERSUS TIME.
	'IFD' - PLOT ICE FORCES AND DEFLECTIONS.
(59-65)	ENTER THE SCALE FACTOR FOR INPUT FORCING FUNCTION.
(66-69)	ENTER THE JOINT TO BE MONITORED TO SELECT THE OUTPUT LOAD CASE.
(70-71)	ENTER THE DEGREE OF FREEDOM. 'DX', 'DY', OR 'DZ' FOR X, Y, OR Z DEFLECTION OR 'RX', 'RY', OR 'RZ' FOR X, Y, OR Z ROTATIONS.
(72-77)	TIME SPAN THAT THE VIBRATION SHOULD BE MONITORED. ENTER THE SHORTEST TIME THAT THE VIBRATION IS REPEATABLE.
(78-80)	ENTER THE NUMBER OF TIME POINTS TO DIVIDE THE TIME SPAN INTO. THIS SHOULD BE SUFFICIENT TO PICK UP THE HIGHEST FREQUENCY OF INTEREST.

[illegible]

JOINT SELECTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IF ANY OUTPUT SELECTED ON THE THLOAD LINE IS FOR OUTPUT FOR SELECTED JOINTS SUCH AS 'JTA', ETC. THIS LINE FOLLOWS THE THLOAD LINE.
(11-74)	ENTER THE JOINT NAMES FOR JOINTS SELECTED FOR OUTPUT. ANY NUMBER OF JOINTS MAY BE SELECTED AND ENTERED IN ANY ORDER. MULTIPLE 'JTNUM' LINES MAY BE USED FOR INPUT WITHOUT RESTRICTIONS.

LINE LABEL	SELECTED JOINTS FOR OUTPUT															
	JOINT 1	JOINT 2	JOINT 3	JOINT 4	JOINT 5	JOINT 6	JOINT 7	JOINT 8	JOINT 9	JOINT 10	JOINT 11	JOINT 12	JOINT 13	JOINT 14	JOINT 15	JOINT 16
JTNUM																
1-- 5	11-->14	15-->18	19-->22	23-->26	27-->30	31-->34	35-->38	39-->42	43-->46	47-->50	51-->54	55-->58	59-->62	63-->66	67-->70	71-->74

FORCED VIBRATION LOAD LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE TIME HISTORY OR PERIODIC LOADING FOR THE VIBRATION ANALYSIS. A SET OF THESE LINES SPECIFIES THE TIME HISTORY LOADING FOR A PARTICULAR LOAD CASE.
(8-11)	ENTER THE JOINT NAME TO WHICH THIS LOAD IS TO BE APPLIED.
(12-16)	FOR TIME HISTORY INPUT, ENTER THE TIME THAT THIS LOAD IS TO BE APPLIED. LEAVE BLANK FOR PERIODIC VIBRATION. THE INTERPOLATION BETWEEN INPUT VALUES DEPENDS ON THE OPTION SELECTED ON THE FVIB LINE.
(17-59)	ENTER THE FORCES AND MOMENTS TO ACT AT THE DEFINED JOINT.
(69-74)	FOR PERIODIC INPUT, ENTER THE PERIOD THAT THIS SET OF FORCES AND MOMENTS IS ACTING. THE FORCES AND MOMENTS WILL BE APPLIED TO THE JOINT AS $F=FO*\cos(2*PI*T/PERIOD+PHASE\ ANGLE)$.
(75-80)	FOR PERIODIC VIBRATION, ENTER THE PHASE ANGLE IN DEGREES.

LINE LABEL	JOINT NAME	TIME HISTORY	JOINT FORCES AND MOMENTS IN GLOBAL COORDINATES						PERIODIC	
		TIME	FORCES			MOMENTS			PERIOD	PHASE ANGLE
			FX	FY	FZ	MX	MY	MZ		
LOAD										
1-- 4	8-->11	12<--16	17<--23	24<--30	31<--37	38<--44	46<--52	53<--59	69<--74	75<--80
DEFAULT		SEC								
ENGLISH		SEC	KIP	KIP	KIP	KIP-IN	KIP-IN	KIP-IN	SEC	DEG
METRIC (KN)		SEC	KN	KN	KN	KN-M	KN-M	KN-M	SEC	DEG
METRIC (KG)			KG	KG	KG	KG-M	KG-M	KG-M	SEC	DEG

LOAD HEADER

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED AND SPECIFIES THAT LOADING DATA (SPECTRAL, TIME HISTORY, OR VIBRATION) FOLLOWS AS DESIGNATED ON THE DROPT LINE.

LINE LABEL	LEAVE BLANK
LOAD	
1-- 4	5-----80

FORCED VIBRATION LOAD CASE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SPECIFY THE TIME HISTORY LOADING USING SACS IV BASIC LOAD CASES.
(8-11)	ENTER THE SACS IV LOAD CASE IDENTIFIER.
(12-16)	ENTER THE TIME THAT THIS LOAD CASE IS TO BE APPLIED. THE INTERPOLATION BETWEEN INPUT VALUES DEPENDS ON THE OPTION SELECTED ON THE FVIB LINE.
(17-23)	ENTER THE LOAD CASE FACTOR FOR THIS TIME POINT.

LINE LABEL	LOAD CASE ID	TIME HISTORY	LOAD CASE FACTOR	LEAVE BLANK
		TIME		
LOADC				
1-- 5	8-->11	12<--16	17<--23	24-----80
DEFAULT				
ENGLISH		SEC		
METRIC		SEC		

STRUCTURAL MODAL DAMPING

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE MODAL DAMPING VALUES. THESE VALUES SHOULD INCLUDE ALL SOURCES OF DAMPING OTHER THAN THE FLUID DAMPING IF AN ITERATIVE SOLUTION IS BEING USED. IN THAT CASE THE FLUID DAMPING IS ACCOUNTED FOR BY CALCULATING THE ACTUAL FLUID FORCES DUE TO MOTION OF THE STRUCTURE AND CALCULATING AN EQUIVALENT LINEAR MODAL DAMPING VALUE WHICH IS ADDED TO THESE VALUES TO OBTAIN THE TOTAL DAMPING. IF THE FLUID DAMPING IS NOT BEING CALCULATED BY THE PROGRAM, THEN THE USER SHOULD INCLUDE THE FLUID DAMPING IN THIS DATA OR ENTER THE FLUID DAMPING ON THE FDAMP LINE.
(11-15)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL IF ALL MODES ARE TO HAVE THE SAME VALUE, OTHERWISE LEAVE THIS FIELD BLANK. IF A VALUE IS ENTERED HERE, LEAVE THE REMAINING FIELDS ON THIS LINE BLANK.
(21-25)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL FOR THE FIRST MODE.
(26-70)	ENTER MODAL DAMPING VALUES FOR THE REMAINING MODES. IF THE NUMBER OF MODES IS GREATER THAN TEN, USE ADDITIONAL LINES WITH THE REMAINING DAMPING VALUES ENTERED IN COLUMNS 21-70.

LINE LABEL	MODAL DAMPING VALUES (PERCENT OF CRITICAL)										
	OVERALL MODAL DAMPING	MODE 1	MODE 2	MODE 3	MODE 4	MODE 5	MODE 6	MODE 7	MODE 8	MODE 9	MODE 10
SDAMP											
1-- 5	11<--15	21<--25	26<--30	31<--35	36<--40	41<--45	46<--50	51<--55	56<--60	61<--65	66<--70

FORCE TIME HISTORY DATA HEADER

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IF THE TIME HISTORY SOURCE IS SPECIFIED AS 'LINE' ON THE FVIB LINE. THE OVERALL PARAMETERS OF THE TIME HISTORY INPUT ARE DESCRIBED HERE.
(8-10)	ENTER THE NUMBER OF JOINTS THAT WILL BE LOADED WITH A TIME HISTORY INPUT.
(11-13)	ENTER THE INPUT FORMAT FOR THE SUBSEQUENT DATA INPUT. THERE ARE TWO OPTIONS, 'NON' FOR THE NONUNIFORM TIME INCREMENTS AND 'UNI' FOR THE UNIFORM TIME INCREMENTS.
(14-20)	ENTER THE TIME INTERVAL BETWEEN INPUT TIME POINTS FOR THE UNIFORM INPUT FORMAT. LEAVE BLANK FOR THE NONUNIFORM INPUT.
(22-25)	ENTER ANY FOUR CHARACTER NAME TO IDENTIFY THE TIME HISTORY INPUT IN THE OUTPUT LISTING.
(31-32)	SELECT INPUT DISPLACEMENT UNITS FROM THE FOLLOWING: 'EN' - ENGLISH 'MN' - METRIC (KILONEWTION FORCE) 'ME' - METRIC (KILOGRAM FORCE)
(33-37)	ENTER TRANSLATION FACTOR. THIS FACTOR WILL APPLY TO ALL TRANSLATIONS.
(38-42)	ENTER ROTATION FACTOR. THIS FACTOR WILL APPLY TO ALL ROTATIONS.

LINE LABEL	NUMBER OF JOINTS HAVING FORCE INPUT	INPUT FORM	TIME INTERVAL (SEC) - 'UNI' ONLY -	TIME HISTORY INPUT NAME	DISPLACEMENT INPUT UNITS	FORCE FACTOR	MOMENT FACTOR	LEAVE BLANK
THDISP								
1-- 7	8-->10	11--13	14<--20	22--25	31--32	33--37	38--42	42-----80
DEFAULT		'NON'			'EN'	1	1	

FORCE TIME HISTORY DATA HEADER

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IF THE TIME HISTORY SOURCE IS SPECIFIED AS 'LINE' ON THE FVIB LINE. THE OVERALL PARAMETERS OF THE TIME HISTORY INPUT ARE DESCRIBED HERE.
(8-10)	ENTER THE NUMBER OF JOINTS THAT WILL BE LOADED WITH A TIME HISTORY INPUT.
(11-13)	ENTER THE INPUT FORMAT FOR THE SUBSEQUENT DATA INPUT. THERE ARE TWO OPTIONS, 'NON' FOR THE NONUNIFORM TIME INCREMENTS AND 'UNI' FOR THE UNIFORM TIME INCREMENTS.
(14-20)	ENTER THE TIME INTERVAL BETWEEN INPUT TIME POINTS FOR THE UNIFORM INPUT FORMAT. LEAVE BLANK FOR THE NONUNIFORM INPUT.
(22-25)	ENTER ANY FOUR CHARACTER NAME TO IDENTIFY THE TIME HISTORY INPUT IN THE OUTPUT LISTING.

LINE LABEL	NUMBER OF JOINTS HAVING FORCE INPUT	INPUT FORM	TIME INTERVAL (SEC) - 'UNI' ONLY -	TIME HISTORY INPUT NAME	LEAVE BLANK
THFORCE					
1-- 7	8-->10	11--13	14<--20	22--25	26-----80
DEFAULT		'NON'			

INTEGRATION CONTROL

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE INTEGRATION PARAMETERS FOR A TIME HISTORY ANALYSIS. IF THIS LINE IS OMITTED, THE START TIME IS SET EQUAL TO THE START TIME OF THE TIME HISTORY INPUT AND THE STOP TIME TO THE END TIME OF THE INPUT TIME HISTORY. OTHER PARAMETERS WILL HAVE THEIR NORMAL DEFAULT VALUES.
(11-20)	ENTER THE START TIME FOR THE BEGINNING OF THE TIME HISTORY INTEGRATION.
(21-30)	ENTER THE END TIME FOR THE INTEGRATION. IF LEFT BLANK, THE PROGRAM WILL USE THE END OF THE INPUT TIME HISTORY.
(31-40)	OUTPUT VALUES INCLUDING OVERTURNING MOMENT AND BASE SHEAR WILL BE CALCULATED AT THIS TIME INTERVAL. IT IS RECOMMENDED THAT THIS VALUE BE NO GREATER THAN ONE FOURTH OF THE SMALLEST NATURAL STRUCTURAL PERIOD OF INTEREST. THE DEFAULT WILL BE SELECTED BY THE PROGRAM BASED ON THE MODE HAVING THE SMALLEST PERIOD.
(41-50)	ENTER THE MINIMUM INTEGRATION TIME STEP ALLOWED DURING THE INTEGRATION PROCESS. SINCE A VARIABLE TIME STEP INTEGRATION PROCEDURE IS BEING USED, A MINIMUM TIME STEP IS NECESSARY TO PREVENT THE PROGRAM FROM REDUCING THE TIME STEP TO AN INFINITESIMAL VALUE. HOWEVER, THIS VALUE SHOULD BE SUFFICIENTLY SMALL TO ALLOW THE INTEGRATION TO PROCEED PAST ANY RAPIDLY CHANGING CONDITIONS DURING THE TIME HISTORY ANALYSIS.
(51-60)	ENTER THE TOLERANCE FACTOR ALLOWED FOR CONTROLLING THE BUILDUP OF ERRORS DURING THE INTEGRATION. THIS FACTOR IS APPLIED TO THE BUILT-IN ERROR CONTROLS SUCH THAT A FACTOR OF 1.0 (DEFAULT) NORMALLY KEEPS THE ERROR BUILDUP WITHIN SATISFACTORY LIMITS. IF IT SHOULD OCCUR THAT THE MINIMUM INTEGRATION TIME STEP LIMIT IS REACHED, IT MAY BECOME NECESSARY TO INCREASE THIS TOLERANCE FACTOR OR TO DECREASE THE MINIMUM ALLOWABLE INTEGRATION TIME STEP.

LINE LABEL	START TIME	END TIME	OUTPUT TIME INTERVAL	MINIMUM INTEGRATION TIME STEP	TOLERANCE FACTOR
TIME					
1-- 4	11<--20	21<--30	31<--40	41<--50	51<--60
DEFAULT				1.00E-08	1
ENGLISH	SEC	SEC	SEC	SEC	
METRIC	SEC	SEC	SEC	SEC	

DYNAMIC RESPONSE OPTIONS LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IN ANY DYNAMIC RESPONSE RUN. IT SPECIFIES OVERALL ANALYSIS PARAMETERS AND TYPE OF ANALYSIS.
(7-10)	ENTER THE TYPE OF ANALYSIS TO BE PERFORMED FROM THE FOLLOWING OPTIONS: 'SPEC' - BASE DRIVEN SPECTRAL ANALYSIS (SPECTRAL EARTHQUAKE) 'TIME' - BASE DRIVEN TIME HISTORY ANALYSIS (TIME HISTORY EARTHQUAKE) 'VIBR' - FORCE DRIVEN ANALYSIS (PERIODIC OR TIME HISTORY) 'WIND' - WIND SPECTRAL ANALYSIS (EXTREME OR FATIGUE ANALYSIS) 'ENGV' - ENGINE VIBRATION ANALYSIS 'TRNF' - REACTION DRIVEN TRANSFER FUNCTION ANALYSIS 'TCLE' - GENERATE INCREMENTAL LOADS FOR COLLAPSE ANALYSIS 'TESL' - GENERATE EQUIVALENT STATIC LOADS FOR LINEAR ANALYSIS 'SHIP' - DYNAMIC SHIP IMPACT ANALYSIS 'DROP' - DYNAMIC DROPPED OBJECT ANALYSIS
(11-14)	ENTER THE NUMBER OF MODES TO BE USED IN THE ANALYSIS. THE DEFAULT IS THE NUMBER OF MODES COMPUTED BY THE PREVIOUS DYNPAC RUN.
(15-16)	ENTER 'EC' IF A LISTING OF THE INPUT DATA IS TO BE INCLUDED IN THE OUTPUT LISTING FILE.
(17-18)	ENTER THE VERTICAL COORDINATE (-X, Y, OR Z, OR +X, Y, OR Z)
(19-24)	ENTER THE MUDLINE ELEVATION IN THE VERTICAL COORDINATE.
(25-27)	ENTER THE SELECTIONS FOR PRINTING THE JOINT ACCELERATIONS, VELOCITIES, AND/OR DISPLACEMENTS. FOR SPECTRAL ANALYSIS ONLY. 'A' - ACCELERATIONS 'V' - VELOCITIES 'D' - DISPLACEMENTS

LINE LABEL	ANALYSIS SELECTION OPTION	NUMBER OF MODES	INPUT ECHO OPTION	VERTICAL COORDINATE	MUDLINE ELEVATION	OUTPUT SELECTION (SPECTRAL ONLY)		
						FIRST SELECTION	SECOND SELECTION	THIRD SELECTION
DROPT								
1-- 5	7--10	11-->14	15--16	17--18	19<--24	25	26	27
DEFAULT				' +Z'	0			
ENGLISH					FT			
METRIC					M			

END LINE

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS REQUIRED AND SIGNALS THE END OF THE RESPONSE CALCULATIONS AND DIRECTS THE PROGRAM TO GENERATE THE OUTPUT FOR THE 'COMBINE' PROGRAM.

LINE LABEL	LEAVE BLANK
END	
1-- 3	4-----80

FLUID MODAL DAMPING

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE FLUID MODAL DAMPING VALUES.
(7- 8)	ENTER 'PC' IF THE PROGRAM IS TO CALCULATE THE FLUID DAMPING. THE DAMPING VALUES WILL BE CALCULATED BASED ON THE NONLINEAR FORCES ON THE STRUCTURE. IN THE CASE OF THE SPECTRAL ANALYSIS, AN EQUIVALENT DAMPING RATIO IS CALCULATED BASED ON A PARTICULAR AMPLITUDE.
(9-10)	ENTER 'NL' IF THE NONLINEAR DAMPING IS TO BE USED FOR THE TIME HISTORY ANALYSIS. OTHERWISE, A LINEAR VALUE WILL BE USED BASED ON A SPECIFIED AMPLITUDE.
(11-15)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL IF ALL MODES ARE TO HAVE THE SAME VALUE. OTHERWISE LEAVE THIS FIELD BLANK. IF A VALUE IS ENTERED HERE, LEAVE THE REMAINING FIELDS ON THIS LINE BLANK.
(16-20)	IF A SPECIFIED AMPLITUDE IS TO BE USED IN THE FLUID DAMPING CALCULATIONS, ENTER THAT VALUE HERE. USED FOR TIME HISTORY ONLY.
(21-25)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL FOR THE FIRST MODE.
(26-70)	ENTER MODAL DAMPING VALUES FOR THE REMAINING MODES. IF THE NUMBER OF MODES IS GREATER THAN TEN, USE ADDITIONAL LINES WITH THE REMAINING DAMPING VALUES ENTERED IN COLUMNS 21-70.

LINE LABEL	FLUID DAMPING SOURCE	NONLINEAR FLUID DAMPING OPTION	OVERALL MODAL DAMPING	FLUID DAMPING AMPLITUDE	*2FLUID MODAL DAMPING VALUES (PERCENT OF CRITICAL)									
					MODE 1	MODE 2	MODE 3	MODE 4	MODE 5	MODE 6	MODE 7	MODE 8	MODE 9	MODE 10
FDAMP														
1-- 5	7-- 8	9--10	11<--15	16<--20	21<--25	26<--30	31<--35	36<--40	41<--45	46<--50	51<--55	56<--60	61<--65	66<--70

FORCED VIBRATION ANALYSIS LOAD

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS REQUIRED FOR A FORCED VIBRATION ANALYSIS. IT DEFINES THE TYPE OF VIBRATION ANALYSIS, SOURCE OF INPUT TIME HISTORY, DAMPING TYPE, LOAD CASE SELECTION, AND OUTPUT OPTIONS. IT IS NECESSARY TO SELECT AN OPTION THAT GENERATES AN OUTPUT LOAD CASE, IF ANY POST-PROCESSING ANALYSIS IS TO BE PERFORMED.
(7-10)	ENTER THE TYPE OF FORCED RESPONSE ANALYSIS: 'THIS' - TIME HISTORY INPUT. 'PERI' - PERIODIC INPUT. 'SPEC' - SPECTRAL INPUT (NOT PRESENTLY AVAILABLE). 'ICE ' - ICE VIBRATION ANALYSIS.
(12-15)	LEAVE BLANK FOR PERIODIC ANALYSIS. FOR TIME HISTORY ANALYSIS, ENTER THE SOURCE OF THE INPUT DATA AS FOLLOWS: 'LINE' - TIME HISTORY IS INPUT ON SUBSEQUENT LINES. 'PREV' - TIME HISTORY FROM PREVIOUS LOAD CASE.
(17-19)	SPECIFY THE DAMPING TYPE: 'SDO' - STRUCTURAL DAMPING ONLY. 'LFD' - LINEARIZED FLUID DAMPING AT A SPECIFIED AMPLITUDE. 'NFD' - NONLINEAR FLUID DAMPING AT THE ACTUAL AMPLITUDE.
(20-27)	ENTER THE DAMPING AMPLITUDE VALUE TO BE USED WITH 'LFD' OPTION
(28-29)	ENTER THE TIME HISTORY INPUT INTERPOLATION SCHEME: 'LN' - LINEAR (USE FOR STEP FUNCTIONS, RAMPS ETC.) 'QD' - QUADRATIC (SMOOTHS INPUT) 'CU' - CUBIC (PROVIDES SMOOTHER INPUT)
(32-58)	OUTPUT OPTION SELECTIONS: (A) LOAD CASE CREATION OPTIONS 'MXM' - CREATE LOAD CASE AT TIME POINT WITH THE MAX OTM. 'MXS' - CREATE LOAD CASE AT TIME POINT WITH THE MAX BASE SHEAR. 'MMS' - CREATE LOAD CASE AT TIME POINT HAVING THE MAXIMUM BASE SHEAR AND AT THE TIME POINT HAVING MAXIMUM OTM. 'ALL' - CREATE LOADS AT ALL TIME POINTS. 'ESL' - GENERATE EQUIVALENT STATIC LOADS (B) MODAL RESPONSE OPTIONS 'PRT' - PRINTS MODAL RESPONSE AT ALL TIME POINTS. 'PLT' - PLOTS MODAL RESPONSE VERSUS TIME. 'PPT' - PRINTS AND PLOTS MODAL RESPONSE VERSUS TIME.

COLUMNS

COMMENTARY

- (C) TIME HISTORY ACCELERATION OPTIONS
 - 'JMA' - OUTPUTS MAX/MIN ACCEL. VALUES FOR EACH DIRECTION AND OUTPUTS THE MAXIMUM ABSOLUTE ACCELERATIONS.
 - 'JPA' - SAME AS 'JMA' PLUS PLOTS TIME HISTORY ACCELERATION OF JOINTS SPECIFIED ON JTNUM INPUT LINES.
 - 'JTA' - SAME AS 'JMA' PLUS PRINTS AND PLOTS TIME HISTORY ACCELERATION OF JOINTS SPECIFIED ON JTNUM INPUT LINES.

- (D) TIME HISTORY VELOCITY OPTIONS
 - 'JMV' - SAME AS 'JMA' EXCEPT FOR VELOCITIES INSTEAD OF ACCELERATIONS.
 - 'JPV' - SAME AS 'JPA' EXCEPT FOR VELOCITIES INSTEAD OF ACCELERATIONS.
 - 'JTV' - SAME AS 'JTA' EXCEPT FOR VELOCITIES INSTEAD OF ACCELERATIONS.

- (E) TIME HISTORY DISPLACEMENT OPTIONS
 - 'JMD' - SAME AS 'JMA' EXCEPT FOR DISPLACEMENTS.
 - 'JPD' - SAME AS 'JPA' EXCEPT FOR DISPLACEMENTS.
 - 'JTD' - SAME AS 'JTA' EXCEPT FOR DISPLACEMENTS.

- (F) PERIODIC VIBRATION DEFLECTION OPTIONS
 - 'MXD' - PRINTS MAX. ABSOLUTE X,Y, AND Z DEFLECTIONS OF EACH JOINT OF THIS PERIODIC FORCED VIBRATION ANALYSIS.
 - 'SMD' - SELECTS MAXIMUM DEFLECTIONS FOR ALL PERIODIC FORCED VIBRATION ANALYSES FOR EACH JOINT.
 - 'DSM' - ABSOLUTE SUM OF MXD VALUES FOR ALL PERIODIC VIBRATION ANALYSES TO PRODUCE MAXIMUM POSSIBLE DISPLACEMENTS AT EACH JOINT IN EACH DIRECTION.

- (G) MISCELLANEOUS PLOT OPTIONS (ENTER ANY OF THE FOLLOWING)
 - 'FLM' - PLOTS OVERTURNING MOMENT VERSUS TIME.
 - 'PLS' - PLOTS BASE SHEAR VERSUS TIME.
 - 'IFD' - PLOT ICE FORCES AND DEFLECTIONS.

(59-65) ENTER THE SCALE FACTOR FOR INPUT FORCING FUNCTION.

(66-69) ENTER THE JOINT TO BE MONITORED TO SELECT THE OUTPUT LOAD CASE.

(70-71) ENTER THE DEGREE OF FREEDOM. 'DX', 'DY', OR 'DZ' FOR X, Y, OR Z DEFLECTION OR 'RX', 'RY', OR 'RZ' FOR X, Y, OR Z ROTATIONS.

(72-77) TIME SPAN THAT THE VIBRATION SHOULD BE MONITORED. ENTER THE SHORTEST TIME THAT THE VIBRATION IS REPEATABLE.

(78-80) ENTER THE NUMBER OF TIME POINTS TO DIVIDE THE TIME SPAN INTO. THIS SHOULD BE SUFFICIENT TO PICK UP THE HIGHEST FREQUENCY OF INTEREST.

[illegible]

FORCED VIBRATION LOAD LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE TIME HISTORY OR PERIODIC LOADING FOR THE VIBRATION ANALYSIS. A SET OF THESE LINES SPECIFIES THE TIME HISTORY LOADING FOR A PARTICULAR LOAD CASE.
(8-11)	ENTER THE JOINT NAME TO WHICH THIS LOAD IS TO BE APPLIED.
(12-16)	FOR TIME HISTORY INPUT, ENTER THE TIME THAT THIS LOAD IS TO BE APPLIED. LEAVE BLANK FOR PERIODIC VIBRATION. THE INTERPOLATION BETWEEN INPUT VALUES DEPENDS ON THE OPTION SELECTED ON THE FVIB LINE.
(17-59)	ENTER THE FORCES AND MOMENTS TO ACT AT THE DEFINED JOINT.
(69-74)	FOR PERIODIC INPUT, ENTER THE PERIOD THAT THIS SET OF FORCES AND MOMENTS IS ACTING. THE FORCES AND MOMENTS WILL BE APPLIED TO THE JOINT AS $F=FO*\cos(2*PI*T/PERIOD+PHASE\ ANGLE)$.
(75-80)	FOR PERIODIC VIBRATION, ENTER THE PHASE ANGLE IN DEGREES.

LINE LABEL	JOINT NAME	TIME HISTORY	JOINT FORCES AND MOMENTS IN GLOBAL COORDINATES						PERIODIC	
		TIME	FORCES			MOMENTS			PERIOD	PHASE ANGLE
			FX	FY	FZ	MX	MY	MZ		
LOAD										
1-- 4	8-->11	12<--16	17<--23	24<--30	31<--37	38<--44	46<--52	53<--59	69<--74	75<--80
DEFAULT		SEC								
ENGLISH		SEC	KIP	KIP	KIP	KIP-IN	KIP-IN	KIP-IN	SEC	DEG
METRIC (KN)		SEC	KN	KN	KN	KN-M	KN-M	KN-M	SEC	DEG
METRIC (KG)			KG	KG	KG	KG-M	KG-M	KG-M	SEC	DEG

LOAD HEADER

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS REQUIRED AND SPECIFIES THAT LOADING DATA (SPECTRAL, TIME HISTORY, OR VIBRATION) FOLLOWS AS DESIGNATED ON THE DROPT LINE.

LINE LABEL	LEAVE BLANK
LOAD	
1-- 4	5-----80

MODE SELECTION LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SELECT MODES TO BE USED IN THE ANALYSIS. THE NUMBER OF MODES SELECTED MUST AGREE WITH THE NUMBER ENTERED IN COLUMNS 11-14 OF THE 'DROPT' LINE. MODE SELECTION IS ONLY USED FOR ENGINE VIBRATION, PERIODIC VIBRATION, OR WIND SPECTRUM ANALYSES.
(7- 9)	ENTER THE FIRST MODE SELECTION. THIS VALUE MUST BE GREATER THAN ZERO AND LESS THAN OR EQUAL TO THE HIGHEST MODE SELECTED IN THE DYNPAC ANALYSIS.
(10-12)	ENTER THE SECOND MODE.
(13-78)	ENTER THE REMAINING MODES DESIRED. IF THE NUMBER OF MODES DESIRED IS GREATER THAN 24, CONTINUE ON A SECOND 'MODSEL' LINE.

LINE LABEL	MODE SELECTIONS																							
	1ST	2ND	3RD	4TH	5TH	6TH	7TH	8TH	9TH	10TH	11TH	12TH	13TH	14TH	15TH	16TH	17TH	18TH	19TH	20TH	21TH	22TH	23TH	24TH
MODSEL																								
1-- 6	7--> 9	10-->12	13-->15	16-->18	19-->21	22-->24	25-->27	28-->30	31-->33	34-->36	37-->39	40-->42	43-->45	46-->48	49-->51	52-->54	55-->57	58-->60	61-->63	64-->66	67-->69	70-->72	73-->75	76-->78

STRUCTURAL MODAL DAMPING

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE MODAL DAMPING VALUES. THESE VALUES SHOULD INCLUDE ALL SOURCES OF DAMPING OTHER THAN THE FLUID DAMPING IF AN ITERATIVE SOLUTION IS BEING USED. IN THAT CASE THE FLUID DAMPING IS ACCOUNTED FOR BY CALCULATING THE ACTUAL FLUID FORCES DUE TO MOTION OF THE STRUCTURE AND CALCULATING AN EQUIVALENT LINEAR MODAL DAMPING VALUE WHICH IS ADDED TO THESE VALUES TO OBTAIN THE TOTAL DAMPING. IF THE FLUID DAMPING IS NOT BEING CALCULATED BY THE PROGRAM, THEN THE USER SHOULD INCLUDE THE FLUID DAMPING IN THIS DATA OR ENTER THE FLUID DAMPING ON THE FDAMP LINE.
(11-15)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL IF ALL MODES ARE TO HAVE THE SAME VALUE, OTHERWISE LEAVE THIS FIELD BLANK. IF A VALUE IS ENTERED HERE, LEAVE THE REMAINING FIELDS ON THIS LINE BLANK.
(21-25)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL FOR THE FIRST MODE.
(26-70)	ENTER MODAL DAMPING VALUES FOR THE REMAINING MODES. IF THE NUMBER OF MODES IS GREATER THAN TEN, USE ADDITIONAL LINES WITH THE REMAINING DAMPING VALUES ENTERED IN COLUMNS 21-70.

LINE LABEL	MODAL DAMPING VALUES (PERCENT OF CRITICAL)										
	OVERALL MODAL DAMPING	MODE 1	MODE 2	MODE 3	MODE 4	MODE 5	MODE 6	MODE 7	MODE 8	MODE 9	MODE 10
SDAMP											
1-- 5	11<--15	21<--25	26<--30	31<--35	36<--40	41<--45	46<--50	51<--55	56<--60	61<--65	66<--70

DYNAMIC RESPONSE OPTIONS LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IN ANY DYNAMIC RESPONSE RUN. IT SPECIFIES OVERALL ANALYSIS PARAMETERS AND TYPE OF ANALYSIS.
(7-10)	ENTER THE TYPE OF ANALYSIS TO BE PERFORMED FROM THE FOLLOWING OPTIONS: 'SPEC' - BASE DRIVEN SPECTRAL ANALYSIS (SPECTRAL EARTHQUAKE) 'TIME' - BASE DRIVEN TIME HISTORY ANALYSIS (TIME HISTORY EARTHQUAKE) 'VIBR' - FORCE DRIVEN ANALYSIS (PERIODIC OR TIME HISTORY) 'WIND' - WIND SPECTRAL ANALYSIS (EXTREME OR FATIGUE ANALYSIS) 'ENGV' - ENGINE VIBRATION ANALYSIS 'TRNF' - REACTION DRIVEN TRANSFER FUNCTION ANALYSIS 'TCLP' - GENERATE INCREMENTAL LOADS FOR COLLAPSE ANALYSIS 'TESL' - GENERATE EQUIVALENT STATIC LOADS FOR LINEAR ANALYSIS 'SHIP' - DYNAMIC SHIP IMPACT ANALYSIS 'DROP' - DYNAMIC DROPPED OBJECT ANALYSIS
(11-14)	ENTER THE NUMBER OF MODES TO BE USED IN THE ANALYSIS. THE DEFAULT IS THE NUMBER OF MODES COMPUTED BY THE PREVIOUS DYNPAC RUN.
(15-16)	ENTER 'EC' IF A LISTING OF THE INPUT DATA IS TO BE INCLUDED IN THE OUTPUT LISTING FILE.
(17-18)	ENTER THE VERTICAL COORDINATE (-X, Y, OR Z, OR +X, Y, OR Z)
(19-24)	ENTER THE MUDLINE ELEVATION IN THE VERTICAL COORDINATE.
(25-27)	ENTER THE SELECTIONS FOR PRINTING THE JOINT ACCELERATIONS, VELOCITIES, AND/OR DISPLACEMENTS. FOR SPECTRAL ANALYSIS ONLY. 'A' - ACCELERATIONS 'V' - VELOCITIES 'D' - DISPLACEMENTS

LINE LABEL	ANALYSIS SELECTION OPTION	NUMBER OF MODES	INPUT ECHO OPTION	VERTICAL COORDINATE	MUDLINE ELEVATION	OUTPUT SELECTION (SPECTRAL ONLY)		
						FIRST SELECTION	SECOND SELECTION	THIRD SELECTION
DROPT								
1-- 5	7--10	11-->14	15--16	17--18	19<--24	25	26	27
DEFAULT				' +Z'	0			
ENGLISH					FT			
METRIC					M			

END LINE

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS REQUIRED AND SIGNALS THE END OF THE RESPONSE CALCULATIONS AND DIRECTS THE PROGRAM TO GENERATE THE OUTPUT FOR THE 'COMBINE' PROGRAM.

LINE LABEL	LEAVE BLANK
END	
1-- 3	4-----80

FORCED VIBRATION ANALYSIS LOAD

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED FOR A FORCED VIBRATION ANALYSIS. IT DEFINES THE TYPE OF VIBRATION ANALYSIS, SOURCE OF INPUT TIME HISTORY, DAMPING TYPE, LOAD CASE SELECTION, AND OUTPUT OPTIONS. IT IS NECESSARY TO SELECT AN OPTION THAT GENERATES AN OUTPUT LOAD CASE, IF ANY POST-PROCESSING ANALYSIS IS TO BE PERFORMED.
(7-10)	ENTER THE TYPE OF FORCED RESPONSE ANALYSIS: 'THIS' - TIME HISTORY INPUT. 'PERI' - PERIODIC INPUT. 'SPEC' - SPECTRAL INPUT (NOT PRESENTLY AVAILABLE). 'ICE' - ICE VIBRATION ANALYSIS.
(12-15)	LEAVE BLANK FOR PERIODIC ANALYSIS. FOR TIME HISTORY ANALYSIS, ENTER THE SOURCE OF THE INPUT DATA AS FOLLOWS: 'LINE' - TIME HISTORY IS INPUT ON SUBSEQUENT LINES. 'PREV' - USE TIME HISTORY FROM PREVIOUS LOAD CASE.
(17-19)	SPECIFY THE DAMPING TYPE: 'SDO' - STRUCTURAL DAMPING ONLY. 'LFD' - LINEARIZED FLUID DAMPING AT A SPECIFIED AMPLITUDE. 'NFD' - NONLINEAR FLUID DAMPING AT THE ACTUAL AMPLITUDE.
(20-27)	ENTER THE DAMPING AMPLITUDE VALUE TO BE USED WITH 'LFD' OPTION.
(28-29)	ENTER THE TIME HISTORY INPUT INTERPOLATION SCHEME: 'LN' - LINEAR (USE FOR STEP FUNCTIONS, RAMPS ETC.) 'QD' - QUADRATIC (SMOOTH INPUT) 'CU' - CUBIC (PROVIDES SMOOTHER INPUT)
(32-58)	OUTPUT OPTION SELECTIONS: (A) LOAD CASE CREATION OPTIONS 'MXM' - CREATE LOAD CASE AT TIME POINT WITH THE MAX OTM. 'MXS' - CREATE LOAD CASE AT TIME POINT WITH THE MAX BASE SHEAR. 'MMS' - CREATE LOAD CASE AT TIME POINT HAVING THE MAXIMUM BASE SHEAR AND AT THE TIME POINT HAVING MAXIMUM OTM. 'ALL' - CREATE LOADS AT ALL TIME POINTS. 'ESL' - GENERATE EQUIVALENT STATIC LOADS (B) MODAL RESPONSE OPTIONS 'PRT' - PRINTS MODAL RESPONSE AT ALL TIME POINTS. 'PLT' - PLOTS MODAL RESPONSE VERSUS TIME. 'PPT' - PRINTS AND PLOTS MODAL RESPONSE VERSUS TIME.

COLUMNS **COMMENTS**

(C) TIME HISTORY ACCELERATION OPTIONS

'JMA' - OUTPUTS MAX/MIN ACCEL. VALUES FOR EACH DIRECTION AND
OUTPUTS THE MAXIMUM ABSOLUTE ACCELERATIONS.

'JPA' - SAME AS 'JMA' PLUS PLOTS TIME HISTORY ACCELERATION OF
JOINTS SPECIFIED ON JTNUM INPUT LINES.

'JTA' - SAME AS 'JMA' PLUS PRINTS AND PLOTS TIME HISTORY
ACCELERATION OF JOINTS SPECIFIED ON JTNUM INPUT LINES.

(D) TIME HISTORY VELOCITY OPTIONS

'JMV' - SAME AS 'JMA' EXCEPT FOR VELOCITIES INSTEAD OF
ACCELERATIONS.

'JPV' - SAME AS 'JPA' EXCEPT FOR VELOCITIES INSTEAD OF
ACCELERATIONS.

'JTV' - SAME AS 'JTA' EXCEPT FOR VELOCITIES INSTEAD OF
ACCELERATIONS.

(E) TIME HISTORY DISPLACEMENT OPTIONS

'JMD' - SAME AS 'JMA' EXCEPT FOR DISPLACEMENTS.

'JPD' - SAME AS 'JPA' EXCEPT FOR DISPLACEMENTS.

'JTD' - SAME AS 'JTA' EXCEPT FOR DISPLACEMENTS.

(F) PERIODIC VIBRATION DEFLECTION OPTIONS

'MXD' - PRINTS MAX. ABSOLUTE X,Y, AND Z DEFLECTIONS OF EACH
JOINT OF THIS PERIODIC FORCED VIBRATION ANALYSIS.

'SMD' - SELECTS MAXIMUM DEFLECTIONS FOR ALL PERIODIC FORCED
VIBRATION ANALYSES FOR EACH JOINT.

'DSM' - ABSOLUTE SUM OF MXD VALUES FOR ALL PERIODIC VIBRATION
ANALYSES TO PRODUCE MAXIMUM POSSIBLE DISPLACEMENTS AT
EACH JOINT IN EACH DIRECTION.

(G) MISCELLANEOUS PLOT OPTIONS (ENTER ANY OF THE FOLLOWING)

'PLM' - PLOTS OVERTURNING MOMENT VERSUS TIME.

'PLS' - PLOTS BASE SHEAR VERSUS TIME.

'IFD' - PLOT ICE FORCES AND DEFLECTIONS.

(59-65) ENTER THE SCALE FACTOR FOR INPUT FORCING FUNCTION.

(66-69) ENTER THE JOINT TO BE MONITORED TO SELECT THE OUTPUT LOAD CASE.

(70-71) ENTER THE DEGREE OF FREEDOM. 'DX', 'DY', OR 'DZ' FOR X, Y, OR
Z DEFLECTION OR 'RX', 'RY', OR 'RZ' FOR X, Y, OR Z ROTATIONS.

(72-77) TIME SPAN THAT THE VIBRATION SHOULD BE MONITORED. ENTER THE
SHORTEST TIME THAT THE VIBRATION IS REPEATABLE.

(78-80) ENTER THE NUMBER OF TIME POINTS TO DIVIDE THE TIME SPAN INTO.
THIS SHOULD BE SUFFICIENT TO PICK UP THE HIGHEST FREQUENCY OF
INTEREST.

[illegible]

ICE LOAD GROUP MODIFICATION

COLUMNS

COMMENTARY

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GENERAL      THIS LINE IS USED TO MODIFY OR ELIMINATE GROUPS OF MEMBERS
              THAT PENETRATE THE ICE SHEET.  FOR EXAMPLE, THE PILES INSIDE
              JACKET LEGS WOULD PENETRATE THE ICE SHEET BUT WOULD NOT BE
              LOADED WITH ICE.

(11-13)      ENTER THE GROUP IDENTIFICATION LABEL TO BE MODIFIED.

(14-20)      ENTER THE EFFECTIVE DIAMETER.  LEAVE BLANK TO REMOVE THIS
              GROUP FROM ICE LOADINGS.

(21-80)      ENTER ANY OTHER GROUP MODIFICATIONS.  REPEAT THIS LINE FOR AS
              MANY GROUP MODIFICATIONS AS DESIRED.

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[illegible]

ICE PARAMETERS

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE ICE PARAMETERS AN ICE VIBRATION ANALYSIS.
(8-13)	ENTER THE THICKNESS OF ICE.
(14-19)	ENTER THE ELASTIC MODULUS OF THE ICE.
(20-25)	ENTER THE STATIC CRUSHING STRENGTH.
(26-31)	ENTER THE RATIO OF THE TOTAL LENGTH (BREAKING+ELASTIC) TO THE ELASTIC LENGTH. THIS VALUE MUST BE GREATER THAN 1.0.
(32-37)	ENTER THE ELEVATION OF THE TOP OF THE ICE.
(38-43)	THE ICE FLOE DIRECTION IS MEASURED FROM X TOWARD Y WHEN Z IS VERTICAL. FOR OTHER VERTICAL AXES, USE THE SIMILAR RIGHT-HAND RULE.
(44-49)	ICE FLOE VELOCITY. IF USING MULTIPLE STEPS, THIS IS THE VELOCITY OF THE FIRST STEP.
(50-55)	DENSITY OF ICE.
(56-61)	THE ICE STIFFNESS PARAMETER IS USED TO ESTIMATE THE STIFFNESS OF THE ICE. THE DEFAULT REPRESENTS AN INFINITE SHEET OF ICE FLOWING PAST A VERTICAL CYLINDER.

COLUMNS	COMMENTARY
(62-67)	ICE VELOCITY STEP SIZE. USED WHEN MULTIPLE ANALYSES ARE BEING PERFORMED TO GET THE VARIATION OF RESULTS WITH ICE VELOCITY.
(68-70)	THE NUMBER OF VELOCITY STEPS. THIS IS LEFT BLANK IF A SINGLE VELOCITY ANALYSIS IS BEING PERFORMED FOR ONE ICE LINE. THE NUMBER OF ANALYSES PERFORMED IS THE NUMBER OF STEPS PLUS ONE.
(71-76)	ENTER THE TIME DURATION THAT THIS PARTICULAR ICE FLOE WILL OCCUR. THIS TIME IS USED TO DETERMINE THE NUMBER OF CYCLES FOR FATIGUE ANALYSIS.

LINE LABEL	THICKNESS	ELASTIC MODULUS	STATIC CRUSHING STRENGTH	BREAKING LENGTH RATIO	TOP OF ICE ELEVATION	ICE FLOE DIRECTION	VELOCITY	DENSITY	STIFFNESS PARAMETER	VELOCITY STEP SIZE	NUMBER VELOCITY STEPS	DURATION OF ICE MOVEMENT
ICE												
1-- 3	08<--13	14<--19	20<--25	26<--31	32<--37	38<--43	44<--49	50<--55	56<--61	62<--67	68-->70	71<--76
DEFAULT				2					0.0315			1
ENGLISH	IN	MPSI	KSI		FT	DEG	FT/SEC	LB/CU.FT		FT/SEC		HOURS
METRIC (KN)	CM	MN/SQ.CM	KN/SQ.CM		M	DEG	M/SEC	TONNE/CU.M		M/SEC		HOURS
METRIC (KG)	CM	MKG/SQ.CM	KG/SQ.CM		M	DEG	M/SEC	TONNE/CU.M		M/SEC		HOURS

LOAD HEADER

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS REQUIRED AND SPECIFIES THAT LOADING DATA (SPECTRAL, TIME HISTORY, OR VIBRATION) FOLLOWS AS DESIGNATED ON THE DROPT LINE.

LINE LABEL	LEAVE BLANK
LOAD	
1-- 4	5-----80

ICE LOAD MEMBER MODIFICATION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO MODIFY OR ELIMINATE MEMBERS THAT PENETRATE THE ICE SHEET. FOR EXAMPLE, THE PILES INSIDE JACKET LEGS WOULD PENETRATE THE ICE SHEET BUT WOULD NOT BE LOADED WITH ICE.
(11-14)	ENTER THE STARTING JOINT OF THE FIRST MEMBER TO BE MODIFIED.
(15-18)	ENTER THE END JOINT OF THE FIRST MEMBER TO BE MODIFIED.
(19-24)	ENTER THE EFFECTIVE DIAMETER. LEAVE BLANK TO REMOVE THIS MEMBER FROM ICE LOADINGS.
(25-80)	ENTER ANY OTHER MEMBER MODIFICATIONS. REPEAT THIS LINE FOR AS MANY MEMBER MODIFICATIONS AS DESIRED.

LINE LABEL	1ST MEMBER			2ND MEMBER			3RD MEMBER			4TH MEMBER			5TH MEMBER		
	START JOINT	END JOINT	EFFECTIVE DIAMETER	START JOINT	END JOINT	EFFECTIVE DIAMETER	START JOINT	END JOINT	EFFECTIVE DIAMETER	START JOINT	END JOINT	EFFECTIVE DIAMETER	START JOINT	END JOINT	EFFECTIVE DIAMETER
MEMMD															
1-- 5	11-->14	15-->18	19<--24	25-->28	29-->32	33<--38	39-->42	43-->46	47<--52	53-->56	57-->60	61<--66	67-->70	71-->74	75<--80
DEFAULT															
ENGLISH			IN			IN			IN			IN			IN
METRIC			CM			CM			CM			CM			CM

STRUCTURAL MODAL DAMPING

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE MODAL DAMPING VALUES. THESE VALUES SHOULD INCLUDE ALL SOURCES OF DAMPING OTHER THAN THE FLUID DAMPING IF AN ITERATIVE SOLUTION IS BEING USED. IN THAT CASE THE FLUID DAMPING IS ACCOUNTED FOR BY CALCULATING THE ACTUAL FLUID FORCES DUE TO MOTION OF THE STRUCTURE AND CALCULATING AN EQUIVALENT LINEAR MODAL DAMPING VALUE WHICH IS ADDED TO THESE VALUES TO OBTAIN THE TOTAL DAMPING. IF THE FLUID DAMPING IS NOT BEING CALCULATED BY THE PROGRAM, THEN THE USER SHOULD INCLUDE THE FLUID DAMPING IN THIS DATA OR ENTER THE FLUID DAMPING ON THE FDAMP LINE.
(11-15)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL IF ALL MODES ARE TO HAVE THE SAME VALUE, OTHERWISE LEAVE THIS FIELD BLANK. IF A VALUE IS ENTERED HERE, LEAVE THE REMAINING FIELDS ON THIS LINE BLANK.
(21-25)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL FOR THE FIRST MODE.
(26-70)	ENTER MODAL DAMPING VALUES FOR THE REMAINING MODES. IF THE NUMBER OF MODES IS GREATER THAN TEN, USE ADDITIONAL LINES WITH THE REMAINING DAMPING VALUES ENTERED IN COLUMNS 21-70.

LINE LABEL	MODAL DAMPING VALUES (PERCENT OF CRITICAL)										
	OVERALL MODAL DAMPING	MODE 1	MODE 2	MODE 3	MODE 4	MODE 5	MODE 6	MODE 7	MODE 8	MODE 9	MODE 10
SDAMP											
1-- 5	11<--15	21<--25	26<--30	31<--35	36<--40	41<--45	46<--50	51<--55	56<--60	61<--65	66<--70

INTEGRATION CONTROL

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE INTEGRATION PARAMETERS FOR A TIME HISTORY ANALYSIS. IF THIS LINE IS OMITTED, THE START TIME IS SET EQUAL TO THE START TIME OF THE TIME HISTORY INPUT AND THE STOP TIME TO THE END TIME OF THE INPUT TIME HISTORY. OTHER PARAMETERS WILL HAVE THEIR NORMAL DEFAULT VALUES.
(11-20)	ENTER THE START TIME FOR THE BEGINNING OF THE TIME HISTORY INTEGRATION.
(21-30)	ENTER THE END TIME FOR THE INTEGRATION. IF LEFT BLANK, THE PROGRAM WILL USE THE END OF THE INPUT TIME HISTORY.
(31-40)	OUTPUT VALUES INCLUDING OVERTURNING MOMENT AND BASE SHEAR WILL BE CALCULATED AT THIS TIME INTERVAL. IT IS RECOMMENDED THAT THIS VALUE BE NO GREATER THAN ONE FOURTH OF THE SMALLEST NATURAL STRUCTURAL PERIOD OF INTEREST. THE DEFAULT WILL BE SELECTED BY THE PROGRAM BASED ON THE MODE HAVING THE SMALLEST PERIOD.
(41-50)	ENTER THE MINIMUM INTEGRATION TIME STEP ALLOWED DURING THE INTEGRATION PROCESS. SINCE A VARIABLE TIME STEP INTEGRATION PROCEDURE IS BEING USED, A MINIMUM TIME STEP IS NECESSARY TO PREVENT THE PROGRAM FROM REDUCING THE TIME STEP TO AN INFINITESIMAL VALUE. HOWEVER, THIS VALUE SHOULD BE SUFFICIENTLY SMALL TO ALLOW THE INTEGRATION TO PROCEED PAST ANY RAPIDLY CHANGING CONDITIONS DURING THE TIME HISTORY ANALYSIS.
(51-60)	ENTER THE TOLERANCE FACTOR ALLOWED FOR CONTROLLING THE BUILDUP OF ERRORS DURING THE INTEGRATION. THIS FACTOR IS APPLIED TO THE BUILT-IN ERROR CONTROLS SUCH THAT A FACTOR OF 1.0 (DEFAULT) NORMALLY KEEPS THE ERROR BUILDUP WITHIN SATISFACTORY LIMITS. IF IT SHOULD OCCUR THAT THE MINIMUM INTEGRATION TIME STEP LIMIT IS REACHED, IT MAY BECOME NECESSARY TO INCREASE THIS TOLERANCE FACTOR OR TO DECREASE THE MINIMUM ALLOWABLE INTEGRATION TIME STEP.

LINE LABEL	START TIME	END TIME	OUTPUT TIME INTERVAL	MINIMUM INTEGRATION TIME STEP	TOLERANCE FACTOR
TIME					
1-- 4	11<--20	21<--30	31<--40	41<--50	51<--60
DEFAULT				1.00E-08	1
ENGLISH	SEC	SEC	SEC	SEC	
METRIC	SEC	SEC	SEC	SEC	

DYNAMIC RESPONSE OPTIONS LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IN ANY DYNAMIC RESPONSE RUN. IT SPECIFIES OVERALL ANALYSIS PARAMETERS AND TYPE OF ANALYSIS.
(7-10)	ENTER THE TYPE OF ANALYSIS TO BE PERFORMED FROM THE FOLLOWING OPTIONS: 'SPEC' - BASE DRIVEN SPECTRAL ANALYSIS (SPECTRAL EARTHQUAKE) 'TIME' - BASE DRIVEN TIME HISTORY ANALYSIS (TIME HISTORY EARTHQUAKE) 'VIBR' - FORCE DRIVEN ANALYSIS (PERIODIC OR TIME HISTORY) 'WIND' - WIND SPECTRAL ANALYSIS (EXTREME OR FATIGUE ANALYSIS) 'ENGV' - ENGINE VIBRATION ANALYSIS 'TRNF' - REACTION DRIVEN TRANSFER FUNCTION ANALYSIS 'TCLP' - GENERATE INCREMENTAL LOADS FOR COLLAPSE ANALYSIS 'TESL' - GENERATE EQUIVALENT STATIC LOADS FOR LINEAR ANALYSIS 'SHIP' - DYNAMIC SHIP IMPACT ANALYSIS 'DROP' - DYNAMIC DROPPED OBJECT ANALYSIS
(11-14)	ENTER THE NUMBER OF MODES TO BE USED IN THE ANALYSIS. THE DEFAULT IS THE NUMBER OF MODES COMPUTED BY THE PREVIOUS DYNPAC RUN.
(15-16)	ENTER 'EC' IF A LISTING OF THE INPUT DATA IS TO BE INCLUDED IN THE OUTPUT LISTING FILE.
(17-18)	ENTER THE VERTICAL COORDINATE (-X, Y, OR Z, OR +X, Y, OR Z)
(19-24)	ENTER THE MUDLINE ELEVATION IN THE VERTICAL COORDINATE.
(25-27)	ENTER THE SELECTIONS FOR PRINTING THE JOINT ACCELERATIONS, VELOCITIES, AND/OR DISPLACEMENTS. FOR SPECTRAL ANALYSIS ONLY. 'A' - ACCELERATIONS 'V' - VELOCITIES 'D' - DISPLACEMENTS

LINE LABEL	ANALYSIS SELECTION OPTION	NUMBER OF MODES	INPUT ECHO OPTION	VERTICAL COORDINATE	MUDLINE ELEVATION	OUTPUT SELECTION (SPECTRAL ONLY)		
						FIRST SELECTION	SECOND SELECTION	THIRD SELECTION
DROPT								
1-- 5	7--10	11-->14	15--16	17--18	19<--24	25	26	27
DEFAULT				' +Z'	0			
ENGLISH					FT			
METRIC					M			

END LINE

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS REQUIRED AND SIGNALS THE END OF THE RESPONSE CALCULATIONS AND DIRECTS THE PROGRAM TO GENERATE THE OUTPUT FOR THE 'COMBINE' PROGRAM.

LINE LABEL	LEAVE BLANK
END	
1-- 3	4-----80

LOAD HEADER

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS REQUIRED AND SPECIFIES THAT LOADING DATA (SPECTRAL, TIME HISTORY, OR VIBRATION) FOLLOWS AS DESIGNATED ON THE DROPT LINE.

LINE LABEL	LEAVE BLANK
LOAD	
1-- 4	5-----80

MODE SELECTION LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SELECT MODES TO BE USED IN THE ANALYSIS. THE NUMBER OF MODES SELECTED MUST AGREE WITH THE NUMBER ENTERED IN COLUMNS 11-14 OF THE 'DROPT' LINE. MODE SELECTION IS ONLY USED FOR ENGINE VIBRATION, PERIODIC VIBRATION, OR WIND SPECTRUM ANALYSES.
(7- 9)	ENTER THE FIRST MODE SELECTION. THIS VALUE MUST BE GREATER THAN ZERO AND LESS THAN OR EQUAL TO THE HIGHEST MODE SELECTED IN THE DYNPAC ANALYSIS.
(10-12)	ENTER THE SECOND MODE.
(13-78)	ENTER THE REMAINING MODES DESIRED. IF THE NUMBER OF MODES DESIRED IS GREATER THAN 24, CONTINUE ON A SECOND 'MODSEL' LINE.

LINE LABEL	MODE SELECTIONS																							
	1ST	2ND	3RD	4TH	5TH	6TH	7TH	8TH	9TH	10TH	11TH	12TH	13TH	14TH	15TH	16TH	17TH	18TH	19TH	20TH	21TH	22TH	23TH	24TH
MODSEL																								
1-- 6	7--> 9	10-->12	13-->15	16-->18	19-->21	22-->24	25-->27	28-->30	31-->33	34-->36	37-->39	40-->42	43-->45	46-->48	49-->51	52-->54	55-->57	58-->60	61-->63	64-->66	67-->69	70-->72	73-->75	76-->78

STRUCTURAL MODAL DAMPING

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE MODAL DAMPING VALUES. THESE VALUES SHOULD INCLUDE ALL SOURCES OF DAMPING OTHER THAN THE FLUID DAMPING IF AN ITERATIVE SOLUTION IS BEING USED. IN THAT CASE THE FLUID DAMPING IS ACCOUNTED FOR BY CALCULATING THE ACTUAL FLUID FORCES DUE TO MOTION OF THE STRUCTURE AND CALCULATING AN EQUIVALENT LINEAR MODAL DAMPING VALUE WHICH IS ADDED TO THESE VALUES TO OBTAIN THE TOTAL DAMPING. IF THE FLUID DAMPING IS NOT BEING CALCULATED BY THE PROGRAM, THEN THE USER SHOULD INCLUDE THE FLUID DAMPING IN THIS DATA OR ENTER THE FLUID DAMPING ON THE FDAMP LINE.
(11-15)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL IF ALL MODES ARE TO HAVE THE SAME VALUE, OTHERWISE LEAVE THIS FIELD BLANK. IF A VALUE IS ENTERED HERE, LEAVE THE REMAINING FIELDS ON THIS LINE BLANK.
(21-25)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL FOR THE FIRST MODE.
(26-70)	ENTER MODAL DAMPING VALUES FOR THE REMAINING MODES. IF THE NUMBER OF MODES IS GREATER THAN TEN, USE ADDITIONAL LINES WITH THE REMAINING DAMPING VALUES ENTERED IN COLUMNS 21-70.

LINE LABEL	MODAL DAMPING VALUES (PERCENT OF CRITICAL)										
	OVERALL MODAL DAMPING	MODE 1	MODE 2	MODE 3	MODE 4	MODE 5	MODE 6	MODE 7	MODE 8	MODE 9	MODE 10
SDAMP											
1-- 5	11<--15	21<--25	26<--30	31<--35	36<--40	41<--45	46<--50	51<--55	56<--60	61<--65	66<--70

SPECTRAL WIND

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS OPTIONAL FOR A SPECTRAL WIND ANALYSIS. IF OMITTED, ALL DEFAULTS WILL BE USED.
(8- 9)	SELECT THE TYPE OF ANALYSIS: 'FT' - FATIGUE ANALYSIS (FATIGUE DATA FILE CREATED). 'EX' - EXTREME EVENT ANALYSIS (COMBINE DATA FILE CREATED).
(11-12)	SELECT THE PRINT LEVEL: 'MN' - MINIMUM PRINT (ONE LINE PER MODE PER WIND). 'MD' - MODERATE PRINT (ONE PAGE PER MODE PER WIND). 'MX' - MAXIMUM PRINT (INCLUDES SPECTRUM FOR EACH MODE).
(14-15)	ENTER 'PL' IF THE GENERALIZED FORCE SPECTRUM IS TO PLOTTED. THIS WILL GENERATE ONE PLOT FOR EACH MODE FOR EACH WIND.
(17-18)	ENTER 'PL' IF THE RESPONSE SPECTRUM IS TO PLOTTED. THIS WILL GENERATE ONE PLOT FOR EACH MODE FOR EACH WIND.
(35-36)	ENTER THE SPATIAL CORRELATION COEFFICIENT OPTION AS FOLLOWS: 'CA' - USE CALCULATED VALUE. 'SK' - DO NOT USE A SPATIAL CORRELATION COEFFICIENT FOR THE SPECTRAL WIND ANALYSIS.
(37-38)	SELECT FROM THE FOLLOWING: 'DR' - SPECIFY SPECTRUM ('WINSPC' LINE) FOR EACH DIRECTION 'VL' - SPECIFY SPECTRUM ('WINSPC' LINE) FOR EACH WIND VELOCITY
(39-44)	ENTER MEAN WIND SPEED AVERAGING TIME. THE DATA IS USED TO CALCULATE THE DYNAMIC AMPLIFICATION FACTOR FOR THE SPECTRAL WIND ANALYSIS.
(45-50)	ENTER THE REFERENCE LENGTH USED FOR THE HARRIS SPECTRUM. THIS IS USED TO CALCULATE THE MODAL RESPONSES FOR THE SPECTRAL WIND ANALYSIS.
(51-56)	ENTER THE ROUGHNESS COEFFICIENT FOR THE HARRIS WIND SPECTRUM.

LINE LABEL	OUTPUT OPTIONS				SPATIAL CORRELATION COEFF. OPTION	GUST SPECTRUM OPTION	WIND AVERAGING TIME	REFERENCE LENGTH	ROUGHNESS COEFFICIENT	LEAVE BLANK
	ANALYSIS TYPE SELECTION	PRINT LEVEL	PLOT GEN. FORCE SPECTRUM	PLOT RESPONSE SPECTRUM						
SPCWIN										
1-- 6	8-- 9	11--12	14--15	17--18	35--36	37--38	39<--44	45<--50	51<--56	57--80
DEFAULT	'FT'	'MN'	NO	NO	'CA'	'DR'	600	1800. METRIC	0.0025	
ENGLISH							SEC	FT		
METRIC							SEC	M		

STATIC + DYNAMIC COMBINATION DATA

COLUMNS	COMMENTARY
GENERAL	THIS DATA IS USED TO SPECIFY HOW THE STATIC LOADS ARE TO BE COMBINED WITH THE DYNAMIC LOADING.
(7)	ENTER 'N' IF THE LOAD COMBINATIONS OF THE STATIC LOADS AND EQUIVALENT STATIC DYNAMIC LOADS ARE NOT TO BE INCLUDED IN THE GENERATED LOAD OUTPUT FILE.
(8-12)	ENTER THE FACTOR TO BE USED ON THE EARTHQUAKE LOADS TO BE COMBINED WITH THE STATIC LOAD CASES FOR BEAM AND PLATE ELEMENT CODE CHECKS.
(13-17)	ENTER THE FACTOR TO BE USED ON THE EARTHQUAKE LOADS TO BE COMBINED WITH THE STATIC LOAD CASES FOR PUNCHING SHEAR CODE CHECKS.
(18-21)	ENTER THE LOAD CASE NAME OF THE FIRST STATIC LOAD CASE TO BE COMBINED WITH THE EARTHQUAKE LOADS. THIS LOAD CASE IS TO BE CREATED IN A SEPARATE STATIC EXECUTION PRIOR TO THE EARTHQUAKE ANALYSIS.
(22-26)	ENTER THE LOAD CASE FACTOR FOR THE FIRST STATIC LOAD CASE.
(27-80)	ENTER THE REMAINING STATIC LOAD CASES AS REQUIRED. UP TO 50 'STCMB' LINES MAY BE USED TO CREATE STATIC COMBINATION CASES.

[illegible]

WIND DIRECTION SPECTRUM

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED FOR A SPECTRAL WIND FATIGUE ANALYSIS AND IS OPTIONAL EXTREME WIND ANALYSIS. EACH WIND DIRECTION IN THE SEASTATE DATA MUST HAVE A CORRESPONDING 'WINSPC' DATA ENTRY.
(7-13)	ENTER THE WIND DIRECTION. THIS ENTRY IS USED TO CROSS CHECK THE 'WINSPC' DATA WITH THE SEASTATE DATA.
(14-20)	ENTER THE FRACTION OF TIME FOR THIS WIND DIRECTION (FATIGUE ONLY).
(22-24)	SELECT THE TYPE OF MEAN WIND DISTRIBUTION. 'WEI' - WEIBULL DISTRIBUTION NOTE: WEIBULL IS THE ONLY WIND DISTRIBUTION AVAILABLE AT THIS TIME.
(26-32)	ENTER THE 'K' WEIBULL DISTRIBUTION PARAMETER.
(33-39)	ENTER THE 'A' WEIBULL DISTRIBUTION PARAMETER.
(40-41)	SELECT FROM THE FOLLOWING GUST SPECTRUM TYPES: 'HR' - HARRIS 'VK' - VON KARMAN 'FR' - FROYA/API/NORSOK/ISO19902
(42-57)	IF THE VON KARMAN SPECTRUM IS SELECTED, ENTER THE PARAMETERS REQUIRED TO CALCULATE THE TURBULENCE INTENSITY FACTOR (TIF) WHERE: $TIF = A + B * WIND \text{ VELOCITY}$
(58-65)	ENTER THE WIND VELOCITY. NOTE: 1 MILE PER HOUR = 0.8690 KNOTS.
(66)	SELECT THE WIND VELOCITY UNITS ENTER 'M' FOR MILES PER HOUR, 'F' FOR FEET PER SECOND, 'K' FOR KNOTS, OR LEAVE BLANK FOR METERS PER SECOND.

LINE LABEL	WIND DIRECTION	FRACTION OF TIME	MEAN WIND DISTRIBUTION TYPE	WEIBULL K PARAMETER	WEIBULL A PARAMETER	GUST SPECTRUM TYPE	VON KARMAN SPECTRUM PARAMETERS		WIND LOAD DATA		LEAVE BLANK
							A VALUE	B VALUE	VELOCITY	VELOCITY UNITS	
WINSPC											
1-- 6	7<--13	14--20	22--24	26<--32	33<--39	40--41	42<--49	50<--57	58<--65	66	67--80
DEFAULT			'WEI'	2	32.8 ENGL	'HR'	0.05	0.0005			
ENGLISH	DEG				FT/SEC			SEC/FT	SEE COLUMN 66		
METRIC	DEG				M/SEC			SEC/M	M/SEC		

DYNAMIC RESPONSE OPTIONS LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IN ANY DYNAMIC RESPONSE RUN. IT SPECIFIES OVERALL ANALYSIS PARAMETERS AND TYPE OF ANALYSIS.
(7-10)	ENTER THE TYPE OF ANALYSIS TO BE PERFORMED FROM THE FOLLOWING OPTIONS: 'SPEC' - BASE DRIVEN SPECTRAL ANALYSIS (SPECTRAL EARTHQUAKE) 'TIME' - BASE DRIVEN TIME HISTORY ANALYSIS (TIME HISTORY EARTHQUAKE) 'VIBR' - FORCE DRIVEN ANALYSIS (PERIODIC OR TIME HISTORY) 'WIND' - WIND SPECTRAL ANALYSIS (EXTREME OR FATIGUE ANALYSIS) 'ENGV' - ENGINE VIBRATION ANALYSIS 'TRNF' - REACTION DRIVEN TRANSFER FUNCTION ANALYSIS 'TCLP' - GENERATE INCREMENTAL LOADS FOR COLLAPSE ANALYSIS 'TESL' - GENERATE EQUIVALENT STATIC LOADS FOR LINEAR ANALYSIS 'SHIP' - DYNAMIC SHIP IMPACT ANALYSIS 'DROP' - DYNAMIC DROPPED OBJECT ANALYSIS
(11-14)	ENTER THE NUMBER OF MODES TO BE USED IN THE ANALYSIS. THE DEFAULT IS THE NUMBER OF MODES COMPUTED BY THE PREVIOUS DYNPAC RUN.
(15-16)	ENTER 'EC' IF A LISTING OF THE INPUT DATA IS TO BE INCLUDED IN THE OUTPUT LISTING FILE.
(17-18)	ENTER THE VERTICAL COORDINATE (-X, Y, OR Z, OR +X, Y, OR Z)
(19-24)	ENTER THE MUDLINE ELEVATION IN THE VERTICAL COORDINATE.
(25-27)	ENTER THE SELECTIONS FOR PRINTING THE JOINT ACCELERATIONS, VELOCITIES, AND/OR DISPLACEMENTS. FOR SPECTRAL ANALYSIS ONLY. 'A' - ACCELERATIONS 'V' - VELOCITIES 'D' - DISPLACEMENTS

LINE LABEL	ANALYSIS SELECTION OPTION	NUMBER OF MODES	INPUT ECHO OPTION	VERTICAL COORDINATE	MUDLINE ELEVATION	OUTPUT SELECTION (SPECTRAL ONLY)		
						FIRST SELECTION	SECOND SELECTION	THIRD SELECTION
DROPT								
1-- 5	7--10	11-->14	15--16	17--18	19<--24	25	26	27
DEFAULT				' +Z'	0			
ENGLISH					FT			
METRIC					M			

END LINE

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS REQUIRED AND SIGNALS THE END OF THE RESPONSE CALCULATIONS AND DIRECTS THE PROGRAM TO GENERATE THE OUTPUT FOR THE 'COMBINE' PROGRAM.

LINE LABEL	LEAVE BLANK
END	
1-- 3	4-----80

LOAD HEADER

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED AND SPECIFIES THAT LOADING DATA (SPECTRAL, TIME HISTORY, OR VIBRATION) FOLLOWS AS DESIGNATED ON THE DROPT LINE.

LINE LABEL	LEAVE BLANK
LOAD	
1-- 4	5-----80

STRUCTURAL MODAL DAMPING

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE MODAL DAMPING VALUES. THESE VALUES SHOULD INCLUDE ALL SOURCES OF DAMPING OTHER THAN THE FLUID DAMPING IF AN ITERATIVE SOLUTION IS BEING USED. IN THAT CASE THE FLUID DAMPING IS ACCOUNTED FOR BY CALCULATING THE ACTUAL FLUID FORCES DUE TO MOTION OF THE STRUCTURE AND CALCULATING AN EQUIVALENT LINEAR MODAL DAMPING VALUE WHICH IS ADDED TO THESE VALUES TO OBTAIN THE TOTAL DAMPING. IF THE FLUID DAMPING IS NOT BEING CALCULATED BY THE PROGRAM, THEN THE USER SHOULD INCLUDE THE FLUID DAMPING IN THIS DATA OR ENTER THE FLUID DAMPING ON THE FDAMP LINE.
(11-15)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL IF ALL MODES ARE TO HAVE THE SAME VALUE, OTHERWISE LEAVE THIS FIELD BLANK. IF A VALUE IS ENTERED HERE, LEAVE THE REMAINING FIELDS ON THIS LINE BLANK.
(21-25)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL FOR THE FIRST MODE.
(26-70)	ENTER MODAL DAMPING VALUES FOR THE REMAINING MODES. IF THE NUMBER OF MODES IS GREATER THAN TEN, USE ADDITIONAL LINES WITH THE REMAINING DAMPING VALUES ENTERED IN COLUMNS 21-70.

LINE LABEL	MODAL DAMPING VALUES (PERCENT OF CRITICAL)										
	OVERALL MODAL DAMPING	MODE 1	MODE 2	MODE 3	MODE 4	MODE 5	MODE 6	MODE 7	MODE 8	MODE 9	MODE 10
SDAMP											
1-- 5	11<--15	21<--25	26<--30	31<--35	36<--40	41<--45	46<--50	51<--55	56<--60	61<--65	66<--70

SPECTRAL WIND

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS OPTIONAL FOR A SPECTRAL WIND ANALYSIS. IF OMITTED, ALL DEFAULTS WILL BE USED.
(8- 9)	SELECT THE TYPE OF ANALYSIS: 'FT' - FATIGUE ANALYSIS (FATIGUE DATA FILE CREATED). 'EX' - EXTREME EVENT ANALYSIS (COMBINE DATA FILE CREATED).
(11-12)	SELECT THE PRINT LEVEL: 'MN' - MINIMUM PRINT (ONE LINE PER MODE PER WIND). 'MD' - MODERATE PRINT (ONE PAGE PER MODE PER WIND). 'MX' - MAXIMUM PRINT (INCLUDES SPECTRUM FOR EACH MODE).
(14-15)	ENTER 'PL' IF THE GENERALIZED FORCE SPECTRUM IS TO PLOTTED. THIS WILL GENERATE ONE PLOT FOR EACH MODE FOR EACH WIND.
(17-18)	ENTER 'PL' IF THE RESPONSE SPECTRUM IS TO PLOTTED. THIS WILL GENERATE ONE PLOT FOR EACH MODE FOR EACH WIND.
(35-36)	ENTER THE SPATIAL CORRELATION COEFFICIENT OPTION AS FOLLOWS: 'CA' - USE CALCULATED VALUE. 'SK' - DO NOT USE A SPATIAL CORRELATION COEFFICIENT FOR THE SPECTRAL WIND ANALYSIS.
(37-38)	SELECT FROM THE FOLLOWING: 'DR' - SPECIFY SPECTRUM ('WINSPC' LINE) FOR EACH DIRECTION 'VL' - SPECIFY SPECTRUM ('WINSPC' LINE) FOR EACH WIND VELOCITY
(39-44)	ENTER MEAN WIND SPEED AVERAGING TIME. THE DATA IS USED TO CALCULATE THE DYNAMIC AMPLIFICATION FACTOR FOR THE SPECTRAL WIND ANALYSIS.
(45-50)	ENTER THE REFERENCE LENGTH USED FOR THE HARRIS SPECTRUM. THIS IS USED TO CALCULATE THE MODAL RESPONSES FOR THE SPECTRAL WIND ANALYSIS.
(51-56)	ENTER THE ROUGHNESS COEFFICIENT FOR THE HARRIS WIND SPECTRUM.

LINE LABEL	OUTPUT OPTIONS				SPATIAL CORRELATION COEFF. OPTION	GUST SPECTRUM OPTION	WIND AVERAGING TIME	REFERENCE LENGTH	ROUGHNESS COEFFICIENT	LEAVE BLANK
	ANALYSIS TYPE SELECTION	PRINT LEVEL	PLOT GEN. FORCE SPECTRUM	PLOT RESPONSE SPECTRUM						
SPCWIN										
1-- 6	8-- 9	11--12	14--15	17--18	35--36	37--38	39<--44	45<--50	51<--56	57--80
DEFAULT	'FT'	'MN'	NO	NO	'CA'	'DR'	600	1800. METRIC	0.0025	
ENGLISH							SEC	FT		
METRIC							SEC	M		

WIND DIRECTION SPECTRUM

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED FOR A SPECTRAL WIND FATIGUE ANALYSIS AND IS OPTIONAL EXTREME WIND ANALYSIS. EACH WIND DIRECTION IN THE SEASTATE DATA MUST HAVE A CORRESPONDING 'WINSPC' DATA ENTRY.
(7-13)	ENTER THE WIND DIRECTION. THIS ENTRY IS USED TO CROSS CHECK THE 'WINSPC' DATA WITH THE SEASTATE DATA.
(14-20)	ENTER THE FRACTION OF TIME FOR THIS WIND DIRECTION (FATIGUE ONLY).
(22-24)	SELECT THE TYPE OF MEAN WIND DISTRIBUTION. 'WEI' - WEIBULL DISTRIBUTION NOTE: WEIBULL IS THE ONLY WIND DISTRIBUTION AVAILABLE AT THIS TIME.
(26-32)	ENTER THE 'K' WEIBULL DISTRIBUTION PARAMETER.
(33-39)	ENTER THE 'A' WEIBULL DISTRIBUTION PARAMETER.
(40-41)	SELECT FROM THE FOLLOWING GUST SPECTRUM TYPES: 'HR' - HARRIS 'VK' - VON KARMAN 'FR' - FROYA/API/NORSOK/ISO19902
(42-57)	IF THE VON KARMAN SPECTRUM IS SELECTED, ENTER THE PARAMETERS REQUIRED TO CALCULATE THE TURBULENCE INTENSITY FACTOR (TIF) WHERE: $TIF = A + B * \text{WIND VELOCITY}$
(58-65)	ENTER THE WIND VELOCITY. NOTE: 1 MILE PER HOUR = 0.8690 KNOTS.
(66)	SELECT THE WIND VELOCITY UNITS ENTER 'M' FOR MILES PER HOUR, 'F' FOR FEET PER SECOND, 'K' FOR KNOTS, OR LEAVE BLANK FOR METERS PER SECOND.

LINE LABEL	WIND DIRECTION	FRACTION OF TIME	MEAN WIND DISTRIBUTION TYPE	WEIBULL K PARAMETER	WEIBULL A PARAMETER	GUST SPECTRUM TYPE	VON KARMAN SPECTRUM PARAMETERS		WIND LOAD DATA		LEAVE BLANK
							A VALUE	B VALUE	VELOCITY	VELOCITY UNITS	
WINSPC											
1-- 6	7<--13	14--20	22--24	26<--32	33<--39	40--41	42<--49	50<--57	58<--65	66	69--80
DEFAULT			'WEI'	2	32.8 ENGL	'HR'	0.05	0.0005			
ENGLISH	DEG				FT/SEC			SEC/FT	SEE COLUMN 66		
METRIC	DEG				M/SEC			SEC/M	M/SEC		